ELEMENTARY PROBABILITY

SECOND EDITION

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7
Continuous Random Variables

He talks at random: sure the man is mad.
W. Shakespeare, Henry VI

7.1 Density and Distribution

Hitherto, we have assumed that a random variable can take any one of only a countable set of values. However, suppose your height is 5 feet or 6 feet (or somewhere in between). Then, previously (however briefly), your height in feet has taken every value in [1, 5], including $\sqrt{2}, e, \pi$, and so on. (Each value can be taken more than once because you are taller in the morning than in the evening.) Thus, if X is the height of a randomly selected member of the population, the state space of X is not countable. There are many other simple examples of variables that may take any one of an uncountable number of values; for example, the brightness of a randomly chosen star, the time until some cell divides, the velocity of a comet, the direction of the wind, and so on. Think of some yourself.

In view of these remarks, we are about to introduce a new class of random variables such that the state space Ω is uncountable, and $X(\omega)$ may take any one of an uncountable number of real values. However, before we embark on this task, it is as well to reassure you that, despite their separate presentation, these new random variables share most of the useful properties of discrete random variables. Also, many of these properties are proved in exactly the same way as in the discrete case and (even better) we are able to use much of the same notation.

Thus, as in the discrete case, we start with a probability function $\mathrm{P}(\cdot)$ defined on a collection $\mathcal{F}$ (the event space) of subsets of Ω (the sample space). Then we think of a random variable X as a real valued function $X(\omega)$ defined for each $\omega \in \Omega$.

Our first requirement (as in the discrete case) is a function that tells us about the relative likelihoods of possible values of X. Happily, we already have such a function; recall the following:

(1) Definition *The* distribution function *F of the random variable X is the function*

(2) $$F(x) = \mathbf{P}(A_x),$$

where A_x *is the event* $A_x = \{\omega : X(\omega) \leq x\}$, $x \in \mathbb{R}$. ▲
We usually write (2) as

(3) $$F(x) = \mathbf{P}(X \leq x),$$

and denote F by $F_X(x)$ *when we want to stress the role of X.*

Notice that for Definition 1 to be meaningful, the event A_x must be in $\mathcal{F}$, so that we know $\mathbf{P}(A_x)$. Thus, $X(\cdot)$ only qualifies to appear in Definition 1 if $A_x \in \mathcal{F}$ for all x. This is true throughout this book; the implications of this so-called *measurability* condition are explored in more advanced books.

(4) Example: Uniform Distribution You devise an experiment in which the outcome is equally likely to be any point Q in the interval [0, 1]. Thus, the sample space is the set of points $(Q : Q \in [0, 1])$. The event space $\mathcal{F}$ will include all intervals in [0, 1]; we omit the proof that such an $\mathcal{F}$ exists, but you should rest assured that it does exist. Define the random variable $X(Q)$ to be the distance from the origin O to Q. From the nature of the experiment, if

$$A_{ab} = \{Q : Q \in (a, b)\}, \quad 0 \leq a \leq b \leq 1,$$

then $\mathbf{P}(A_{ab}) = b - a$. Hence, X has distribution function sketched in Figure 7.1

$$F_X(x) = \mathbf{P}(A_{0x}) = \begin{cases} 0 & \text{if} \quad x < 0 \\ x & \text{if} \quad 0 \leq x \leq 1 \\ 1 & \text{if} \quad x > 1. \end{cases}$$ ●

In the future, the underlying sample space will make few appearances. We tend to think of the possible values of X as the sample space, as we did for discrete random variables. (It can be proved that this is a permissible view.)

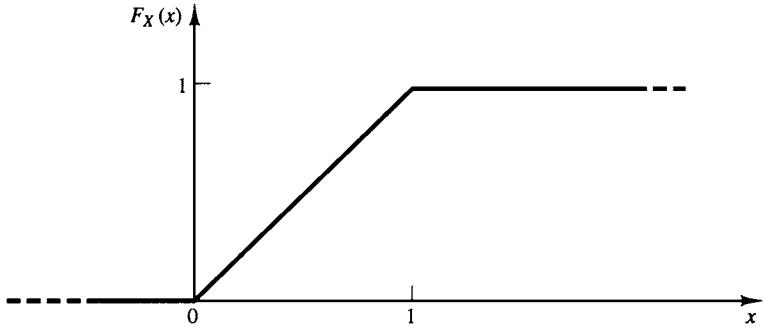


Figure 7.1 The distribution function $F_X(x)$ of a random variable X distributed uniformly on (0, 1).

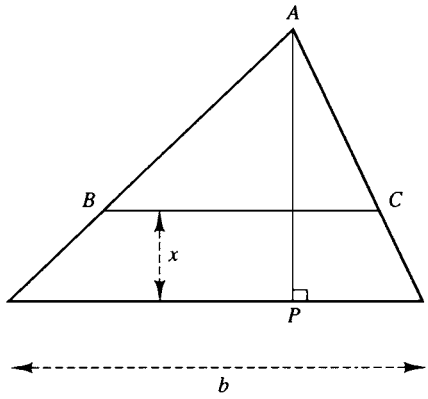


Figure 7.2 The altitude AP is of length h.

(5) Example A point Q is picked at random in a triangle of area a, with base of length b. Let X be the perpendicular distance from Q to the base. What is the distribution function $F_X(x)$? [See figure 7.2 for a sketch of the triangle].

Solution Let the height of the triangle AP be h. The event $X > x$ occurs when Q lies inside the triangle ABC. For reasons of symmetry,

$$\mathbf{P}(Q \in \text{ABC}) = (\text{area of ABC})/a = \left(\frac{h-x}{h}\right)^2.$$

Hence,

$$F_X(x) = \mathbf{P}(X \le x) = 1 - \mathbf{P}(Q \in ABC) = \begin{cases} 0 & x < 0 \\ 1 - \left(\dfrac{h-x}{h}\right)^2 & 0 \le x \le h \\ 1 & x > h. \end{cases} \quad \bullet$$

We summarize the basic properties of $F(x)$ in the following:

(6) Theorem *Let X have distribution function $F(x)$. Then $0 \le F(x) \le 1$ for all x, and*

(7) $$\mathbf{P}(x < X \le y) = F(y) - F(x) \ge 0, \text{ for all } x \le y.$$

(8) *If $h > 0$, then*

$$\lim_{h \to 0} F(x+h) = F(x).$$

(9) *If $\mathbf{P}(|X| < \infty) = 1$, then*

$$\lim_{x \to \infty} F(x) = 1 \quad \text{and} \quad \lim_{x \to -\infty} F(x) = 0.$$

In plain words, this theorem says that as x increases, $F(x)$ is nondecreasing, continuous on the right, and lies between 0 and 1. It can be shown conversely that any function with these properties is the distribution function of some random variable.

Proof The first result follows from (2) because $0 \leq \mathbf{P}(A_x) \leq 1$. To show (7), note that

$$\{\omega: x < X \leq y\} = A_y \cap A_x^c = A_y \backslash A_x.$$

Because $A_x \subseteq A_y$ for $x \leq y$, we have

$$\mathbf{P}(x < X \leq y) = \mathbf{P}(A_y) - \mathbf{P}(A_x) = F(y) - F(x) \geq 0$$

by the nonnegativity of $\mathbf{P}(\cdot)$. To prove (8), we use the continuity of $\mathbf{P}(\cdot)$, see (1.5.4.). Let $(h_k; k \geq 1)$ be any sequence decreasing to zero, and let $A_{(k)}$ be the event that $X \leq x + h_k$. Then

$$\begin{aligned}\lim_{h \to 0} F(x+h) &= \lim_{k \to \infty} F(x + h_k) = \lim_{k \to \infty} \mathbf{P}(A_{(k)}) \\ &= \mathbf{P}(\lim_{k \to \infty} A_{(k)}) = \mathbf{P}(A_x) = F(x),\end{aligned}$$

as required. Finally, for (9),

$$\lim_{x \to \infty} F(x) = \lim_{n \to \infty} \mathbf{P}(A_n) = \mathbf{P}(\Omega) \quad \text{by (1.5.4).}$$

The last part is proved similarly. ■

Although the distribution function has not played a very active role so far in this book, it now assumes a greater importance. One reason for this is Theorem 6 above, which shows that $F(x)$ really is a function that can tell us how likely X is to be in some simple subset of the real line. Another reason is the following simple corollary of Example 5.

(10) Corollary *If $F(x)$ is continuous, then for all x* $\mathbf{P}(X = x) = 0$.

Proof

$$\begin{aligned}\mathbf{P}(X = x) &= \lim_{n \to \infty} \mathbf{P}\left(x - \frac{1}{n} < X \leq x\right) \qquad \text{by (1.5.4)} \\ &= \lim_{n \to \infty} \left(F(x) - F\left(x - \frac{1}{n}\right)\right) \quad \text{by Theorem 6} \\ &= 0 \quad \text{because } F \text{ is continuous.}\end{aligned}$$

■

(11) Example Let X be uniformly distributed on $(0, 1)$. Then

$$F(x) = \begin{cases} 0 & x \leq 0 \\ x & 0 < x < 1 \\ 1 & x \geq 1 \end{cases}$$

is clearly continuous, so that $\mathbf{P}(X = x) = 0$ for all x. ■

If F_X is continuous, then X is known as a continuous random variable. We now define a particularly important class of continuous random variables.

(12) Definition *Let X have distribution function F. If the derivative*

$$\frac{dF}{dx} = F'(x) \tag{13}$$

exists at all but a finite number of points, and the function f defined by

$$f(x) = \begin{cases} F'(x) & \text{where } F'(x) \text{ exists} \\ 0, & \text{elsewhere,} \end{cases} \tag{14}$$

satisfies

$$F(x) = \int_{-\infty}^{x} f(v)\,dv, \tag{15}$$

then X is said to be a continuous random variable with density $f(x)$. ▲

It follows from (15) that if X has density $f(x)$, then for $C \subseteq \mathbb{R}$ the Key Rule is:

$$\mathbf{P}(X \in C) = \int_C f(x)dx,$$

when both sides exist. [In line with the remark following (3), they exist if $\{\omega: X(\omega) \in C\} \in \mathcal{F}$.]

Example 4 Revisited: Uniform Density In Example 4, we found that if X was chosen uniformly at random in [0, 1], then

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x & 0 \le x \le 1 \\ 1 & x > 1. \end{cases}$$

It follows that X has a density

$$f_X(x) = \begin{cases} 1 & 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$ ●

Figures 7.3 to 7.6 illustrate the density and distribution of random variables Y and Z, which are uniformly distributed on (a, b) and $(a, b) \cup (c, d)$, respectively.

Example (5) Revisited For the point Q picked at random in a triangle, where X is the perpendicular distance from Q to the base, we found

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - \left(\dfrac{h-x}{h}\right)^2 & 0 \le x \le h \\ 1 & x > h. \end{cases}$$

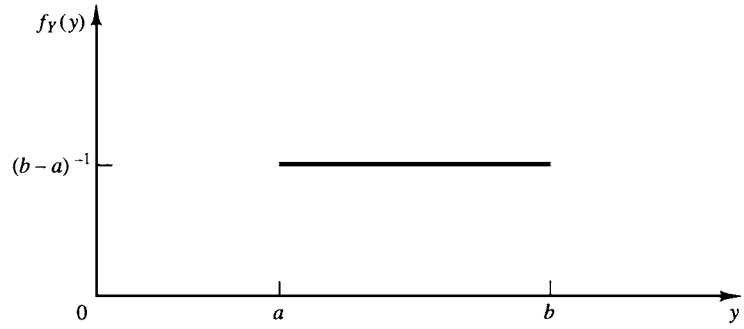


Figure 7.3 The density function $f(y)$ of a random variable distributed uniformly on (a, b).

It follows that X has density

$$f_X(x) = \begin{cases} 2(h-x)/h^2 & 0 < x < h \\ 0 & \text{otherwise.} \end{cases}$$

●

There are continuous random variables that do not have a density, but none appear in this text. Therefore, whenever X is said to be continuous here, it always follows that X has a density f.

Let us consider a basic example. In the discrete case, the geometric distribution is of the form $F(x) = 1 - q^x$, if $x = 1, 2, \ldots$ It is natural to consider the analogous distribution in the continuous case, which turns out to be equally (if not more) important.

(16) Example: Exponential Density Let X have distribution

$$F(x) = \max\{0, 1 - e^{-\lambda x}\},$$

where $\lambda > 0$. Then F is continuous for all x, and also differentiable except at $x = 0$, so

$$F'(x) = \begin{cases} 0 & \text{if } x < 0, \\ \lambda e^{-\lambda x} & \text{if } x > 0. \end{cases}$$

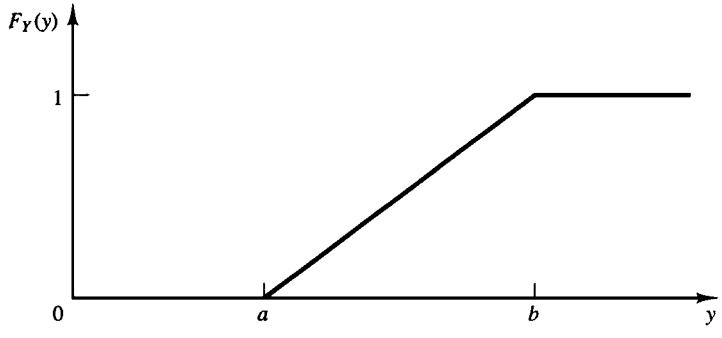


Figure 7.4 The distribution function $F(y)$ of a random variable distributed uniformly on (a, b).

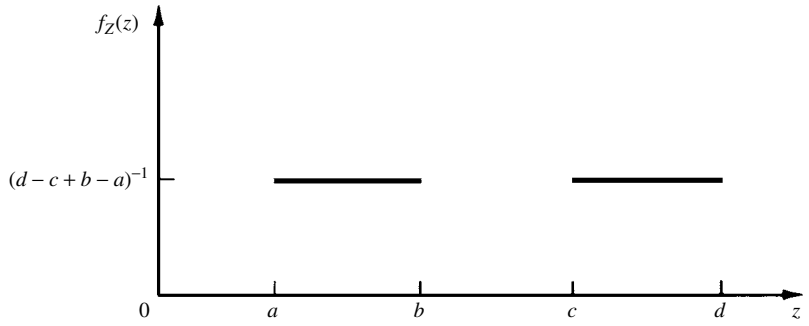


Figure 7.5 The density function of a random variable distributed uniformly on $(a, b) \cup (c, d)$, where $a < b < c < d$.

Now let

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x > 0, \\ 0 & x \leq 0. \end{cases}$$

Then of course, for $x > 0$,

$$F(x) = 1 - e^{-\lambda x} = \int_0^x \lambda e^{-\lambda v} dv = \int_\infty^x f(v) dv,$$

and, for $x < 0$,

$$F(x) = 0 = \int_{-\infty}^x f(v) dv.$$

Hence, $f(x)$ is a density of X. See Figures 7.7 and 7.8. ●

Notice that $F'(0)$ does not exist, and also that the function

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

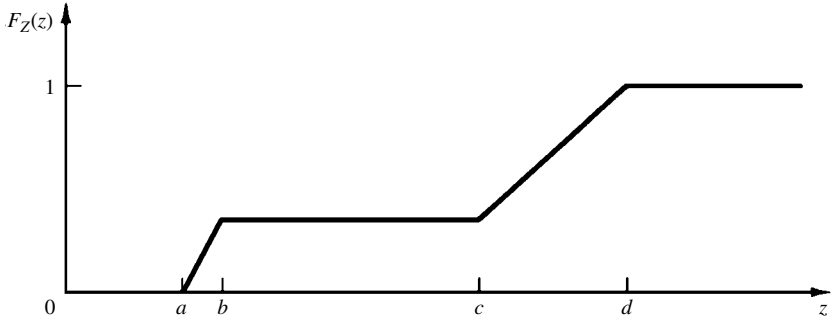


Figure 7.6 The distribution function of a random variable distributed uniformly on $(a, b) \cup (c, d)$, where $a < b < c < d$.

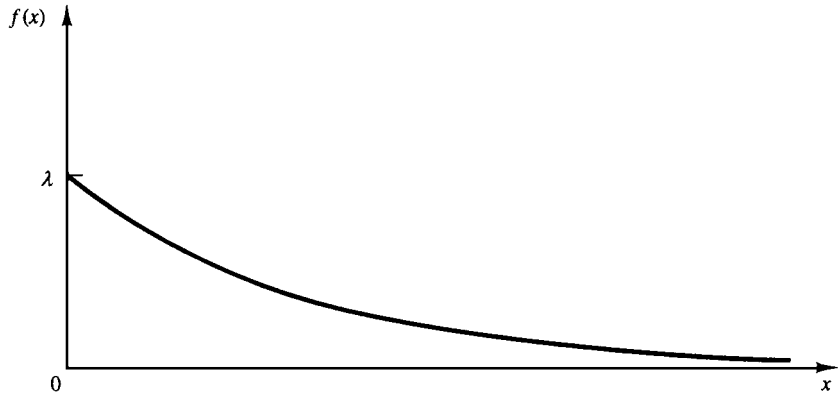


Figure 7.7 The density of an exponential random variable with parameter λ; $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$.

would also satisfy $F(x) = \int_{-\infty}^{x} f(v)dv$. This illustrates the fact that (15) does not uniquely determine $f(x)$ given $F(x)$. However, this problem can only arise at a finite number of points and (to some extent) it does not matter what value we give $f(x)$ at any of these exceptional points. Usually we make it zero, but consider the following:

(17) Example: Two-Sided Exponential Distribution Let X have distribution

$$F(x) = \begin{cases} pe^{\lambda x} & \text{if } x < 0 \\ 1 - (1-p)e^{-\lambda x} & \text{if } x \geq 0, \end{cases}$$

where $0 < p < 1$ and $\lambda > 0$. Then $F(x)$ is continuous and

$$F'(x) = \begin{cases} \lambda p e^{\lambda x} & \text{if } x < 0 \\ \lambda(1-p)e^{-\lambda x} & \text{if } x > 0. \end{cases}$$

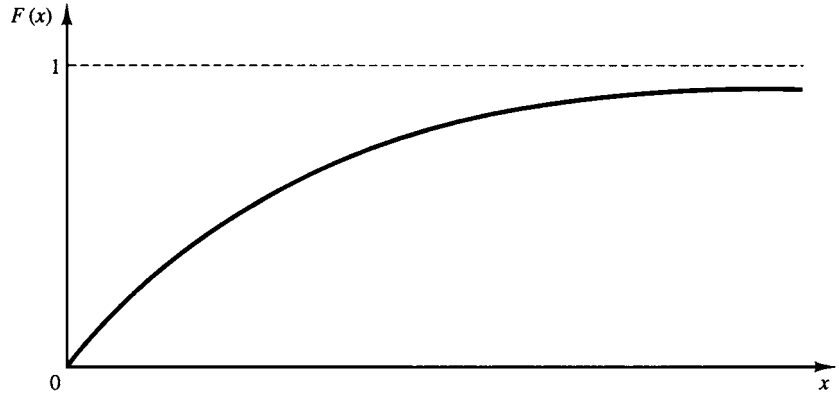


Figure 7.8 The distribution of an exponential random variable with parameter λ; $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$.

A suitable density is

$$f(x) = \begin{cases} \lambda p e^{\lambda x} & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ \lambda(1-p)e^{-\lambda x} & \text{if } x > 0. \end{cases}$$

However, if $p = \frac{1}{2}$, it is tempting to set $f(0) = \frac{\lambda}{2}$ and write $f(x) = \frac{1}{2}\lambda e^{-\lambda|x|}$ for all x. The point is that it really does not matter very much. ●

Finally, we note the obvious facts that for any density f,

$$f(x) \geq 0, \tag{18}$$

and if $\mathbf{P}(|X| < \infty) = 1$, then

$$\int_{-\infty}^{\infty} f(v)dv = 1. \tag{19}$$

It is straightforward to see that any function with these properties is the density of some random variable, and so any integrable nonnegative function can be used to form a density.

Example The function $g(x) = x^2 - x + 1$ is easily seen to be nonnegative for all x, and

$$\int_{+a}^{+b} g(v)dv = \frac{1}{3}(b^3 - a^3) - \frac{1}{2}(b^2 - a^2) + b - a = c(a, b), \quad \text{say.}$$

Hence, the function

$$f(x) = \begin{cases} c(a,b)^{-1} g(x) & \text{if } a < x < b \\ 0 & \text{elsewhere} \end{cases}$$

is a density function. ●

(20) Example: Cauchy Distribution Show that for an appropriate choice of the constant $c(a, b)$, the function

$$f(x) = \begin{cases} c(a,b)(1+x^2)^{-1} & \text{if } a < x < b \\ 0 & \text{elsewhere} \end{cases}$$

is a density function. Show that $c(-\infty, \infty) = 1/\pi$ and that $c(-1, 1) = c(0, \infty) = 2/\pi$.

Solution Trivially, $f \geq 0$ if $c \geq 0$. Now recall that

$$\frac{d}{dx}\tan^{-1} x = (1+x^2)^{-1}.$$

Thus, $\int_a^b f(x)dx = 1$ if and only if $c(a, b)^{-1} = \tan^{-1} b - \tan^{-1} a$. In particular,

$$c(-\infty, \infty)^{-1} = \frac{\pi}{2} + \frac{\pi}{2} = \pi,$$

as required, and

$$c(-1, 1)^{-1} = \frac{\pi}{4} + \frac{\pi}{4} = c(0, \infty)^{-1}.$$

●

In general then, given a nonnegative function $g(x)$, the function

$$f(x) = g(x)\left(\int_{-\infty}^{\infty} g(v)dv\right)^{-1}$$

is a density, if the integral exists. To discover whether it does, the following technique is useful. If we can find a constant b such that for all $n > 0$, $\int_0^n g(v)dv < b < \infty$, then

$$\int_0^{\infty} g(v)dv = \lim_{n\to\infty} \int_0^n g(v)dv$$

exists by monotone convergence.

(21) Example: Normal Density Show that

$$f = c\exp\left(-\frac{1}{2}x^2\right) \quad \text{for all } x \in \mathbb{R}$$

can be a density.

Solution For any $n > 1$,

$$\int_{-n}^{n} \exp\left(-\frac{1}{2}v^2\right) dv < 2\left(\int_0^1 dv + \int_1^n e^{-v}dv\right) < 2(1 + e^{-1}).$$

Hence, $c^{-1} = \int_{-\infty}^{\infty} \exp(-\frac{1}{2}v^2)dv$ exists, and f is a density for this c. ●

Remark In fact, it can be shown that

(22) $$c^{-1} = (2\pi)^{1/2}.$$

The proof of this is not quite trivial; we give it in Example 8.3.8. Also, note that there are other normal densities; the one in Example 21 is called the *standard* normal density denoted by $N(0, 1)$, and by

(23) $$\phi(x) = (2\pi)^{-1/2}\exp(-\frac{1}{2}x^2).$$

Its distribution is $\Phi(x)$, given by

(24) $$\Phi(x) = \int_{-\infty}^{x} \phi(v)dv.$$

(25) Example: Gamma Distribution Show that for $\alpha, \lambda, x > 0$, the function $f(x) = c\lambda^{\alpha}x^{\alpha-1}e^{-\lambda x}$ can be a density. When α is a positive integer, show that $c^{-1} = (\alpha - 1)!$

Solution First, we show that the integral $\int_0^\infty x^{\alpha-1}e^{-\lambda x}dx$ exists. Because $x^{\alpha-1}e^{-\frac{1}{2}\lambda x} \to 0$ as $x \to \infty$, there is some $m < \infty$ such that for $x > m$, $x^{\alpha-1}e^{-\frac{1}{2}\lambda x} < 1$. Hence, for $n > m$,

$$\begin{aligned}\int_0^n v^{\alpha-1}e^{-\lambda v}dv &< \int_0^m v^{\alpha-1}e^{-\lambda v}dv + \int_m^n e^{-\frac{1}{2}\lambda v}dv \\ &< \int_0^m v^{\alpha-1}e^{-\lambda v}dv + 2\lambda^{-1}e^{-\frac{1}{2}\lambda m} = b \text{ (say).}\end{aligned}$$

Hence, $c^{-1} = \int_0^\infty \lambda^\alpha x^{\alpha-1}e^{-\lambda x}dx$ exists, and if α is a positive integer then repeated integration by parts gives $c^{-1} = (\alpha-1)!\int_0^\infty \lambda e^{-\lambda v}dv = (\alpha-1)!$ ●

This density is known as the *gamma density* with parameters α and λ. When α is not an integer, the integral above defines a function of α known as the *gamma function*, and denoted by

$$\Gamma(\alpha) = \int_0^\infty \lambda^\alpha v^{\alpha-1}e^{-\lambda v}dv = \int_0^\infty u^{\alpha-1}e^{-u}du. \tag{26}$$

The density function in this case is

$$f(x) = \frac{1}{\Gamma(\alpha)}\lambda^\alpha x^{\alpha-1}e^{-\lambda x} \quad \text{for } x > 0. \tag{27}$$

In particular, when $\lambda = \frac{1}{2}$ and $\alpha = \frac{n}{2}$, the density

$$f(x) = \frac{1}{\Gamma(\frac{n}{2})}\lambda^{\frac{n}{2}}x^{\frac{n}{2}-1}e^{\frac{-1}{2}x} \tag{28}$$

is known as the $\chi^2(n)$ density, and it is referred to as *chi-squared* with n degrees of freedom.

We conclude with another way of producing densities.

(29) Example: Mixtures Let $f_1(x)$ and $f_2(x)$ be density functions, and let $f_3(x) = \gamma f_1(x) + (1-\gamma)f_2(x)$, where $0 \le \gamma \le 1$. Then $f_3(x) \ge 0$, and

$$\int f_3 = \gamma \int f_1 + (1-\gamma)\int f_2 = 1.$$

Hence, f_3 is a density, and is said to be a *mixture* of f_1 and f_2. ●

For example, the two-sided exponential density of Example 17 may now be seen as a mixture of $f_1 = \lambda e^{\lambda x}(x < 0)$ and $f_2 = \lambda e^{-\lambda x}(x > 0)$ with $\gamma = p$.

7.2 Functions of Random Variables

Suppose that X and Y are random variables such that $Y = g(X)$, where $g(.)$ is some given function. If we know the density of X, can we find the distribution of Y? In general terms, the answer is straightforward because, by the properties of densities and

distributions,

$$F(y) = \mathbf{P}(Y \leq y) = \mathbf{P}(g(X) \leq y) = \int_C f_X(v)dv, \tag{1}$$

where $C = \{v : g(v) \leq y\}$. Then, if $F(y)$ is continuous and differentiable, we can go on to find the density of Y, if it exists. Here are some simple examples of this idea in practice.

(2) Example Let X be uniformly distributed on $(0, 1)$ with density

$$f(x) = \begin{cases} 1 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

If $Y = -\lambda^{-1} \log X$, where $\lambda > 0$, what is the density of Y?

Solution First, we seek the distribution of Y:

$$\begin{aligned} F_Y(y) &= \mathbf{P}(-\lambda^{-1} \log X \leq y) = \mathbf{P}(\log X \geq -\lambda y) \\ &= \mathbf{P}(X \geq \exp(-\lambda y)) = \begin{cases} 1 - e^{-\lambda y} & \text{for } y \leq 0 \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

Hence, the derivative exists except at $y = 0$, and

$$f_Y(y) = \begin{cases} \lambda e^{-\lambda y} & \text{if } y > 0 \\ 0 & \text{if } y \leq 0. \end{cases}$$

This is the exponential density with parameter λ. ●

Some care is required if $g(.)$ is not one–one.

(3) Example Let X be uniformly distributed on $[-1, 1]$. Find the density of $Y = X^r$ for nonnegative integers r.

Solution First, note that X has distribution function $F(x) = \frac{1}{2}(1 + x)$ for $-1 \leq x \leq 1$. Now, if r is odd, then the function $g(x) = x^r$ maps the interval $[-1, 1]$ onto itself in one–one correspondence. Hence, routinely:

$$\mathbf{P}(Y \leq y) = \mathbf{P}(X^r \leq y) = \mathbf{P}(X \leq y^{1/r}) = \frac{1}{2}(1 + y^{1/r}) \quad \text{for } -1 \leq y \leq 1,$$

and Y has density

$$f(y) = \frac{1}{2r} y^{\frac{1}{r}-1}, \quad -1 \leq y \leq 1.$$

If r is even, then $g(x) = x^r$ takes values in $[0, 1]$ for $x \in [-1, 1]$. Therefore,

$$\mathbf{P}(Y \leq y) = \mathbf{P}(0 \leq X^r \leq y) = \mathbf{P}(-y^{1/r} \leq X \leq y^{1/r}) = y^{1/r} \quad \text{for } 0 \leq y \leq 1.$$

Hence, Y has density

$$f(y) = \frac{1}{r} y^{\frac{1}{r}-1}, \quad 0 \leq y \leq 1.$$

Finally, if $r = 0$, then $X^r = 1$, $F_Y(y)$ is not continuous (having a jump from 0 to 1 at $y = 1$) and so Y does not have a density in this case. Obviously, Y is discrete, with $\mathbf{P}(Y = 1) = 1$. ●

(4) Example Let X have the standard normal distribution with density

$$f(x) = (2\pi)^{-\frac{1}{2}} \exp\left(-\frac{1}{2}x^2\right).$$

Find the density of $Y = \sigma X + \mu$ for given constants μ and $\sigma \neq 0$. Also, find the density of $Z = X^2$.

Solution Adopting the by now familiar technique:

$$\mathbf{P}(\sigma X + \mu \leq y) = \mathbf{P}(\sigma X \leq y - \mu) = \begin{cases} \mathbf{P}\left(X \leq \dfrac{y-\mu}{\sigma}\right) & \text{if } \sigma > 0 \\ \mathbf{P}\left(X \geq \dfrac{y-\mu}{\sigma}\right) & \text{if } \sigma < 0 \end{cases} \tag{5}$$

$$= \begin{cases} F_X\left(\dfrac{y-\mu}{\sigma}\right) & \text{if } \sigma > 0 \\ 1 - F_X\left(\dfrac{y-\mu}{\sigma}\right) & \text{if } \sigma < 0 \end{cases}$$

Hence, differentiating (5) with respect to y,

$$f_Y(y) = \frac{1}{|\sigma|} f_X\left(\frac{y-\mu}{\sigma}\right) = \frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}} \exp\left(-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2\right). \tag{6}$$

Second,

$$\mathbf{P}(X^2 \leq z) = \mathbf{P}(X \leq \sqrt{z}) - \mathbf{P}(X \leq -\sqrt{z}) = F_X(\sqrt{z}) - F_X(-\sqrt{z}).$$

Differentiating now gives

$$f_Z(z) = \frac{1}{2\sqrt{z}} f_X(\sqrt{z}) + \frac{1}{2\sqrt{z}} f_X(-\sqrt{z}) = \frac{1}{\sqrt{2\pi z}} \exp\left(-\frac{1}{2}z\right). \tag{7}$$ ●

Remark The density given by (6) is known as the *normal* density with parameters μ and σ^2, sometimes denoted by $N(\mu, \sigma^2)$. The standard normal density of Example 7.1.21 was $N(0, 1)$ because $\phi(x)$ has $\mu = 0$ and $\sigma = 1$.

The density given by (7) is the gamma density of (7.1.23) with parameters $\frac{1}{2}$ and $\frac{1}{2}$. This is known as the chi-squared density with parameter 1, sometimes denoted by $\chi^2(1)$. This is a special case of (7.1.28).

(8) Example: Inverse Functions Let X have distribution function $F(x)$, where $F(x)$ is continuous and strictly increasing. Let $g(x)$ be a function satisfying $F(g) = x$. Because

$F(x)$ is continuous and strictly increasing, this defines $g(x)$ uniquely for every x in $(0, 1)$. The function $g(.)$ is called the inverse function of $F(.)$ and is often denoted by

$$g(x) = F^{-1}(x).$$

Clearly, F is the inverse function of g, that is

$$g(F(x)) = F(g(x)) = x, \tag{9}$$

and $g(x)$ is an increasing function.

(a) Use this function to show that $Y = F(X)$ is uniformly distributed on $(0, 1)$.
(b) Show that if U is uniform on $(0, 1)$, then $Z = F^{-1}(U)$ has distribution $F(z)$.

Solution (a) As usual, we seek the distribution function

$$\begin{aligned}\mathbf{P}(Y \leq y) &= \mathbf{P}(F(X) \leq y) = \mathbf{P}(g(F(X)) \leq g(y)) = \mathbf{P}(X \leq g(y)) \qquad \text{by (9)} \\ &= F(g(y)) = y \quad \text{by(9)}.\end{aligned}$$

(b) Again,

$$\begin{aligned}\mathbf{P}(F^{-1}(U) \leq z) &= \mathbf{P}(F(g(U)) \leq F(z)) = \mathbf{P}(U \leq F(z)) \qquad \text{by (9)} \\ &= F(z).\end{aligned}$$

●

Although we have introduced them separately, discrete and continuous variables do have close links. Here are some examples to show this.

(10) Example: Step Functions Let X have distribution function $F(x)$ and density f. Define the function $S : \mathbb{R} \to \mathbb{Z}$ by

$$S(X) = k, \text{ if } k \leq X < k + 1, \tag{11}$$

where k is any integer. Then $S(X)$ is an integer valued discrete random variable with mass function

$$f_S(k) = \mathbf{P}(k \leq X < k + 1) = \int_k^{k+1} f(v)dv. \tag{12}$$

Obviously, $\mathbf{P}(S(X) \leq X) = 1$ and $F_S(x) \geq F_X(x)$, and

$$|S(X) - X| \leq 1. \tag{13}$$

●

Now equation (13) shows that the integer valued $S(X)$ is, in some sense, a rough approximation to the continuous random variable X. It is easy to get much better approximations as follows.

(14) Example: Discrete Approximation As usual, X has density $f(x)$; suppose also that $X > 0$. For fixed n, with $0 \leq r \leq 2^n - 1$ and $k \geq 0$, define

$$S_n(X) = k + r2^{-n} \text{ if } k + r2^{-n} \leq X < k + (r + 1)2^{-n}.$$

Then $S_n(X)$ is a discrete random variable taking values in $(k + r2^{-n}; k \geq 0, 0 \leq r \leq 2^n - 1)$, with

$$\mathbf{P}(S_n(X) = k + r2^{-n}) = \int_{k+r2^{-n}}^{k+(r+1)2^{-n}} f(v)dv.$$

Again, we have $S_n(X) \leq X$, but this time, by the construction,

$$|S_n(X) - X| \leq 2^{-n}. \tag{15}$$

Thus, by choosing n large enough, we can find a discrete random variable $S_n(X)$ such that $|X - S_n(X)|$ is as small as we please. ●

In fact, it can be shown that we can find a simple random variable (taking only a finite number of values) that is arbitrarily close to X, but in a weaker sense than (15). (See Problem 12.)

7.3 Simulation of Random Variables

A random variable is a mathematical concept (having no other existence) that is suggested by the outcomes of real experiments. Thus, tossing a coin leads us to define an $X(.)$ such that $X(H) = 1$, $X(T) = 0$, and X is the number of heads. The coin exists, X is a concept. A natural next step, having developed theorems about mathematical coins (e.g., the arc-sine laws) is to test them against reality. However, the prospect of actually tossing a large enough number of coins to check the arc-sine laws is rather forbidding.

Luckily, we have machines to do large numbers of boring and trivial tasks quickly, namely, computers. These can be persuaded to produce many numbers $(u_i; i \geq 1)$ that are sprinkled evenly and "randomly" over the interval (0, 1). The word randomly appears in quotations because each u_i is not really random. Because the machine was programmed to produce it, the outcome is known in advance, but such numbers behave for many practical purposes as though they were random. They are called pseudorandom numbers.

Now if we have a pseudorandom number u from a collection sprinkled uniformly in (0, 1), we can look to see if $u < \frac{1}{2}$, in which case we call it "heads", or $u > \frac{1}{2}$ in which case we call it "tails." This process is called *simulation*; we have simulated tossing a coin.

Different problems produce different random variables, but computers find it easiest to produce uniform pseudorandom numbers. We are thus forced to consider appropriate transformations of uniform random variables, and therefore many of the results of Section 7.2 find concrete applications when we seek to simulate random variables. A natural first question (before "how") is *why* might we want to simulate such random variables? Some examples should suffice to answer this question.

(1) Example: Epidemic An infection is introduced into a population. For each individual the incubation period is a random variable X, the infectious period is a random variable Y, and the number of further individuals infected is a random variable N, depending on Y and the behaviour of the infected individual. What happens? Unfortunately, exact solutions

to such problems are rare and, for many diseases (e.g., the so-called "slow viruses" or prions), X and Y are measured in decades so experiments are impractical. However, if we could simulate X and Y and the infection process N, then we could produce one simulated realization (not a real realization) of the epidemic. With a fast computer, we could do this many times and gain a pretty accurate idea of how the epidemic would progress (if our assumptions were correct). ●

(2) Example: Toll Booths Motorists are required to pay a fee before entering a toll road. How many toll booths should be provided to avoid substantial queues? Once again an experiment is impractical. However, simple apparatus can provide us with the rates and properties of traffic on equivalent roads. If we then simulate the workings of the booth and test it with the actual traffic flows, we should obtain reasonable estimates of the chances of congestion. ●

Because of the ready availability of large numbers of uniform pseudorandom numbers, interest is concentrated on finding transformations that then yield random variables of arbitrary type. We have seen several in Section 7.2. Here is another idea.

Example: Composition The pseudorandom variable U is uniformly distributed on $(0, 1)$. Show how to simulate a random variable with density

$$f_X = \frac{1}{4}(x^{-\frac{1}{2}} + (1-x)^{-\frac{1}{2}}), \quad 0 < x < 1.$$

Solution Recall that if U is uniform on $(0, 1)$ then U^2 has density $f_1(x) = \frac{1}{2}x^{-\frac{1}{2}}$. Next consider

$$\mathbf{P}(1 - U^2 \leq x) = \mathbf{P}(U \geq (1-x)^{\frac{1}{2}}) = 1 - (1-x)^{\frac{1}{2}}.$$

Hence, $1 - U^2$ has density $f_2(x) = \frac{1}{2}(1-x)^{-\frac{1}{2}}$. Now toss a coin (real or simulated), and write

$$X = \begin{cases} U^2 & \text{if it's heads} \\ 1 - U^2 & \text{if it's tails.} \end{cases}$$

Then

$$f_X(x) = \frac{1}{2}f_1(x) + \frac{1}{2}f_2(x) = \frac{1}{4}(x^{-\frac{1}{2}} + (1-x)^{-\frac{1}{2}}),$$

as required. ●

We describe other methods of simulation as the necessary ideas are developed.

7.4 Expectation

Random variables with a density may have an expected value, similar to random variables with a mass function.

(1) **Definition** *Let X have density $f(x)$. If $\int_{-\infty}^{\infty} |v| f(v) dv < \infty$, then X has an* expected value, *which is given by*

(2) $$\mathbf{E}(X) = \int_{-\infty}^{\infty} v f(v) dv.$$ ▲

(3) **Example: Uniform Density** Let X be uniformly distributed on (a, b). Then

$$\mathbf{E}(X) = \int_a^b \frac{v}{b-a} dv = \frac{1}{2}(b-a).$$ ●

(4) **Example: Exponential Density** Let X have density $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Then

$$\mathbf{E}(X) = \int_0^{\infty} v\lambda e^{-\lambda v} dv = \lambda^{-1}.$$ ●

(5) **Example: Normal Density** Let X have the $N(\mu, \sigma^2)$ density. Then

$$\begin{aligned}
\mathbf{E}(X) &= \frac{1}{\sigma(2\pi)^{\frac{1}{2}}} \int_{-\infty}^{\infty} v \exp(-(v-\mu)^2/(2\sigma^2)) dv \\
&= \frac{1}{\sigma(2\pi)^{\frac{1}{2}}} \int_{-\infty}^{\infty} (v-\mu) \exp\left(-\frac{1}{2}\left(\frac{v-\mu}{\sigma}\right)^2\right) dv \\
&\quad + \frac{\mu}{\sigma(2\pi)^{\frac{1}{2}}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}\left(\frac{v-\mu}{\sigma}\right)^2\right) dv \\
&= \frac{1}{(2\pi)^{\frac{1}{2}}} \int_{-\infty}^{\infty} u \exp\left(-\frac{1}{2}u^2\right) du + \frac{\mu}{(2\pi)^{\frac{1}{2}}} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}u^2\right) du
\end{aligned}$$

on making the substitution $u = (v-\mu)/\sigma$ in both integrands. The first integrand is an odd function, so the integral over $\mathbb{R}$ is zero. The second term is μ by Example 7.1.21 and 7.1.22. Hence, $\mathbf{E}(X) = \mu$. ●

Expectation may be infinite, as the next example shows.

(6) **Example: Pareto Density** Let X have density $f(x) = (\alpha-1)x^{-\alpha}$ for $x \geq 1$ and $\alpha > 1$. Then if $\alpha \leq 2$, the expected value of X is infinite because

$$\mathbf{E}(X) = \lim_{n\to\infty} \int_1^n \frac{(\alpha-1)v}{v^{\alpha}} dv = (\alpha-1) \lim_{n\to\infty} \int_1^n \frac{1}{v^{\alpha-1}} dv,$$

which diverges to ∞ for $\alpha - 1 \leq 1$. However, for $\alpha > 2$,

$$\mathbf{E}(X) = \int_1^{\infty} \frac{(\alpha-1)}{v^{\alpha-1}} dv = \frac{(\alpha-1)}{(\alpha-2)}.$$ ●

Then again, the expectation of X may not exist, as the next example shows.

(7) Example: Cauchy Density Let X have density

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty.$$

Because $\int_0^a v(\pi(1+v^2))^{-1}dv$ diverges as a $a \to -\infty$ and as a $a \to +\infty$, X does not have an expected value. ●

It is appropriate to give a moment to considering why we define $\mathbf{E}(X)$ by Definition 1. This definition is at least plausible, by analogy with the definition

$$\mathbf{E}(X) = \sum_{v=-\infty}^{\infty} vf(v),$$

in the discrete case. Of course, Definition 1 is much more than just a plausible analogy, but a complete account of expectation is well beyond our scope. However, we can use Example 7.2.14 to give a little more justification for Definition 1. Let $k + r2^{-n} = a(k, r, n)$. Recall from (7.2.15) that $|S_n - X| < 2^{-n}$. Now by definition, because $S_n(X)$ is discrete,

$$\begin{aligned}\mathbf{E}(S_n(X)) &= \sum_{k,r} a(k,r,n) \int_{a(k,r,n)}^{a(k,r+1,n)} f(v)dv \\ &= \sum_{k,r} \int_{a(k,r,n)}^{a(k,r+1,n)} (vf(v) + (a(k,r,n) - v)f(v))dv.\end{aligned}$$

Because $|a(k,r,n) - v| < 2^{-n}$, it can be shown (with more work, which we omit) that

$$\mathbf{E}(S_n(X)) = \int_{-\infty}^{\infty} vf(v)dv + \epsilon_n = \mathbf{E}(X) + \epsilon_n,$$

where $\epsilon_n \to 0$ as $n \to \infty$. An explicit demonstration may be helpful here.

(8) Example Let X be uniform on (0, 1) with mean value $\mathbf{E}(X) = \frac{1}{2}$. Then $S_n(X)$ is uniformly distributed on $\{0, 2^{-n}, 2.2^{-n}, 3.2^{-n}, \ldots, 1 - 2^{-n}\}$. Therefore,

$$\begin{aligned}\mathbf{E}(S_n(X)) &= \sum_{r=0}^{2^n-1} r2^{-n}.2^{-n} = \frac{1}{2}(2^n - 1)2^{-n} \\ &= \mathbf{E}(X) - 2^{-(n+1)} \to \mathbf{E}(X), \quad \text{as } n \to \infty.\end{aligned}$$ ●

Thus, our definitions of expectation for discrete and continuous variables are at least consistent in some way. In more advanced books, a single definition of $\mathbf{E}(.)$ is given, which is shown to yield our definitions as special cases.

Next we return to considering functions of random variables. Suppose we are given random variables Y and X related by $Y = g(X)$. What is $\mathbf{E}(Y)$?

If we know the density of X, then we may be able to find $\mathbf{E}(Y)$ by first discovering $f_Y(y)$, if it exists. This is often an unattractive procedure. We may do much better to use the following theorem, which we state without proof.

(9) **Theorem** *Let random variables X and Y satisfy $Y = g(X)$, where X has density $f(x)$. Then Y has an expected value if $\int_{-\infty}^{\infty} |g(v)|f(v)dv < \infty$, and in this case,*

(10) $$\mathbf{E}(Y) = \int_{-\infty}^{\infty} g(v)f(v)dv.$$

The proof of this is straightforward but long. An heuristic discussion of the type above shows that if we represent the distribution of X as a limit of discrete distributions, and then formally proceed to this limit in Theorem 4.3.4, equation (10) is the result. Again, this only makes (10) plausible, it does not provide the proof, which is beyond our scope.

This important result implies that the useful consequences of Theorem 4.3.4 remain true for random variables with a density. In particular, Theorem 4.3.6 remains true; the proofs of most parts are just typographical variants of the proofs in the discrete case; just replace $\sum$ by $\int$.

We describe one important and less trivial case in detail, namely, the analogy of Theorem 4.3.11.

(11) **Theorem: Tail integral** *Let the nonnegative random variable X have density f, distribution F, and finite expected value $\mathbf{E}(X)$. Then*

(12) $$\mathbf{E}(X) = \int_0^{\infty} (1 - F(x))dx.$$

Proof For any finite y, we may integrate by parts to obtain

(13) $$\int_0^y xf(x)dx = -x(1 - F(x))|_0^y + \int_0^y (1 - F(x))dx.$$

But we have

$$y(1 - F(y)) = y\int_y^{\infty} f(x)dx \leq \int_y^{\infty} xf(x)dx \to 0$$

as $y \to \infty$, because $\mathbf{E}(X) < \infty$. Hence, we can let $y \to \infty$ in (13) to prove the theorem. ■

We can use this to prove a useful special case of Theorem 9.

(14) **Example** Let the nonnegative random variable X have density f, and let $g(X) \geq 0$. Show that $\mathbf{E}(g(X)) = \int_0^{\infty} g(v)f(v)dv$.

Solution

$$\begin{aligned}\mathbf{E}(g(X)) &= \int_0^{\infty} \mathbf{P}(g(X)) \geq v)\, dv \quad \text{by (12)} \\ &= \int_0^{\infty} \int_{x:g(x)\geq v} f(x)dxdv = \int_0^{\infty} f(x)\int_0^{g(x)} dvdx = \int_0^{\infty} f(x)g(x)dx,\end{aligned}$$

as required. The interchange in the order of integration is justified by a theorem on double integrals, which we omit. ●

The various moments of a random variable with a density are defined just as they were for discrete random variables, that is to say: $\mu_k = \mathbf{E}(X^k)$, and

$$\sigma_k = \mathbf{E}((X - \mathbf{E}(X))^k).$$

(15) Example: Normal Density Let X have the density $N(0, \sigma^2)$. Find μ_k for all k.

Solution If k is odd, then $x^k \exp(-x^2/(2\sigma^2))$ is an odd function. Hence, $\mu_k = 0$ if k is odd. If $k = 2n$, then integrating by parts gives

$$\begin{aligned}\mu_{2n} &= \frac{1}{\sigma(2\pi)^{\frac{1}{2}}} \int_{-\infty}^{\infty} v^{2n} \exp(-v^2/(2\sigma^2)) dv \\ &= \frac{1}{\sigma(2\pi)^{\frac{1}{2}}} \left(-v^{2n-1}\sigma^2 \exp(-v^2/(2\sigma^2))|_{-\infty}^{\infty} \right. \\ &\quad \left. + \int_{-\infty}^{\infty} (2n-1)\sigma^2 v^{2n-2} \exp(-v^2/(2\sigma^2)) dv \right) \\ &= (2n-1)\sigma^2 \mu_{2n-2} = \sigma^{2n} \frac{(2n)!}{2^n n!}\end{aligned}$$

on iterating and observing that $\mu_0 = 1$. Hence, in particular, $\mu_2 = \sigma^2$. ●

Finally, and thankfully, we are pleased to record that the expectation $\mathbf{E}(X)$ of a continuous random variable X has the same useful basic properties that we established for the discrete case in Section 4.6. For convenience, we recall them here.

(16) Theorem *Let a and b be constants, and let g and h be functions. Then:*

(i) If $g(X)$ and $h(X)$ have finite mean, then

$$\mathbf{E}(g(X) + h(X)) = \mathbf{E}(g(X)) + \mathbf{E}(h(X)).$$

(ii) If $\mathbf{P}(a \le X \le b) = 1$, then $a \le \mathbf{E}(X) \le b$.
(iii) If h is nonnegative, then for $a > 0$, $\mathbf{P}(h(X) \ge a) \le \mathbf{E}(h(X)/a)$.
(iv) **Jensen's inequality** *If g is convex then $\mathbf{E}(g(X)) \ge g(E(X))$.*

Proof The proof is an exercise for you. When $h(x) = x^2$ in (iii), we have:

(17) **Chebyshov's inequality:** $\mathbf{P}(|X| \ge a) \le \mathbf{E}X^2/a^2$. ■

7.5 Moment Generating Functions

In dealing with integer valued discrete random variables, we found the probability generating function exceptionally useful (see Chapter 6). It would be welcome to have such

a useful workhorse available for random variables with densities. Of course, if X has a density then $\mathbf{P}(X = x) = 0$, so we cannot expect the probability generating function to be of much use. Fortunately, another function will do the job.

(1) Definition *If X has density f, then X has* moment generating function $M_X(t)$ *given by*

(2) $$M_X(t) = \mathbf{E}(e^{tX}) = \int_{-\infty}^{\infty} e^{tv} f(v)dv. \quad \blacktriangle$$

We are only interested in $M_X(t)$ for those values of t for which it is finite; this includes $t = 0$, of course. It is particularly pleasant when $M_X(t)$ exists in a neighbourhood of zero, but it is beyond our scope to explain all the reasons for this.

(3) Example: Uniform Density Let X be uniform on $[0, a]$. Find $\mathbf{E}(e^{tX})$. Where does it exist?

Solution

$$\mathbf{E}(e^{tX}) = \int_0^a \frac{1}{a} e^{tv} dv = \left[\frac{1}{at} e^{tv}\right]_0^a = \frac{e^{at} - 1}{at}.$$

This exists for all t, including $t = 0$, where it takes the value 1. ●

(4) Example: Gamma Density Recall from (7.1.24) that the gamma function $\Gamma(\alpha)$ is defined for any $\alpha > 0$ and $\lambda > 0$ by

(5) $$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} \lambda^\alpha e^{-\lambda x} dx.$$

Hence,

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x \geq 0,$$

is the density of a random variable x. Find $\mathbf{E}(e^{tX})$. Where does it exist?

Solution

$$\mathbf{E}(e^{tX}) = \int_0^\infty e^{tv} \frac{\lambda^\alpha}{\Gamma(\alpha)} v^{\alpha-1} e^{-\lambda v} dv = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty v^{\alpha-1} e^{-(\lambda - t)v} dv.$$

The integral exists if $\lambda > t$, and then making the substitution $(\lambda - t)v = u$ gives

(6) $$M_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^\alpha \int_0^\infty \frac{u^{\alpha-1}}{\Gamma(\alpha)} e^{-u} du = \left(\frac{\lambda}{\lambda - t}\right)^\alpha \quad \text{by (5)},$$

for $-\infty < t < \lambda$. ●

(7) Example: Normal Density Let X be a standard normal random variable. Then

$$\begin{aligned}\sqrt{2\pi}M_X(t) &= \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}x^2 + tx\right) dx = \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}(x-t)^2 + \frac{1}{2}t^2\right) dx \\ &= e^{\frac{1}{2}t^2} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}v^2\right) dv, \quad \text{setting } x - t = v, \\ &= e^{\frac{1}{2}t^2}\sqrt{2\pi}.\end{aligned}$$

So $M_X(t) = e^{\frac{1}{2}t^2}$. Now by (7.2.4) if Y is $N(\mu, \sigma^2)$, $M_Y(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$. ●

You may ask, why is $M_X(t)$ called the moment generating function? The answer lies in the following formal expansion.

$$\mathbf{E}(e^{tX}) = \mathbf{E}\left(\sum_{k=0}^{\infty} \frac{X^k t^k}{k!}\right) = \sum_{k=0}^{\infty} \mathbf{E}\frac{(X^k)t^k}{k!} = \sum_{k=0}^{\infty} \frac{\mu_k t^k}{k!}. \tag{8}$$

Thus, provided the interchange of expectation and summation at (8) is justified, we see that $M_X(t)$ is the (exponential) generating function of the moments μ_k.

Note that the word "exponential" is always omitted in this context, and that the required interchange at (8) is permissible if $M_X(t)$ exists in an interval that includes the origin.

You may also ask, do we always know the density $f_X(x)$, if we know $M_X(t)$? After all, the probability generating function uniquely determines the corresponding mass function. Unfortunately, the answer is no in general because densities not uniquely determined by their moments do exist. However, none appear here; every density in this book is uniquely determined by its moment generating function (if it has one). We state the following *inversion theorem* without proof.

(9) Theorem *If X has moment generating function $M(t)$, where for some $a > 0$, $M(t) < \infty$ for $|t| < a$, then the distribution of X is determined uniquely.*

Furthermore, $M(t) = \sum_{k=0}^{\infty} \frac{1}{k!} t^k \mathbf{E}(X^k)$.

The moment generating function is especially useful in dealing with sequences of random variables; the following theorem is the basis of this assertion. We state it without proof.

(10) Theorem: Continuity Theorem *Let $(F_n(x); n \geq 1)$ be a sequence of distribution functions with corresponding moment generating functions $(M_n(t); n \geq 1)$ that exist for $|t| < b$. Suppose that as $n \to \infty$ $M_n(t) \to M(t)$ for $|t| \leq a < b$, where $M(t)$ is the m.g.f. of the distribution $F(x)$. Then, as $n \to \infty$, $F_n(x) \to F(x)$ at each point x where $F(x)$ is continuous.*

The main application of this theorem arises when $M(t) = e^{\frac{1}{2}t^2}$ and $F(x) = \Phi(x)$, as we see in Chapter 8 when we come to the celebrated central limit theorem. Here is a preliminary note.

Note: The O–o Notation In considering limits of sequences of functions, we quite often produce large and unwieldy expressions of which only one or two terms remain in the limit. Rather than keep a precise record of the essentially irrelevant terms, it is convenient to have a special compact notation for them.

Definition *If $g(n)$ and $h(n)$ are two functions of n, then we write*

$$h(n) = O(g(n)) \quad \text{as } n \to \infty$$

if $|h(n)/g(n)| < c$ for all large enough n and some finite constant c. ▲

For example, as $n \to \infty$,

$$n^2 + \log n = O(n^2) \quad \text{with } c = 2$$

and

$$n^2 + n^{\frac{3}{2}} = O(n^2) \quad \text{with } c = 2.$$

Observe that this is an abuse of notation (= being the abused symbol) because it does *not* follow from these two examples that $\log n = n^{\frac{3}{2}}$. Also, if $h(n) = O(g(n))$ and $k(n) = O(g(n))$, then $h(n) + k(n) = O(g(n))$.

A similar definition holds for small values of the argument.

Definition *If $g(x)$ and $h(x)$ are two functions of x, then we write $h(x) = O(g(x))$ as $x \to 0$, if $|h(x)/g(x)| < c$ for all small enough x and some constant c.* ▲

Often, an even cruder representation will suffice.

Definition *If $g(x)$ and $h(x)$ are two functions of x, then we write $h(x) = o(g(x))$ as $x \to \infty$, if $\lim_{x\to\infty}(h(x)/g(x)) = 0$. Likewise, $h(x) = o(g(x))$ as $x \to 0$ if $\lim_{x\to 0}(h(x)/g(x)) = 0$.* ▲

For example, $x^2 = o(x)$ as $x \to 0$ and $x = o(x^2)$ as $x \to \infty$. For another example, $x + x \log x + x^2 = o(1)$ as $x \to 0$.

We use this new notation in the following famous result.

(11) Example: de Moivre–Laplace Theorem For each $n \geq 1$, let X_n be a binomial random variable with parameters n and p. Let $q = 1 - p$, and define

$$Y_n = \frac{X_n - np}{(npq)^{\frac{1}{2}}}.$$

Show that as $n \to \infty$

$$\mathbf{P}(Y_n \leq x) \to \Phi(x) = \int_{-\infty}^{x} (2\pi)^{-\frac{1}{2}} e^{-y^2/2} dy.$$

Solution We use Theorem 10. First calculate the moment generating function

$$\mathbf{E}(e^{tY_n}) = \mathbf{E}\left(\exp\left(\frac{t(X_n - np)}{(npq)^{\frac{1}{2}}}\right)\right) = \left[\mathbf{E}\exp\left(\frac{t(X_1 - p)}{(npq)^{\frac{1}{2}}}\right)\right]^n \tag{12}$$
$$= \left[p\exp\left(\frac{qt}{(npq)^{\frac{1}{2}}}\right) + q\exp\left(\frac{-pt}{(npq)^{\frac{1}{2}}}\right)\right]^n.$$

Next we expand the two exponential terms in (12) to give

$$\mathbf{E}(e^{tY_n}) = \left(1 + \frac{t^2}{2n} + O\left(n^{-\frac{3}{2}}\right)\right)^n. \tag{13}$$

Now we recall the useful result that says that, for constant a,

$$\lim_{n\to\infty}\left(1 + \frac{a}{n} + o(n^{-1})\right)^n = e^a. \tag{14}$$

Applying this to (13) shows that

$$\lim_{n\to\infty}\mathbf{E}(e^{tY_n}) = e^{\frac{1}{2}t^2},$$

which is the m.g.f. of the standard normal distribution, as required. [More demanding readers should note that they can prove (14) by first taking logarithms.] ●

The appearance of the normal distribution in these circumstances is one of the most remarkable results in the theory of probability. The first proof, due to de Moivre, was greatly improved by Laplace. Their methods were different from those used here, relying on fairly precise direct estimates of the binomial probabilities. We outline a modern version of their proof in Example 7.20.

7.6 Conditional Distributions

Just as in the discrete case, it is often necessary to consider the distribution of a random variable X conditional upon the occurrence of some event A. By definition of conditional probability, we have

$$F_{X|A}(x) = \mathbf{P}(X \le x|A) = \mathbf{P}(\{\omega\colon X \le x\} \cap A)/\mathbf{P}(A) \tag{1}$$
$$= \mathbf{P}(X \le x; A)/\mathbf{P}(A), \text{ say.}$$

(Obviously, A has to be in $\mathcal{F}$, the event space.) The case that arises most commonly is when A is an event of the form

$$A = \{\omega\colon a < X \le b\}; \tag{2}$$

that is, we seek the distribution of X conditional on its lying in some subset of its range.

(3) Example Let $a < b < c < d$. Let X be uniform on (a, d), and let $A = \{\omega\colon b < X(\omega) \le c\}$.

Then, by (1),

$$\mathbf{P}(X \leq x|A) = \frac{\mathbf{P}(X \leq x; b < X \leq c)}{\mathbf{P}(b < X \leq c)} = \begin{cases} \dfrac{x-b}{d-a} \Big/ \left(\dfrac{c-b}{d-a}\right) & \text{if } b < x \leq c \\ 0 & \text{otherwise} \end{cases}$$
$$= \frac{x-b}{c-b} \quad \text{for } b < x \leq c.$$

Thus, the distribution of X given A is just uniform on (b, c). More generally, it is easy to see that a uniform random variable, constrained to lie in any subset A of its range, is uniformly distributed over the subset A. ●

Because $\mathbf{P}(X \leq x|A)$ is a distribution, it may have an expectation. For example, suppose that X has density f, and A is given by (2). Then, by (1),

$$\mathbf{P}(X \leq x|A) = \frac{F(x) - F(a)}{F(b) - F(a)} \quad \text{for } a < x \leq b, \tag{4}$$

and differentiating yields the conditional density

$$f_{X|A}(x) = \begin{cases} \dfrac{f(x)}{F(b) - F(a)} & \text{if } a < x \leq b \\ 0 & \text{otherwise.} \end{cases} \tag{5}$$

Notice that $\int_a^b f_{X|A}(v)dv = 1$, as it must. Then we may define the *conditional expectation*

$$\begin{aligned} \mathbf{E}(X|A) &= \int_a^b \frac{vf(v)}{F(b) - F(a)}\, dv \\ &= a + \int_a^b \frac{F(b) - F(v)}{F(b) - F(a)}\, dv, \quad \text{on integrating by parts,} \\ &= \int_0^\infty (1 - F_{X|A}(v))dv \end{aligned} \tag{6}$$

on using (4). Notice that this is in agreement with Theorem 7.4.11, as of course it must be.

(7) Example: Exponential Density and Lack-of-Memory Let X be exponentially distributed with parameter λ. Show that

$$\mathbf{P}(X > s + t|X > s) = e^{-\lambda t} = \mathbf{P}(X > t). \tag{8}$$

Find $\mathbf{E}(X|X > s)$ and $\mathbf{E}(X|X \leq s)$.

Solution Trivially,

$$\mathbf{P}(X > s + t|X > s) = \mathbf{P}(X > s + t)/\mathbf{P}(X > s) = e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}.$$

Hence,

$$\mathbf{E}(X|X > s) = s + \int_0^\infty e^{-\lambda t} dt = s + \mathbf{E}(X). \tag{9}$$

We remark that the remarkable identity (8) is known as the *lack-of-memory property* of the exponential distribution. Finally,

$$\mathbf{E}(X|X \le s) = \int_0^s \frac{\mathbf{P}(s \ge X > v)}{\mathbf{P}(s \ge X)} dv = \frac{1}{\lambda} - \frac{s}{e^{\lambda s} - 1}. \qquad \bullet$$

7.7 Ageing and Survival

Many classic examples of continuous random variables arise as waiting times or survival times. For instance, the time until the cathode-ray tube in your television fails, the time until you are bitten by a mosquito after disembarking in the tropics, the time until a stressed metal component fails due to fatigue. For definiteness, we consider the *lifetime* T of some *device* or component. The device is said to *fail* at time T.

It is often useful to quantify the ageing process of a device; in particular, we may want to compare a device of given age with a new one. (We are all familiar with the fact that it is not necessarily always a good thing to replace a working component with a new one. This fact is embodied in the popular saying: "If it works, don't fix it") Let T have distribution F and density f. The following quantities turn out to be of paramount importance in comparing devices of different ages.

The *survival* function

$$\bar{F}(t) = 1 - F(t) = \mathbf{P}(T > t). \tag{1}$$

The *hazard* function

$$H(t) = -\log(1 - F(t)). \tag{2}$$

The *hazard rate* function

$$r(t) = \frac{f(t)}{\bar{F}(t)} = \frac{f(t)}{1 - F(t)} = \frac{dH(t)}{dt}. \tag{3}$$

The last equality explains why $r(t)$ is called the hazard rate. Integrating (3) yields

$$\exp\left(-\int_0^t r(s)ds\right) = \bar{F}(t). \tag{4}$$

Before we explain the significance of these quantities, you are warned that terminology in the literature of ageing is quite chaotic. Note that:

(i) The survival function is also known as the survivor function, reliability function, or hazard function.
(ii) The hazard function is also known as the log-survivor function.
(iii) The hazard rate function is also known as the failure rate function, mortality function, or hazard function.

Beware!

Now let A_t denote the event that $T > t$. Then

$$F_{T|A_t}(s+t) = \mathbf{P}(T \leq s+t | T > t) = \frac{F(s+t) - F(t)}{1 - F(t)}.$$

This is the probability that the device fails during $(t, t+s)$, given that it has not failed by time t. Now

$$\lim_{s \to 0} \frac{1}{s} F_{T|A_t}(s+t) = (1 - F(t))^{-1} \lim_{s \to 0} \frac{F(t+s) - F(t)}{s} = \frac{f(t)}{1 - F(t)} = r(t). \tag{5}$$

Thus, $r(t)$ may be thought of as the "intensity" of the probability that a device aged t will fail.

(6) Example: Exponential Life If T has an exponential density, then $\bar{F}(t) = e^{-\lambda t}$, $H(t) = \lambda t$, and $r(t) = \lambda$. This constant hazard rate is consonant with the lack-of-memory property mentioned in Example 7.6.7. Roughly speaking, the device cannot remember how old it is, and so the failure intensity remains constant. ●

We see that intuitively there is a distinction between devices for which $r(t)$ increases, essentially they are "wearing out," and those for which $r(t)$ decreases, they are "bedding in."

A simple and popular density in this context is the Weibull density, which can exhibit both types of behaviour.

(7) Example: Weibull Life If T has density

$$f(t) = \alpha t^{\alpha - 1} \exp(-t^\alpha), \quad t > 0, \alpha > 0,$$

then it has distribution $F(t) = 1 - \exp(-t^\alpha)$. Hence, $\bar{F}(t) = \exp(-t^\alpha)$, and so

$$\frac{\mathbf{P}(T > t+s | T > s)}{\mathbf{P}(T > t)} = \exp(-(t+s)^\alpha + s^\alpha + t^\alpha),$$

which is > 1 or < 1 according as $\alpha < 1$ or $\alpha > 1$. [To see this, just consider the stationary value of $x^\alpha + (1-x)^\alpha - 1$ at $x = \frac{1}{2}$.] Hence, if $\alpha < 1$, the chance of lasting a further time t (conditional on $T > s$) increases with s. However, if $\alpha > 1$, this chance decreases with s. ●

The behaviour of $r(t)$ is not the only measure of comparison between new and old devices. There is a large hierarchy of measures of comparison, which we display formally as follows [in the notation of (1)–(3)].

(8) Definition

(i) If $r(t)$ increases, then T is (or has) increasing failure rate, *denoted by IFR.*
(ii) If $\frac{H(t)}{t}$ increases, then T is (or has) increasing failure rate average, *denoted by IFRA.*
(iii) If for all $s \geq 0, t \geq 0$,

$$H(s+t) \geq H(s) + H(t),$$

then T is new better than used, *denoted by NBU.*

(iv) If for all $t \geq 0$

$$\mathbf{E}(T) \geq \mathbf{E}(T - t | A_t),$$

then T *is* new better than used in expectation, *denoted by NBUE.*

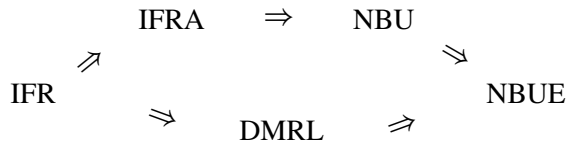
(v) If for all $0 \leq s < t < \infty$

$$\mathbf{E}(T - s | A_s) \geq \mathbf{E}(T - t | A_t),$$

then T *has (or is)* decreasing mean residual life, *denoted by DMRL.* ▲

The random variable T may also be decreasing failure rate (DFR), decreasing failure rate on average (DFRA), new worse than used (NWU), new worse than used in expectation (NWUE), or increasing mean residual life (IMRL). All these are defined in the obvious way, analogous to (i)–(v).

It can be shown that the following relationships hold between these concepts:



Some of these implications are trivial, and some are established in Example 7.17 below. These ideas are linked to another concept, that of stochastic ordering.

7.8 Stochastic Ordering

As in Section 7.7, let T be a nonnegative random variable. In general, let $R(s)$ be a random variable whose distribution is that of $T - s$ given that $T > s$, namely,

(1) $$F_R(x) = \mathbf{P}(T - s \leq x | T > s).$$

We refer to $R(s)$ as the *residual life* (of T at s).

The above example shows that if T has the exponential density, then its residual life is also exponentially distributed with constant mean.

More generally, $F_{R(s)}$ may depend on s, and more significantly it may do so in a systematic way; the following definition is relevant here.

(2) Definition *Let X and Y be random variables. If*

(3) $$\bar{F}_X(x) \geq \bar{F}_Y(x) \quad \textit{for all } x,$$

then X is said to be stochastically larger than Y. ▲

Now we can supply a connection with the ideas of the preceding section (7.7).

(4) Example If T is a random variable with residual life $R(s)$, $s > 0$, show that T has increasing failure rate if and only if $R(s)$ is stochastically larger than $R(t)$ for all $s < t$.

Solution First, we find

$$\mathbf{P}(R(t) > x) = \mathbf{P}(T - t > x | T > t) = \bar{F}(t + x)/\bar{F}(t) \tag{5}$$
$$= \exp\left(-\int_0^{t+x} r(s)ds\right) \exp\left(\int_0^t r(s)ds\right) \quad \text{by (7.7.4)}$$
$$= \exp\left(-\int_0^{t+x} r(s)ds\right).$$

Differentiating (5) with respect to t, we have

$$\frac{\partial}{\partial t}\mathbf{P}(R(t) > x) = (r(t) - r(t + x)) \exp\left(-\int_t^{t+x} r(s)ds\right). \tag{6}$$

Because $\exp(-\int r ds)$ is positive, and $r(t) - r(t + x)$ is positive or negative according as T is DFR or IFR, the result follows. ●

Finally, we have the useful:

(7) Theorem *If X is stochastically larger than Y, then $\mathbf{E}(X) \geq \mathbf{E}(Y)$.*

Proof We prove this when $X \geq 0$ and $Y \geq 0$. (The general result is left as an exercise.) From Theorem 7.4.11,

$$\mathbf{E}(X) = \int_0^\infty \bar{F}_X(x)dx \geq \int_0^\infty \bar{F}_Y(x)dx \quad \text{by hypothesis,}$$
$$= \mathbf{E}(Y).$$

■

7.9 Random Points

Picking a point Q at random in the interval (0, 1) yielded the uniform density (of the length OQ). It is intuitively attractive to consider problems that involve picking one or more points at random in other nice geometric figures, such as discs, squares, triangles, spheres, and so on. Indeed this idea is so natural that mathematicians had already started doing this kind of thing in the eighteenth century, and one of the most celebrated articles on the subject is that of M.W. Crofton in the 1885 edition of the *Encyclopaedia Britannica*. Such questions also have applications in statistics.

Confining ourselves to two dimensions for definiteness, suppose a point Q is picked at random in a region R of area $|R|$. Then it is natural to let the probability $\mathbf{P}(S)$, that Q lies in a set $S \subseteq R$, be given by

$$\mathbf{P}(S) = \frac{|S|}{|R|}, \tag{1}$$

where, now, $|S|$ denotes the area of S. It follows from the properties of area that $\mathbf{P}(.)$ has the required properties of a probability function, and we can proceed to solve various simple problems using elementary geometry. The following is typical.

(2) Example A point Q is picked at random in the unit square. What is the probability ν that it is nearer to the centre O of the square than to its perimeter?

Solution By symmetry, we need to consider only the sector $0 \le y \le x \le \frac{1}{2}$. Then the point (x, y) is nearer to O than the perimeter if $\sqrt{x^2 + y^2} < \frac{1}{2} - x$; that is, if in this sector,

$$x < \frac{1}{4} - y^2, \text{ for } 0 \le y \le \frac{1}{2}(\sqrt{2} - 1).$$

Hence, the area is given by an integral and

$$v = 8 \int_0^{\frac{1}{2}(\sqrt{2}-1)} \left(\frac{1}{4} - y^2 - y \right) dy = \frac{4}{3}\sqrt{2} - \frac{5}{3}. \quad \bullet$$

An equally trivial but much more important example is the following.

(3) Example Let $f = f(x)$ be an integrable function with $0 \le f(x) \le 1$ for $0 \le x \le 1$. Let Q be picked at random in the unit square, and let A_v be the set of points such that $0 \le x \le v$ and $0 \le y \le f(x)$. Then, from (1),

$$\mathbf{P}(Q \in A_v) = \int_0^v f(x)dx. \quad (4) \quad \bullet$$

This trivial result has at least two important applications. The first we have met already in Example 5.8.9.

(5) Example: Hit-or-Miss Monte Carlo Integration Let $f(x)$ and Q be as defined in Example 3, and declare Q a *hit* if Q lies below $f(x)$, $0 \le x \le 1$. Then the probability of a hit is $\mathbf{P}(A_1) = \int_0^1 f(x)dx$. Now we pick a sequence of such points $Q_1, Q_2, \ldots$ and let X_n be the number of hits. If points are picked independently, then X_n is a binomial random variable with parameters n and $\mathbf{P}(A_1)$, and we have shown that as $n \to \infty$, for $\epsilon > 0$,

$$\mathbf{P}\left(|n^{-1}X_n - \int_0^1 f(x)dx| > \epsilon \right) \to 0.$$

This therefore offers a method for evaluating the integral $\int_0^1 f(x)dx$. In practice, one would be unlikely to use this method in one dimension, but you might well use the analogous method to evaluate $\int f(\boldsymbol{x})d\boldsymbol{x}$, where $\boldsymbol{x}$ is a vector in (say) 11 dimensions. $\bullet$

(6) Example: Simulation With Q and $f(x)$ defined as above, consider the probability that Q lies in A_v given that it is a hit. By definition, this has probability

$$\mathbf{P}(A_v | A_1) = \frac{\int_0^v f(x)dx}{\int_0^1 f(x)dx}.$$

By inspection, the function $F(v) = \mathbf{P}(A_v | A_1)$ is the distribution function of the x-coordinate of Q given that it is a hit. This procedure therefore offers a method of simulating

a random variable X with density function

$$f_X(x) = \frac{f(x)}{\int_0^1 f(x)dx}. \tag{7}$$

You can just pick a point Q and, if it is a hit, let its x-coordinate be X. ●

A natural next step is to consider events defined jointly by a number of points picked independently in a region R. One famous example is Sylvester's problem: for four points picked at random in R, what is the probability that one of them lies in the triangle formed by the other three? This is too difficult for us, but we can consider an amusing simpler problem to illustrate a few of the basic ideas.

(8) Example: Two Points in a Disc Let $\lambda(r)$ be the expected value of the distance $L(r)$ between two points Q_1 and Q_2, each distributed uniformly (and independently) over a disc of radius r. Show that

$$\lambda(r) = \frac{128r}{45\pi}. \tag{9}$$

Solution This can be done by a brutal integration; here is a better way, discovered by M.V. Crofton in 1885.

Consider a disc of radius $x + h$, which we may think of as a disc D of radius x, surrounded by an annulus A of width h. Then, if Q_1 and Q_2 are dropped at random on to the disc of radius $x + h$, we have (using independence and the properties of the uniform density) that

$$\mathbf{P}(Q_1 \in D \cap Q_2 \in D) = \left(\frac{\pi x^2}{\pi(x+h)^2}\right)^2 = 1 - \frac{4h}{x} + o(h). \tag{10}$$

Also,

$$\mathbf{P}(Q_1 \in D \cap Q_2 \in A) = \frac{\pi x^2}{\pi(x+h)^2}\left(1 - \frac{\pi x^2}{\pi(x+h)^2}\right) = \frac{2h}{x} + o(h)$$

and $\mathbf{P}(Q_1 \in A \cap Q_2 \in A) = o(h)$. Hence, by conditional expectation,

$$\begin{aligned}\lambda(x+h) = {} & \mathbf{E}(L(x+h)|Q_1 \in D; Q_2 \in D)\left(1 - \frac{4h}{x} + o(h)\right) \\ & + 2\mathbf{E}(L(x+h)|Q_1 \in D; Q_2 \in A)\left(\frac{2h}{x} + o(h)\right) + o(h).\end{aligned} \tag{11}$$

Now $\mathbf{E}(L(x+h)|Q_1 \in D; Q_2 \in A)$ is just the mean distance of a random point Q_1 in a disc of radius x, from a point Q_2 on its circumference (plus a quantity that is $o(h)$). Hence, taking plane polar coordinates with Q_2 as origin:

$$\begin{aligned}\mathbf{E}(L(x+h)|Q_1 \in D; Q_2 \in A) &= \frac{1}{\pi x^2}\int_{-\pi/2}^{\pi/2}\int_0^{2x\cos\theta} v^2 dv d\theta + o(h) \\ &= \frac{32x}{9\pi} + o(h).\end{aligned}$$

Returning to (11), note that $\mathbf{E}(L(x+h)|Q_1 \in D; Q_2 \in D) = \lambda(x)$; hence, rearranging (11) and letting $h \to 0$ gives

$$\frac{d\lambda(x)}{dx} = \lim_{h\to 0} \frac{1}{h}\left(\lambda(x+h) - \lambda(x)\right) = \frac{-4}{x}\lambda(x) + \frac{128}{9\pi}.$$

Integrating this, and observing that $\lambda(0) = 0$, we have

$$\lambda(x) = \frac{128x}{45\pi}. \qquad \bullet$$

Using the same idea, and with a lot more toil, we can find the density of L.

The next natural step is to pick lines (or other objects) at random and ask how they divide up the region R in random tessellations or coverings. This is well beyond our scope, but the trivial Example 7.18 illustrates some of the problems.

7.10 Review and Checklist for Chapter 7

We introduced the class of random variables having a density $f_X(x)$ and a distribution $F_X(x)$. These take one of an uncountable number of values in $\mathbb{R}$ and are called "absolutely continuous." The familiar ideas of expectation, conditioning, functions, and generating functions are explored in this new context, together with some applications.

SYNOPSIS OF FORMULAE:

Key Rule: $\mathbf{P}(X \in B) = \int_{x\in B} f_X(x)dx.$

Distribution and density:

$$F_X(x) = \int_{-\infty}^{x} f_X(y)dy = \mathbf{P}(X \le x).$$

For small h, $\mathbf{P}(x < X \le x + h) \simeq f_X(x)h$, and if $F(x)$ is differentiable $\frac{dF}{dx} = f(x)$.
$F(x)$ is nondecreasing; $\lim_{x\to-\infty} F(x) = 0$;

$$\lim_{x\to+\infty} F(x) = \int_{-\infty}^{\infty} f(u)du = 1, \qquad \text{if } X \text{ is proper.}$$

Mixture: If f and g are densities, then so is

$$h = \lambda f + (1-\lambda)g, 0 \le \lambda \le 1.$$

Functions: If continuous random variables X and Y are such that $Y = g(X)$ for some function $g(.)$ that is differentiable and strictly increasing, then

$$f_Y(y) = f_X(g^{-1}(y))\frac{d}{dy}[g^{-1}(y)],$$

where $g^{-1}(.)$ is the inverse function of g. In general, we can write

$$f_Y(y) = \frac{d}{dy} \int_{x:g(x)\leq y} f_X(x)dx,$$

and proceed by ad hoc arguments.

Expectation: A random variable X has an expected value $\mathbf{E}X$ provided that $\int_{-\infty}^{\infty} |x| f_X(x)dx < \infty$, and then

$$\mathbf{E}X = \int_{-\infty}^{\infty} x f_X(x)dx.$$

When $X > 0$,

$$\mathbf{E}X = \int_0^{\infty} \mathbf{P}(X > x)dx.$$

If random variables X and Y are such that $Y = g(X)$ and X is continuous, then Y has an expected value if $\int_{-\infty}^{\infty} |g(x)| f_X(x)dx < \infty$ and

$$\mathbf{E}Y = \mathbf{E}g(X) = \int_{-\infty}^{\infty} g(x) f_X(x)dx.$$

Moments: In particular, if $g(X) = X^r$, this yields the rth moment μ_r of X. When $X > 0$,

$$\mathbf{E}X^r = \int_0^{\infty} r x^{r-1} \mathbf{P}(X > x)dx.$$

When $g(X) = \exp(tX)$, this yields the m.g.f.,

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x)dx.$$

Conditioning: Any event B in Ω may condition a random variable X on Ω, leading to a conditional distribution and density,

$$F_{X|B}(x|B) = \mathbf{P}(X \leq x|B)$$

and

$$f_{X|B}(x|B) = \frac{d}{dx} F_{X|B}(x|B),$$

when the derivative exists, with the

Key Rule:

$$\mathbf{P}(X \in A|B) = \int_{x\in A} f(x|B)dx.$$

Such conditioned random variables may have an expectation if

$$\int_{-\infty}^{\infty} |x| f_{X|B}(x|B)dx < \infty,$$

Table 7.1. *Continuous random variables and their associated characteristics*

X	$f(x)$	$\mathbf{E}X$	var X	m.g.f.
Uniform	$(b-a)^{-1}, a \le x \le b$	$\frac{1}{2}(b+a)$	$\frac{1}{12}(b-a)^2$	$\frac{e^{bt}-e^{at}}{t(b-a)}$
Exponential	$\lambda e^{-\lambda x}, x \ge 0$	λ^{-1}	λ^{-2}	$\lambda/(\lambda - t)$
Normal	$(2\pi)^{-1/2}\sigma^{-1} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}$,	μ	σ^2	$\exp(\mu t + \frac{1}{2}\sigma^2 t^2)$
$N(\mu, \sigma^2)$	$-\infty < x < \infty$			
Gamma	$x^{r-1}e^{-\lambda x}\frac{\lambda^r}{(r-1)!}, x \ge 0$	$r\lambda^{-1}$	$r\lambda^{-2}$	$\left(\frac{\lambda}{\lambda - t}\right)^r$
Laplace	$\frac{1}{2}\lambda \exp(-\lambda\lvert x\rvert)$	0	$2\lambda^{-2}$	$\frac{\lambda^2}{\lambda^2 - t^2}$
Cauchy	$\{\pi(1+x^2)\}^{-1}$			

and then

$$\mathbf{E}(X|B) = \int_{-\infty}^{\infty} x f_{X|B}(x|B)dx.$$

Table 7.1 gives some useful continuous random variables with their elementary properties.

Checklist of Terms for Chapter 7

7.1 distribution function
density
standard normal density $\phi(x)$
mixture
7.2 functions
inverse function
7.3 simulation
composition
7.4 expected value
expectation of functions
tail integral for expectation
7.5 moment generating function
continuity theorem
O–o notation
de Moivre–Laplace theorem
7.6 conditional distribution
conditional density
conditional expectation
lack-of-memory property
7.7 hazard function
hazard rate
7.8 stochastically larger
7.9 geometrical probability

Conclude that for fixed finite a and b, as $n \to \infty$,

$$\mathbf{P}(a < Y_n < b) \to \frac{1}{(2\pi)^{\frac{1}{2}}} \int_a^b e^{-\frac{1}{2}y^2} dy = \Phi(b) - \Phi(a).$$

(9) Exercise Let $(S_n; n \geq 0)$ be a simple random walk with $S_0 = 0$. Given that $S_{2n} = 2j$, show that the probability that the last visit to the origin was at the $2r$th step is

$$f_r = \frac{j}{n-r} \binom{n+j}{r} \binom{n-j}{r} \Big/ \binom{2n}{2r}, \quad 0 \leq x < n.$$

Show that if r, j, and n all increase in such a way that $\frac{r}{n} \to x$, $j/\sqrt{n} \to y$, then

$$\lim_{r \to \infty} f_r = \frac{y}{\sqrt{\pi x}} \frac{1}{(1-x)^{\frac{3}{2}}} \exp\left(\frac{-xy^2}{1-x}\right), \quad 0 < x < 1.$$

Remark The result of Exercise 6, which is known as Stirling's formula, was established by de Moivre in 1730. The formula actually proved by Stirling in 1730 was

$$n! \left(n + \frac{1}{2}\right)^{-(n+\frac{1}{2})} e^{n+\frac{1}{2}} \to (2\pi)^{\frac{1}{2}}, \quad \text{as } n \to \infty. \tag{10}$$

(11) Exercise Prove (10).

PROBLEMS

1 Let $f(x) = c(\alpha, \beta)(x - \alpha)(\beta - x)$.
For what values of x and $c(\alpha, \beta)$ can f be a density function?

2 Let X have distribution $F(x)$. Show that $\mathbf{P}(X = x) > 0$ if and only if $F(x)$ is discontinuous at x.

3 The *beta function* $B(a, b)$ is given by $B(a, b) = \int_0^1 v^{a-1}(1-v)^{b-1} dv; a > 0, b > 0$. The beta distribution has density

$$f(x) = \frac{1}{B(a, b)} x^{a-1}(1-x)^{b-1} \quad \text{for } 0 < x < 1.$$

If X has the beta distribution, show that $\mathbf{E}(X) = B(a+1, b)/B(a, b)$. What is var (X)?

4 For what value of c is $f = c(\sin x)^{\alpha}(\cos x)^{\beta}; 0 < x < \pi/2$, a density function?

5 What is the distribution function of the random variable having the beta density with $a = b = \frac{1}{2}$?

6 Let X have the density $f = \exp(-x - \exp(-x))$ for $x \in \mathbb{R}$. What is the distribution function of X?

7 Let X be exponentially distributed with parameter λ. What is the density of $Y = e^{aX}$? For what values of λ does $\mathbf{E}(Y)$ exist?

8 Let X have the gamma density with parameters α and λ. Show that

$$\mu_k = \alpha(\alpha+1)\ldots(\alpha+k-1)\lambda^{-k}, \quad \text{and } \operatorname{var}(X) = \frac{\alpha}{\lambda^2}.$$

9 Let X have the standard normal density, and $a > 0$. Show that

$$\mathbf{P}(X > x + ax^{-1} | X > x) \to e^{-a} \quad \text{as } x \to \infty.$$

10 (a) Let X have the standard normal density. Show that $|X|$ has distribution function $F = 2\Phi(x) - 1$, for $x > 0$.

(b) Let X have distribution $F(x)$. What is the distribution of $|X|$? What is the density of $|X|$ if it exists?

11 An elastic string has modulus of elasticity λ and natural length l_0. A mass m is attached to one end, the other being fixed. The period of oscillation of the mass when slightly displaced is $2\pi\sqrt{ml_0/\lambda}$. Suppose that the modulus of elasticity is uniformly distributed on $[a, b]$. What is the density and expectation of the period?

12 Let X have density $f(x)$. Construct a simple random variable $S_n(X)$ such that given $\epsilon > 0$, $\mathbf{P}(|S_n(X) - X| > \epsilon) < 2^{-n}$. (Assume X is proper.)

13 If X is exponentially distributed find the m.g.f. of X, $\mathbf{E}(e^{tX})$.

14 A point Q is chosen at random inside an equilateral triangle of unit side. Find the density of the perpendicular distance X to the nearest side of the triangle.

15 For what value of c is $\mathbf{E}((X - c)^2)$ least?

16 Suppose a machine's lifetime T has hazard rate $\lambda\sqrt{t}$, where $\lambda > 0$. Find $\mathbf{P}(T > t)$.

17 Suppose that X has distribution function

$$F(x) = 1 - \exp\left(-\int_0^x g(u)du\right)$$

for some function $g(.)$. Show that this is possible if and only if $g(u) \geq 0$ and $\int_0^\infty g(u)du = \infty$.

18 What are the cumulants of the normal density?

19 What are the cumulants of the exponential density?

20 You have two independent random variables, each uniform on (0, 1). Explain how you would use them to obtain a random variable X with density

$$f(x) = \frac{3}{5}\left(1 + x + \frac{1}{2}x^2\right) \quad \text{for } 0 \leq x \leq 1.$$

21 Define $I(a, b) = \int_0^\infty \exp(-a^2u^2 - b^2u^{-2})du$ for $a, b > 0$. Show that:
(a) $I(a, b) = a^{-1}I(1, ab)$; (b) $\frac{\partial I}{\partial b} = -2I(1, ab)$; (c) $I(a, b) = \frac{\sqrt{\pi}}{2a}e^{-2ab}$.

22 Use the result of Problem 21 to find the m.g.f. of the following densities:
(a) $f(x) = \alpha x^{-\frac{1}{2}} \exp(-\beta/x - \gamma x)$ for $x > 0; \beta, \gamma > 0$. What is α?
(b) $f(x) = (2\pi x^3)^{-\frac{1}{2}} \exp(-(2x)^{-1})$ for $x > 0$.

23 Let X be a standard normal random variable. What is the density of X^{-2}?

24 Let X be a standard normal random variable. Find the m.g.f. and density of X^2.

25 What is the moment generating function of the two-sided exponential density? Where is it defined?

26 Let U be uniform on (0, 1). Show that, if $[a]$ denotes the integer part of a, and $0 < p < 1$,

$$X = 1 + \left[\frac{\log U}{\log(1 - p)}\right]$$

has a geometric distribution.

27 Let U be uniform on (0, 1). Show how to use U to simulate a random variable with density

$$f(x) = \frac{24}{25}\left(1 + \frac{1}{2}\left(x - \frac{1}{2}\right)^2\right); \quad 0 \leq x \leq 1.$$

28 Let P and Q be two points chosen independently and uniformly in $(0, a)$. Show that the distance between P and Q has density $2(a - x)a^{-2}$ for $0 < x < a$.

29 **Continuous mixture** Let $f(\theta, x)$ be the exponential density with parameter θ for $0 \leq x$. Let $g(\theta) = \nu e^{-\nu\theta}$ for $\theta \geq 0$, $\nu > 0$. Show that $\int_0^\infty f(\theta, x)g(\theta)d\theta$ is a density.

30 Let X_n be a Poisson random variable with parameter n. Show that as $n \to \infty$, $\mathbf{P}(X_n \leq n + \sqrt{n}x) \to \Phi(x)$.

31 Let U be uniform on $(0, 1)$. Show how to simulate a random variable X with the Pareto distribution given by $F(x) = 1 - x^{-d}; x > 1, d > 1$.

32 Let X_α have gamma density with parameters α and 1, so $f_\alpha(x) = \frac{x^{\alpha-1}e^{-x}}{(\alpha-1)!}$. Let $\phi_\alpha(x)$ be the density of $Y_\alpha = (X_\alpha - \alpha)\alpha^{-\frac{1}{2}}$. Show that as $\alpha \to \infty$, $\phi_\alpha(x) \to \phi(x)$, where $\phi(x)$ is the standard normal density.

33 Let X have mean μ and variance σ^2. Show that $\mathbf{P}(|X - \mu| \leq a\sigma) \geq 1 - a^{-2}$, for any $a > 0$. Let Z be a standard normal random variable, and define $Y = \alpha + \beta Z + \gamma Z^2$. Find the mean and variance of Y, and show that

$$\mathbf{P}\left(|Y - \alpha - \gamma| \leq \frac{1}{2}\alpha\right) \geq 1 - 4\alpha^{-2}(\beta^2 + 2\gamma^2).$$

34 Verify the assertions of Theorem 7.4.16.

35 Use the indicator function $I(X > x)$ to show that, for a random variable X that is nonnegative,

(a) $\mathbf{E}X = \int_0^\infty \mathbf{P}(X > x)\,dx.$

(b) $\mathbf{E}X^r = \int_0^\infty rx^{r-1}\mathbf{P}(X > x)\,dx.$

(c) $\mathbf{E}e^{\theta X} = 1 + \theta\int_0^\infty e^{\theta x}\mathbf{P}(X > x)\,dx.$

(d) When $X \geq 0$ is integer valued,

$$\sum_{k=0}^\infty s^k\mathbf{P}(X > k) = \frac{1 - G_X(s)}{1 - s}.$$

36 Let X be a standard normal random variable with density $\phi(x)$ and distribution $\Phi(x)$. Show that for $x > 0$,

$$\frac{x}{x^2 + 1}\phi(x) \leq 1 - \Phi(x) \leq \frac{1}{x}\phi(x).$$

[*Hint*: Consider $f(x) = xe^{-x^2}/2 - (x^2 + 1)[1 - \Phi(x)]$.]

8
Jointly Continuous Random Variables

> Some instances of correlation are quite whimsical: thus cats which are entirely white and have blue eyes are generally deaf.
>
> *Charles Darwin, Origin of Species*

8.1 Joint Density and Distribution

It is often necessary to consider the joint behaviour of several random variables, which may each take an uncountable number of possible values. Just as for discrete random vectors, we need to define a variety of useful functions and develop the appropriate machinery to set them to work. For simplicity in definitions and theorems, we start by considering a pair of random variables (X, Y) taking values in $\mathbb{R}^2$. This theoretical outline can be easily extended to larger collections of random variables $(X_1, X_2, \ldots, X_n)$ taking values in $\mathbb{R}^n$, with a correspondingly greater expenditure of notation and space.

As usual, we should start with a sample space Ω, an event space $\mathcal{F}$, and a probability function $\mathbf{P}$, such that for all x and y

$$A_{xy} = \{\omega: X \leq x, Y \leq y\} \in \mathcal{F}.$$

Then

(1) $$F(x, y) = \mathbf{P}(A_{xy}) = \mathbf{P}(X \leq x, Y \leq y)$$

is the joint distribution function of X and Y. In fact, we suppress this underlying structure, and begin with random variables X and Y having joint distribution $F(x, y)$ given by (1). A special class of such jointly distributed random variables is of great importance.

(2) Definition *Let $F(x, y)$ be a joint distribution. Suppose that $\frac{\partial^2 F}{\partial x \partial y}$ exists and is nonnegative, except possibly on a finite collection of lines in $\mathbb{R}^2$. Suppose further that the function $f(x, y)$ defined by*

$$f(x, y) = \begin{cases} \dfrac{\partial^2 F}{\partial x \partial y} & \text{where this exists} \\ 0 & \text{elsewhere,} \end{cases}$$

satisfies

$$F(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v)\, du\, dv. \tag{3}$$

Then X and Y, being random variables having the (joint) distribution F, are said to be (jointly) continuous with (joint) density function $f(x, y)$. ▲

The words "joint" and "jointly" are often omitted to save time and trees. Sometimes we write $f_{X,Y}(x, y)$ and $F_{X,Y}(x, y)$ to stress the role of X and Y, or to avoid ambiguity.

(4) Example: Uniform Distribution Suppose you pick a point Q at random in the rectangle $R = (x, y : 0 < x < a, 0 < y < b)$. Then from the properties of the uniform distribution [see (7.8.1)], we have

$$F(x, y) = \begin{cases} 1 & \text{if } x \geq a, y \geq b \\ \dfrac{xy}{ab} & \text{if } 0 \leq x \leq a, 0 \leq y \leq b \\ \dfrac{y}{b} & \text{if } x \geq a, 0 \leq y \leq b \\ \dfrac{x}{a} & \text{if } 0 \leq x \leq a, y \geq b \\ 0 & \text{elsewhere.} \end{cases} \tag{5}$$

Differentiating wherever possible gives

$$\frac{\partial^2 F}{\partial x \partial y} = \begin{cases} \dfrac{1}{ab} & \text{if } 0 < x < a, 0 < y < b \\ 0 & \text{if } x < 0 \text{ or } x > a \\ & \text{or } y < 0 \text{ or } y > b. \end{cases} \tag{6}$$

Hence, the function

$$f(x, y) = \begin{cases} \dfrac{1}{ab} & \text{if } 0 < x < a, 0 < y < b \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

satisfies (3), and is the density of X and Y. It is uniformly distributed over the rectangle R. Furthermore, if A is a subset of R with area $|A|$, then using (7.8.1) (and a theorem about double integrals), we have

$$\mathbf{P}((X, Y) \in A) = \frac{|A|}{ab} = \int \int_{(x,y)\in A} f(x, y)\, dx dy. \tag{8}$$

●

In fact, a version of the useful relationship (8) holds true for all densities $f(x, y)$. This is important enough to state formally as a theorem, which we do not prove.

(9) **Theorem** *If X and Y have density $f(x, y)$ and $\mathbf{P}((X, Y) \in A)$ exists, then it is given by the Key Rule:*

$$\mathbf{P}((X, Y) \in A) = \int\int_{(x,y)\in A} f(x, y)\, dx dy.$$

Note that the condition that the probability exists is equivalent to saying that $\{\omega: (X(\omega), Y(\omega)) \in A\} \in \mathcal{F}$. This is another demonstration of the fact that, although we can just about suppress $(\Omega, \mathcal{F}, \mathbf{P})$ at this elementary level, further rigorous progress is not possible without bringing the underlying probability space into play. The attractive result (9) may then be proved.

Here is a simple example of Theorem 9 in use.

(10) Example Let X and Y have density

$$f(x, y) = \begin{cases} 8xy & \text{if } 0 < y < x < 1 \\ 0 & \text{elsewhere.} \end{cases}$$

What are $\mathbf{P}(2X > 1, 2Y < 1)$ and $\mathbf{P}(X + Y > 1)$? Find $F(x, y)$.

Solution Notice that the constraints $2X > 1, 2Y < 1$ require that $(X, Y) \in S$, where S is the square with vertices $(\frac{1}{2}, 0)$, $(1, 0)$, $(\frac{1}{2}, \frac{1}{2})$, $(1, \frac{1}{2})$. Hence,

$$\mathbf{P}(2X > 1, 2Y < 1) = 8\int_{\frac{1}{2}}^{1}\int_{0}^{\frac{1}{2}} f(x, y)\, dy dx = 8\int_{\frac{1}{2}}^{1} x\, dx \int_{0}^{\frac{1}{2}} y\, dy = \frac{3}{8}.$$

Likewise, $X + Y > 1$ if $(X, Y) \in T$, where T is the triangle with vertices $(\frac{1}{2}, \frac{1}{2})$, $(1, 0)$, $(1, 1)$. Hence,

$$\mathbf{P}(X + Y > 1) = 8\int_{\frac{1}{2}}^{1}\int_{1-x}^{x} xy\, dy\, dx = \frac{5}{6}.$$

Finally,

$$F(x, y) = \int_{0}^{y}\int_{v}^{x} 8uv\, du\, dv = 2x^2y^2 - y^4. \qquad \bullet$$

The geometric problems of Section 7.9 can now be reformulated and generalized in this new framework. Obviously, "picking a point Q at random in some region R" is what we would now describe as picking (X, Y) such that X and Y are jointly uniform in R. More generally, we can allow (X, Y) to have joint density $f(x, y)$ in R.

Example: More Triangles The random variables X and Y have joint density

$$f(x, y) = \begin{cases} cx^a & \text{if } x < 1, y < 1, x + y > 1, a > -1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is c?
(b) What is $F(x, y)$?

(c) Show that it is possible to construct a triangle with sides $X, Y, 2 - X - Y$, with probability one.
(d) Show that the angle opposite to the side of length Y is obtuse with probability

$$p_0 = c \int_0^1 \frac{x^{a+1} - x^{a+2}}{2 - x} dx.$$

(e) When $a = 0$, show that $p_0 = 3 - 4 \log 2$.

Solution (a) Because $\int \int f(x, y)\, dx\, dy = 1$, this entails

$$c^{-1} = \int_0^1 x^a \int_{1-x}^1 dy\, dx = (a + 2)^{-1}.$$

(b) Using (3) gives

$$F(x, y) = \int_{1-y}^x \int_{1-u}^y cu^a\, dv\, du = \frac{a+2}{a+1} y x^{a+1} + x^{a+2} - \frac{a+2}{a+1} x^{a+1} + \frac{1}{a+1}(1 - y)^{a+2}.$$

(c) Three such lengths form a triangle if $X + Y > 2 - X - Y$, $X + 2 - X - Y > Y$, and $Y + 2 - X - Y > X$. But these constraints are just those that define the region in which $f(x, y)$ is nonzero and $\int \int f\, dx\, dy = 1$.
(d) if θ is the angle opposite Y, then

$$\cos \theta = \frac{X^2 + (2 - X - Y)^2 - Y^2}{2X(2 - X - Y)} < 0,$$

if θ is an obtuse angle. Hence, in this case,

$$Y > \frac{X^2 - 2X + 2}{2 - X} = g(X), \quad \text{say.}$$

Now $g(x) \geq 1 - x$ (with equality only at $x = 0$). Hence, p_0 is given by

$$\mathbf{P}(\theta \text{ is obtuse}) = \mathbf{P}(Y > g(X)) = \int_0^1 \int_{g(x)}^1 f(x, y)\, dy dx = c \int_0^1 (1 - g(x)) x^a\, dx = c \int_0^1 \frac{x - x^2}{2 - x} x^a\, dx.$$

(e) When $a = 0$,

$$p_0 = 2 \int_0^1 \frac{x - x^2}{2 - x}\, dx = 2 \int_0^1 \frac{2x - x^2}{2 - x} + \frac{2 - x}{2 - x} - \frac{2}{2 - x}\, dx = 3 - 4 \log 2. \quad \bullet$$

Next we record that, as a result of (1) and Theorem 9, $f(x, y)$ and $F(x, y)$ have the following elementary properties, analogous to those of f and F in the discrete case.

First, $F(x, y)$ is obviously nondecreasing in x and y. More strongly, we have

$$\begin{aligned} 0 &\leq \mathbf{P}(a < X \leq b, c \leq Y \leq d) \\ &= \mathbf{P}(a < X \leq b, Y \leq d) - \mathbf{P}(a < X \leq b, Y \leq c) \\ &= F(b, d) - F(a, d) - F(b, c) + F(a, c). \end{aligned} \tag{11}$$

Second, if X and Y are finite with probability 1, then

$$1 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u, v)\, du\, dv = \lim_{x,y \to \infty} F(x, y). \tag{12}$$

Third, knowledge of $F(x, y)$ and $f(x, y)$ will also provide us with the separate distributions and densities of X and Y. Thus, the *marginals* are:

$$F_X(x) = \mathbf{P}(X \leq x) = \lim_{y \to \infty} \mathbf{P}(X \leq x, Y \leq y) = \int_{-\infty}^{\infty} \int_{-\infty}^{x} f(u, v)\, du dv, \tag{13}$$

and

$$f_X(x) = \frac{d}{dx} F_X(x) = \int_{-\infty}^{\infty} f(x, v)\, dv. \tag{14}$$

Likewise,

$$f_Y(y) = \int_{-\infty}^{\infty} f(u, y)\, du \tag{15}$$

and

$$F_Y(y) = \lim_{x \to \infty} F(x, y). \tag{16}$$

Here are some examples to illustrate these properties. Note that in future we will specify $f(x, y)$ only where it is nonzero.

(17) Example Verify that the function $f(x, y) = 8xy$ for $0 < y < x < 1$ is a density. For what value of c is $f(x, y) = cxy$ for $0 < x < y < 1$, a density? Find the density of X in the second case.

Solution Because $f > 0$ and

$$\int_0^1 \int_0^x 8xy\, dy dx = \int_0^1 4x^3\, dx = 1,$$

f is indeed a density. By symmetry, $c = 8$ in the second case also, and we have

$$f_X(x) = \int_0^1 8xy\, dy = 4x(1 - x^2). \qquad \bullet$$

(18) Example The function $H(x, y) = 1 - e^{-(x+y)}$ for $x > 0$, $y > 0$, is nondecreasing in x and y, and $0 \leq H \leq 1$. Is it a distribution?

Solution No, because $\frac{\partial^2 H}{\partial x \partial y}$ exists and is negative in $x > 0$, $y > 0$. Alternatively, note that

$$H(1,1) - H(1,0) - H(0,1) + H(0,0) = 2e^{-1} - 1 - e^{-2} < 0,$$

which cannot (as it should) be the value of $\mathbf{P}(0 < X \leq 1, 0 < Y \leq 1)$. ●

(19) Example: Bivariate Normal Density Verify that when $\sigma, \tau > 0$,

$$f(x,y) = \frac{1}{2\pi\sigma\tau(1-\rho^2)^{\frac{1}{2}}} \exp\left[-\frac{1}{2(1-\rho^2)} \left(\frac{x^2}{\sigma^2} - \frac{2\rho xy}{\sigma\tau} + \frac{y^2}{\tau^2} \right) \right]$$

is a density for $|\rho| < 1$, and find the marginal densities $f_X(x)$ and $f_Y(y)$.

Solution From (14), if $f(x,y)$ is a density, we have

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x,y)\,dy \\ &= \frac{1}{2\pi\sigma\tau(1-\rho^2)^{\frac{1}{2}}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2(1-\rho^2)} \left(\left(\frac{y}{\tau} - \frac{\rho x}{\sigma} \right)^2 + \frac{x^2}{\sigma^2} - \frac{\rho^2 x^2}{\sigma^2} \right) \right] dy \end{aligned}$$

Now setting $\frac{y}{\tau} - \frac{\rho x}{\sigma} = u$, and recalling that

$$\int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2(1-\rho^2)} \right) \tau\, du = (2\pi(1-\rho^2))^{\frac{1}{2}} \tau$$

yields

$$f_X(x) = \frac{1}{(2\pi)^{\frac{1}{2}}\sigma} \exp\left(-\frac{x^2}{2\sigma^2} \right).$$

This is the $N(0, \sigma^2)$ density, and so f satisfies (12) and is nonnegative. It is therefore a density. Interchanging the roles of x and y in the above integrals shows that $f_Y(y)$ is the $N(0, \tau^2)$ density. See Example 8.20 for another approach. ●

8.2 Change of Variables

We have interpreted the random vector (X, Y) as a random point Q picked in $\mathbb{R}^2$ according to some density $f(x,y)$, where (x,y) are the Cartesian coordinates of Q. Of course, the choice of coordinate system is arbitrary; we may for some very good reasons choose to represent Q in another system of coordinates (u, v), where (x,y) and (u,v) are related by $u = u(x,y)$ and $v = v(x,y)$. What now is the joint density of $U = u(X,Y)$ and $V = v(X,Y)$?

Equally, given a pair of random variables X and Y, our real interest may well lie in some function or functions of X and Y. What is their (joint) distribution?

As we have remarked above, at a symbolic or formal level, the answer is straightforward. For U and V above, and $A = \{x, y: u(x, y) \le w, v(x, y) \le z\}$ then, by Theorem 8.1.9,

$$F_{U,V}(w, z) = \int\int_A f_{X,Y}(x, y)\, dx\, dy.$$

The problem is to turn this into a more tractable form.

Fortunately, there are well-known results about changing variables within a multiple integral that provide the answer. We state without proof a theorem for a transformation T satisfying the following conditions. Let C and D be subsets of $\mathbb{R}^2$. Suppose that T given by

$$T(x, y) = (u(x, y), v(x, y))$$

maps C one–one onto D, with inverse T^{-1} given by

$$T^{-1}(u, v) = (x(u, v), y(u, v)),$$

which maps D one–one onto C. We define the so-called *Jacobian* J as

$$J(u, v) = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u},$$

where the derivatives are required to exist and be continuous in D. Then we have the following result.

(1) **Theorem** *Let X and Y have density $f(x, y)$, which is zero outside C. Then $U = u(X, Y)$ and $V = v(X, Y)$ have joint density*

$$f_{U,V}(u, v) = f_{X,Y}(x(u, v), y(u, v))|J(u, v)| \quad \text{for } (u, v) \in D.$$

Here are some examples of this theorem in use.

(2) **Example** Suppose $Q = (X, Y)$ is uniformly distributed over the circular disc of radius 1. Then X and Y have joint density

$$f(x, y) = \frac{1}{\pi} \quad \text{for } x^2 + y^2 \le 1. \tag{3}$$

However, it seems more natural to use polar rather than Cartesian coordinates in this case. These are given by $r = (x^2 + y^2)^{\frac{1}{2}}$ and $\theta = \tan^{-1}(y/x)$, with inverse $x = r\cos\theta$ and $y = r\sin\theta$. They map $C = \{x, y: x^2 + y^2 \le 1\}$ one–one onto $D = \{r, \theta: 0 \le r \le 1, 0 < \theta \le 2\pi\}$.

In this case,

$$J(r, \theta) = \frac{\partial x}{\partial r}\frac{\partial y}{\partial \theta} - \frac{\partial x}{\partial \theta}\frac{\partial y}{\partial r} = r\cos^2\theta + r\sin^2\theta = r.$$

Hence, the random variables $R = r(X, Y)$ and $\Theta = \theta(X, Y)$ have joint density given by

$$f_{R,\Theta}(r, \theta) = \frac{r}{\pi} \quad \text{for } 0 \le r \le 1, 0 < \theta \le 2\pi. \tag{4}$$

Notice that $f(r, \theta)$ is not uniform, as was $f(x, y)$. ●

(5) Example Let $Q = (X, Y)$ be uniformly distributed over the ellipse C with boundary satisfying

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

of area $|C|$. What is $\mathbf{P}(X > Y, X > -Y)$?

Solution Here the transformation $x = ar\cos\theta$ and $y = br\sin\theta$ maps the ellipse one–one onto the circular disc with radius 1. Furthermore, $J = abr$, Now X and Y have density

$$f(x, y) = \frac{1}{|C|}, \quad \text{for } (x, y) \in C,$$

so R and Θ have joint density

$$f(r, \theta) = \frac{abr}{|C|}, \quad \text{for } 0 \le r < 1, 0 < \theta \le 2\pi.$$

Hence, $|C| = \pi ab$, and

$$\mathbf{P}(X > Y, X > -Y) = \mathbf{P}\left(-1 < \frac{Y}{X} < 1\right) = \mathbf{P}\left(-1 < \frac{b}{a}\tan\Theta < 1\right) = \frac{1}{\pi}\tan^{-1}\frac{a}{b},$$

because Θ is uniform on $(0, 2\pi)$. ●

8.3 Independence

As usual, independence is an extremely important property; its definition is by now familiar.

(1) Definition *Jointly distributed random variables are* independent *if, for all x and y,*

$$\mathbf{P}(X \le x, Y \le y) = \mathbf{P}(X \le x)\mathbf{P}(Y \le y).$$

In terms of distributions, this is equivalent to the statement that

(2) $$F(x, y) = F_X(x)F_Y(y).$$ ▲

For random variables with a density, it follows immediately by differentiating that

(3) $$f(x, y) = f_X(x)f_Y(y)$$

if X and Y are independent. Using the basic property of densities (Theorem 8.1.9) now further shows that, if $C = (x, y: x \in A, y \in B)$ and X and Y are independent, then

(4) $$\int\int_C f(x, y)\,dxdy = \int_A f_X(x)\,dx \int_B f_Y(y)\,dy.$$

(Assuming of course that the integrals exist.)

Finally, if the random variables U and V satisfy $U = g(X)$, $V = h(Y)$, and X and Y are independent, then U and V are independent. To see this, just let

(5) $$A = (x: g(x) \le u) \quad \text{and} \quad B = (g: h(y) \le v),$$

and the independence follows from (4) and (2). An important and useful converse is the following.

(6) **Theorem** *If X and Y have density $f(x, y)$, and for all x and y it is true that*

$$f(x, y) = f_X(x) f_Y(y),$$

then X and Y are independent.

The proof follows immediately from a standard theorem on multiple integrals (just consider $\int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v)\, du\, dv$) and we omit it.

(7) **Example: Uniform Distribution** Let X and Y have the uniform density over the unit circular disc C, namely,

$$f(x, y) = \pi^{-1} \quad \text{for } (x, y) \in C.$$

(a) Are X and Y independent?
(b) Find $f_X(x)$ and $f_Y(y)$.
(c) If $X = R \cos \Theta$, and $Y = R \sin \Theta$, are R and Θ independent?

Solution (a) The set $\{x, y: x \leq -1/\sqrt{2}, y \leq -1/\sqrt{2}\}$ lies outside C, so

$$F\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right) = 0.$$

However, the intersection of the set $\{x: x \leq -1/\sqrt{2}\}$ with C has nonzero area, so

$$F_X\left(-\frac{1}{\sqrt{2}}\right) F_Y\left(-\frac{1}{\sqrt{2}}\right) > 0.$$

Therefore, X and Y are not independent.

(b) By (8.1.14),

$$f_X(x) = \int_{-1}^{1} f(x, y)\, dy = \frac{1}{\pi} \int_{-(1-x^2)^{\frac{1}{2}}}^{(1-x^2)^{\frac{1}{2}}} dy = \frac{2}{\pi}(1 - x^2)^{\frac{1}{2}}.$$

Likewise,

$$f_Y(y) = \frac{2}{\pi}(1 - y^2)^{\frac{1}{2}}.$$

(c) By Example 8.2.4, R and Θ have joint density

$$f_{R,\Theta}(r, \theta) = \frac{r}{\pi}, \quad \text{for } 0 \leq r < 1, 0 < \theta \leq 2\pi.$$

Hence,

$$f_\Theta(\theta) = \int_0^1 f(r, \theta)\, dr = \frac{1}{2\pi}; \quad 0 < \theta \leq 2\pi,$$

and

$$f_R(r) = \int_0^{2\pi} f(r, \theta)\, d\theta = 2r; \quad 0 \le r \le 1.$$

Hence, $f(r, \theta) = f_\Theta(\theta) f_R(r)$, and so R and Θ are independent. ●

Example: Bertrand's Paradox Again Suppose we choose a random chord of a circle C radius a, as follows. A point P is picked at random (uniformly) inside C. Then a line through P is picked independently of P at random [i.e., its direction Θ is uniform on $(0, 2\pi)$]. Let X be the length of the chord formed by the intersection of this line with the circle. Show that

$$\mathbf{P}(X > a\sqrt{3}) = \frac{1}{3} + \frac{\sqrt{3}}{2\pi}.$$

Solution Let R be the distance from the centre of the circle to P; by the above, R has distribution given by $\mathbf{P}(R/a \le r) = r^2; 0 \le r \le 1$. Now $X > a\sqrt{3}$, if and only if $2R \sin\Theta < a$, as you can see by inspecting Figure 8.1. Hence,

$$\begin{aligned}\mathbf{P}(X > a\sqrt{3}) &= \frac{2}{\pi}\int_0^{\pi/2} \mathbf{P}\left(R < \frac{a}{2\sin\theta}\right) d\theta \\ &= \frac{2}{\pi}\int_0^{\pi/6} d\theta + \frac{2}{\pi}\int_{\pi/6}^{\pi/2} \frac{1}{4}\operatorname{cosec}^2\theta\, d\theta = \frac{1}{3} + \frac{\sqrt{3}}{2\pi}.\end{aligned}$$

Compare this with the results of Example 7.13. ●

(8) Example: Normal Densities Let X and Y be independent with common density $f(x) = k\exp(-\frac{1}{2}x^2)$ for all x.

(a) Show that $k = (2\pi)^{-\frac{1}{2}}$.
(b) Show that $X^2 + Y^2$ and $\tan^{-1}(Y/X)$ are independent random variables.
(c) If $a > 0 < b < c$ and $0 < \alpha < \frac{1}{2}\pi$, find the probability that $b < (X^2 + Y^2)^{\frac{1}{2}} < c$ and $\frac{1}{4}\pi < \tan^{-1}(Y/X) < \frac{1}{2}\pi$, given that $(X^2 + Y^2)^{\frac{1}{2}} < a$, $Y > 0$, and $\tan^{-1}(Y/X) < \frac{1}{3}\pi$.

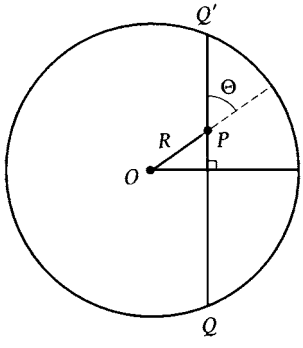


Figure 8.1 Bertrand's paradox.

Solution Because X and Y are independent, they have joint density

$$f(x, y) = k^2 \exp\left(-\frac{1}{2}(x^2 + y^2)\right).$$

Make the change of variables to polar coordinates, so that by Theorem 8.2.1 the random variables $R = (X^2 + Y^2)^{\frac{1}{2}}$ and $\Theta = \tan^{-1}(Y/X)$ have joint density

$$f(r, \theta) = k^2 r \exp\left(-\frac{1}{2}r^2\right) \quad \text{for } 0 \le r < \infty, 0 < \theta \le 2\pi.$$

Hence, R has density

$$f_R(r) = r \exp\left(-\frac{1}{2}r^2\right) \quad \text{for } 0 \le r < \infty,$$

and Θ has density

$$f_\Theta(\theta) = k^2 \quad \text{for } 0 < \theta \le 2\pi.$$

It follows immediately that

(a) $k^2 = (2\pi)^{-1}$.
(b) $f(r, \theta) = f_R(r) f_\Theta(\theta)$, so that Θ and R are independent by Theorem 6. Hence, Θ and R^2 are independent.
(c) Finally, note that

$$\mathbf{P}\left(\frac{1}{4}\pi < \Theta < \frac{1}{2}\pi, 0 < \Theta < \frac{1}{3}\pi\right) = \mathbf{P}\left(\frac{1}{4}\pi < \Theta < \frac{1}{3}\pi\right) = \frac{\pi}{12} \tag{9}$$

and

$$\begin{aligned}\mathbf{P}(b < R < c, R < a) &= \begin{cases} F_R(c) - F_R(b) & \text{if } c < a \\ F_R(a) - F_R(b) & \text{if } b < a \le c \\ 0 & \text{otherwise} \end{cases} \\ &= F_R((a \wedge c) \vee b) - F_R(b), \end{aligned} \tag{10}$$

where $x \wedge y = \min\{x, y\}$ and $x \vee y = \max\{x, y\}$. Now, because R and Θ are independent,

$$\begin{aligned}&\mathbf{P}\left(b < R < c, \frac{1}{4}\pi < \Theta < \frac{1}{2}\pi \,|\, R < a, 0 < \Theta < \frac{1}{3}\pi\right) \\ &= \frac{\mathbf{P}(b < R < c, R < a)}{\mathbf{P}(R < a)} \frac{\mathbf{P}\left(\frac{1}{4}\pi < \Theta < \frac{1}{2}\pi, 0 < \Theta < \frac{1}{3}\pi\right)}{\mathbf{P}\left(0 < \Theta < \frac{1}{3}\pi\right)} \\ &= \frac{1}{4F_R(a)}(F_R((a \wedge c) \vee b) - F_R(b)), \quad \text{by (9) and (10).} \end{aligned}$$

●

8.4 Sums, Products, and Quotients

We now return to the question of the distribution of functions of random vectors, and take a brief look at some particularly important special cases. Of these, the most important is the sum of two random variables.

Theorem *Let X and Y have joint density $f(x, y)$. Show that if $Z = X + Y$, then*

$$f_Z(z) = \int_{-\infty}^{\infty} f(u, z-u)\, du, \tag{1}$$

and that if X and Y are independent, then

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(u) f_Y(z-u)\, du. \tag{2}$$

Proof First notice that by (8.3.3) the result (2) follows immediately from (1) when X and Y are independent. Turning to the proof of (1), we give two methods of solution.

I Let A be the region in which $u + v \leq z$. Then

$$\begin{aligned}\mathbf{P}(Z \leq z) &= \int\int_{(u,v)\in A} f(u, v)\, du\, dv = \int_{-\infty}^{\infty}\int_{-\infty}^{z-u} f(u, v)\, dv du \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{z} f(u, w-u)\, dw du \quad \text{on setting } v = w - u.\end{aligned}$$

Now differentiating with respect to z gives $f_Z(z) = \int_{-\infty}^{\infty} f(u, z-u)\, du$.

II This time we use the change of variable technique of Section 8.2. Consider the transformation $z = x + y$ and $u = x$, with inverse $x = u$ and $y = z - u$. Here $J = 1$. This satisfies the conditions of Theorem 8.2.1, and so $U = u(X, Y)$ and $Z = z(X, Y)$ have joint density $f(u, z-u)$. We require the marginal density of Z, which is of course just (1). ■

(3) Example Let X and Y have the bivariate normal distribution of Example 8.1.19,

$$f(x, y) = \frac{1}{2\pi\sigma\tau(1-\rho^2)^{1/2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left(\frac{x^2}{\sigma^2} - \frac{2\rho xy}{\sigma\tau} + \frac{y^2}{\tau^2}\right)\right).$$

Find the density of $aX + bY$ for constants a and b.

Remark: Note from 8.3.6 that X and Y are independent if and only if $\rho = 0$.

Solution The joint density of $U = aX$ and $V = bY$ is

$$g(u, v) = \frac{1}{ab} f\left(\frac{u}{a}, \frac{v}{b}\right).$$

Hence, by the above theorem, the density of $Z = U + V = aX + bY$ is

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{ab} f\left(\frac{u}{a}, \frac{z-u}{b}\right) du. \tag{4}$$

Rearranging the exponent in the integrand we have, after a little manipulation,

$$\begin{aligned} &\frac{-1}{2(1-\rho^2)}\left(\frac{u^2}{a^2\sigma^2} - \frac{2\rho u(z-u)}{ab\sigma\tau} + \frac{(z-u)^2}{b^2\tau^2}\right) \\ &\quad = \frac{-1}{2(1-\rho^2)}\left(\alpha\left(u - \frac{\beta}{\alpha}z\right)^2 + \frac{z^2}{\alpha} - \frac{(1-\rho^2)}{a^2b^2\sigma^2\tau^2}\right), \end{aligned}$$

where

$$\alpha = \frac{1}{a^2\sigma^2} + \frac{2\rho}{ab\sigma\tau} + \frac{1}{b^2\tau^2}, \quad \text{and} \quad \beta = \frac{\rho}{ab\sigma\tau} + \frac{1}{b^2\tau^2}.$$

Setting $u = v + \frac{\beta}{\alpha}z$ in the integrand, we evaluate

$$\int_{-\infty}^{\infty} \exp\left(-\frac{\alpha v^2}{2(1-\rho^2)}\right) dv = \left(\frac{2\pi(1-\rho^2)}{\alpha}\right)^{\frac{1}{2}}.$$

Hence, after a little more manipulation, we find that

$$f_Z(z) = \frac{1}{(2\pi\xi^2)^{\frac{1}{2}}} \exp\left(-\frac{z^2}{2\xi^2}\right),$$

where $\xi^2 = a^2\sigma^2 + 2\rho ab\sigma\tau + b^2\tau^2$. That is to say, Z is $N(0, \xi^2)$. ●

One important special case arises when $\rho = 0$, and X and Y are therefore independent. The above result then shows we have proved the following.

(5) Theorem *Let X and Y be independent normal random variables having the densities $N(0, \sigma^2)$ and $N(0, \tau^2)$. Then the sum $Z = aX + bY$ has the density $N(0, a^2\sigma^2 + b^2\tau^2)$. (See Example 8.20 for another approach.)* ■

Next we turn to products and quotients.

(6) Theorem *Let X and Y have joint density $f(x, y)$. Then the density of $Z = XY$ is*

$$f(z) = \int_{-\infty}^{\infty} \frac{1}{|u|} f\left(u, \frac{z}{u}\right) du, \tag{7}$$

and the density of $W = \dfrac{X}{Y}$ is

$$f(w) = \int_{-\infty}^{\infty} |u| f(uw, u)\, du.$$

Proof We use Theorem 8.2.1 again. Consider the transformation $u = x$ and $z = xy$, with inverse $x = u$ and $y = z/u$. Here,

$$J(u, z) = \begin{vmatrix} 1 & 0 \\ \dfrac{-z}{u^2} & \dfrac{1}{u} \end{vmatrix} = u^{-1}.$$

This satisfies the conditions of Theorem 8.2.1, and so $U = X$ and $Z = XY$ have joint density

$$f(u, z) = \frac{1}{|u|} f\left(u, \frac{z}{u}\right). \tag{8}$$

The result (7) follows immediately as it is the marginal density of Z obtained from $f(u, z)$.

Alternatively, it is possible to derive the result directly by the usual plod, as follows:

$$\begin{aligned} \mathbf{P}(XY \le z) &= \mathbf{P}\left(X > 0, Y \le \frac{z}{X}\right) + \mathbf{P}\left(X < 0, Y \ge \frac{z}{X}\right) \\ &= \int_{-\infty}^{0} \int_{z/u}^{\infty} f(u, v)\, dv\, du + \int_{0}^{\infty} \int_{-\infty}^{z/u} f(u, v)\, dv\, du \\ &= \int_{-\infty}^{0} \int_{-\infty}^{z} f\left(u, \frac{t}{u}\right) \frac{dt}{(-u)} du + \int_{0}^{\infty} \int_{-\infty}^{z} f\left(u, \frac{t}{u}\right) \frac{dt}{u} du \\ &= \int_{-\infty}^{z} \int_{-\infty}^{\infty} f\left(u, \frac{t}{u}\right) \frac{du}{|u|} dt. \end{aligned}$$

The required density is obtained by comparison of this expression with (7.1.15).

Now we turn to the quotient $W = X/Y$. First, let $V = 1/Y$. Then, by definition,

$$F_{X,V}(x, v) = \mathbf{P}(X \le x, V \le v) = \mathbf{P}\left(X \le x, Y \ge \frac{1}{v}\right) = \int_{-\infty}^{x} \int_{1/v}^{\infty} f(s, t)\, ds dt.$$

Hence, on differentiating, the joint density of X and Y^{-1} is given by

$$f_{X,V}(x, v) = \frac{1}{v^2} f\left(x, \frac{1}{v}\right).$$

Now $W = XV$, so by the first part,

$$f_W(w) = \int_{-\infty}^{\infty} \frac{1}{|u|} \frac{u^2}{w^2} f\left(u, \frac{u}{w}\right) du = \int_{-\infty}^{\infty} |v| f(vw,\ v)\, dv$$

on setting $u = vw$ in the integrand. Alternatively, of course, you can obtain this by using Theorem 8.2.1 directly via the transformation $w = x/y$ and $u = y$, or you can proceed via the routine plod. ■

As usual, here are some illustrative examples.

(9) Example Let X and Y be independent with respective density functions $f_X(x) = xe^{-\frac{x^2}{2}}$ for $x > 0$ and $f_Y(y) = \pi^{-1}(1 - y^2)^{-\frac{1}{2}}$ for $|y| < 1$.

Show that XY has a normal distribution.

Solution When X and Y are independent, we have $f(x, y) = f_X(x)f_Y(y)$, and Theorem 6 takes the special form

$$f(z) = \int_{-\infty}^{\infty} \frac{1}{|u|} f_X(u) f_Y\left(\frac{z}{u}\right) du = \int_{u>z} \frac{1}{|u|} u e^{-\frac{u^2}{2}} \pi^{-1} \left(1 - \frac{z^2}{u^2}\right)^{-\frac{1}{2}} du$$
$$= \frac{1}{\pi} \int_z^{\infty} \frac{e^{-\frac{u^2}{2}}}{(u^2 - z^2)^{\frac{1}{2}}} u \, du.$$

Now we make the substitution $u^2 = z^2 + v^2$ to find that

$$f(z) = \frac{1}{\pi} e^{-\frac{z^2}{2}} \int_0^{\infty} e^{-\frac{v^2}{2}} dv = \frac{e^{-\frac{z^2}{2}}}{(2\pi)^{\frac{1}{2}}},$$

which is the $N(0, 1)$ density. ●

(10) Example Let X and Y have density $f(x, y) = e^{-x-y}$ for $x > 0$, $y > 0$. Show that $U = X/(X + Y)$ has the uniform density on $(0, 1)$.

Solution To use Theorem 6, we need to know the joint density of X and $V = X + Y$. A trivial application of Theorem 8.2.1 shows that X and V have density $f(x, v) = e^{-v}$ for $0 < x < v < \infty$. Hence, by Theorem 6,

$$f(u) = \int_0^{\infty} v e^{-v} dv, \quad \text{for } 0 < uv < v$$
$$= 1 \qquad\qquad , \quad \text{for } 0 < u < 1.$$

Alternatively, we may use Theorem 8.2.1 directly by considering the transformation

$$u = \frac{x}{x + y}, \quad v = x + y,$$

with

$$x = uv, \quad y = v(1 - u)$$

and $|J| = v$. Hence, $U = X/(X + Y)$ and $V = X + Y$ have density $f(u, v) = ve^{-v}$, for $v > 0$ and $0 < u < 1$. The marginal density of U is 1, as required. ●

8.5 Expectation

Suppose that the random variable $Z = g(X, Y)$ has density $f(z)$. Then, by definition,

$$\mathbf{E}(Z) = \int_{-\infty}^{\infty} z f(z) \, dz$$

provided that $\mathbf{E}(|Z|) < \infty$. However, suppose we know only the joint density $f(x, y)$ of X and Y. As we have discovered above, finding the density of $g(X, Y)$ may not be a trivial matter. Fortunately, this task is rendered unnecessary by the following result, which we state without proof.

(1) Theorem *If X and Y have joint density $f(x, y)$ and*

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |g(u, v)| f(u, v)\, du dv < \infty,$$

then

(2)
$$\mathbf{E}(g(X, Y)) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(u, v) f(u, v)\, du dv.$$

This useful result has the same pleasing consequences as did the corresponding result (Theorem 5.3.1) for discrete random variables.

(3) Corollary *Let X and Y have finite expectations. Then*

(i) $\mathbf{E}(aX + bY) = a\mathbf{E}(X) + b\mathbf{E}(Y)$, for any constants a and b.
(ii) If $\mathbf{P}(X \leq Y) = 1$, then $\mathbf{E}(X) \leq \mathbf{E}(Y)$.

Suppose further that $\mathbf{E}(X^2)$ and $\mathbf{E}(Y^2)$ are finite. Then

(iii) $\mathbf{E}(X) \leq \mathbf{E}(|X|) \leq (\mathbf{E}(X^2))^{\frac{1}{2}}$,
(iv) $\mathbf{E}(XY) \leq (\mathbf{E}(X^2)\mathbf{E}(Y^2))^{\frac{1}{2}}$.

Recall that this last result is the Cauchy–Schwarz inequality.

Finally, suppose that $\mathbf{E}(g(X))$ and $\mathbf{E}(h(Y))$ are finite, and that X and Y are independent. Then

(v) $\mathbf{E}(g(X)h(Y)) = \mathbf{E}(g(X))\mathbf{E}(h(Y))$.

We omit the proofs of these results. Generally speaking, the proofs follow the same line of argument as in the discrete case, with the difference that those proofs used results about rearrangement of sums, whereas these proofs use standard results about multiple integrals.

Definition *Some important expectations deserve special mention. Just as we did for discrete random variables, we define the* covariance *as*

$$\operatorname{cov}(X, Y) = \mathbf{E}((X - \mathbf{E}(X))(Y - \mathbf{E}(Y))),$$

and the correlation *as*

$$\rho(X, Y) = \frac{\operatorname{cov}(X, Y)}{(\operatorname{var}(X)\operatorname{var}(Y))^{\frac{1}{2}}}.$$ ▲

Remark When X and Y are independent, then it follows from Corollary 3(v) that $\operatorname{cov}(X, Y) = \rho(X, Y) = 0$, but not conversely. There is an important exception to this, in that bivariate normal random variables *are* independent if and only if $\rho(X, Y) = 0$. See Examples 8.11 and 8.20 for details

(4) Example Let Θ be uniformly distributed over $(0, \alpha)$. Find $\operatorname{cov}(\sin\Theta, \cos\Theta)$, and show that for $\alpha = k\pi$, $(k \neq 0)$, $\sin\Theta$ and $\cos\Theta$ are uncorrelated and not independent.

Solution Routine calculations proceed thus:

$$\mathbf{E}(\sin\Theta\cos\Theta) = \frac{1}{\alpha}\int_0^\alpha \frac{1}{2}\sin 2\theta d\theta = \frac{1}{4\alpha}(1-\cos 2\alpha),$$

and $\mathbf{E}(\sin\Theta) = (1/\alpha)(1-\cos\alpha)$, and $\mathbf{E}(\cos\Theta) = (1/\alpha)\sin\alpha$. Hence,

$$\operatorname{cov}(\sin\Theta, \cos\Theta) = \frac{1}{4\alpha}(1-\cos 2\alpha) - \frac{1}{\alpha^2}\sin\alpha(1-\cos\alpha).$$

This covariance is zero whenever $\alpha = k\pi$, $(k \neq 0)$, and so for these values of α, $\rho = 0$. However, $\sin\Theta$ and $\cos\Theta$ are not independent. This is obvious because $\sin\Theta = \cos(\frac{\pi}{2} - \Theta)$, but we can verify it formally by noting that

$$\mathbf{P}\left(\sin\Theta > \frac{3}{4}, \cos\Theta > \frac{3}{4}\right) = 0 \neq \mathbf{P}\left(\sin\Theta > \frac{3}{4}\right)\mathbf{P}\left(\cos\Theta > \frac{3}{4}\right). \quad \bullet$$

Generating functions have been so useful above that it is natural to introduce them again now. Because we are dealing with jointly continuous random variables, the obvious candidate for our attention is a joint moment generating function.

(5) Definition *Let X and Y be jointly distributed. The* joint moment generating function *of X and Y is*

$$M_{X,Y}(s,t) = \mathbf{E}(e^{sX+tY}). \quad \blacktriangle$$

If this exists in a neighbourhood of the origin, then it has the same attractive properties as the ordinary m.g.f. That is, it determines the joint distribution of X and Y uniquely, and also it does yield the moments, in that

(6)
$$\frac{\partial^{m+n}}{\partial s^m \partial t^n} M(s,t)\Big|_{s=t=0} = \mathbf{E}(X^m Y^n).$$

Furthermore, just as joint p.g.f.s factorize for independent discrete random variables, it is the case that joint m.g.f.s factorize for independent continuous random variables. That is to say,

(7)
$$M_{X,Y}(s,t) = M_X(s)M_Y(t)$$

if and only if X and Y are independent.

We offer no proofs for the above statements, as a proper account would require a wealth of analytical details. Nevertheless, we will use them freely, as required.

(8) Example Let X and Y have density $(x, y) = e^{-y}$, $0 < x < y < \infty$. Then

$$M_{X,Y}(s,t) = \int_0^\infty \int_x^\infty e^{sx+ty-y} dy dx = ((1-t)(1-s-t))^{-1}.$$

Hence, differentiating, we obtain

$$\left.\frac{\partial M}{\partial s}\right|_{s=t=0} = 1, \left.\frac{\partial M}{\partial t}\right|_{s=t=0} = 2, \left.\frac{\partial^2 M}{\partial s \partial t}\right|_{s=t=0} = 3.$$

Thus, $\operatorname{cov}(X, Y) = 1$. ●

(9) Example Let X and Y be independent and identically distributed normal random variables. Let $M = \frac{1}{2}(X + Y)$ and $V = (X - M)^2 + (Y - M)^2$. Show that M and V are independent.

Solution We will use an obvious extension of (7). Let $\mathbf{E}(X) = \mu$ and $\operatorname{var}(X) = \sigma^2$. Then consider the joint moment generating function of M, $X - M$, and $Y - M$:

$$\begin{aligned}\mathbf{E}(\exp(sM + t(X - M) + u(Y - M))) &= \mathbf{E}\left(\exp\left(\frac{1}{2}(s + t + u)X + \frac{1}{2}(s - t - u)Y\right)\right)\\ &= \exp\left(\frac{\mu}{2}(s + t + u) + \frac{\sigma^2}{8}(s + t + u)^2\right)\exp\left(\frac{\mu}{2}(s - t - u) + \frac{\sigma^2}{8}(s - t - u)^2\right),\end{aligned}$$

because X and Y are independent,

$$= \exp\left(\mu s + \frac{\sigma^2}{4}s^2\right)\exp\left(\frac{\sigma^2}{4}(t + u)^2\right). \tag{10}$$

Hence, M is independent of the random vector $(X - M, Y - M)$, and so M is independent of V. ●

Remark This remarkable property of the normal distribution extends to any independent collection $(X_i; 1 \le i \le n)$ of $N(\mu, \sigma^2)$ random variables, and is known as the independence of sample mean and sample variance property.

(11) Example Let X and Y be independent and identically distributed with mean zero, variance 1, and moment generating function $M(t)$, which is thrice differentiable at 0. Show that if $X + Y$ and $X - Y$ are independent, then X and Y are normally distributed.

Solution By the independence of X and Y,

$$\begin{aligned}M(s + t)M(s - t) &= \mathbf{E}(e^{(s+t)X})\mathbf{E}(e^{(s-t)Y}) = \mathbf{E}(e^{s(X+Y)+t(X-Y)})\\ &= \mathbf{E}(e^{s(X+Y)})\mathbf{E}(e^{t(X-Y)}),\end{aligned} \tag{12}$$

by the independence of $X + Y$ and $X - Y$

$$= (M(s))^2 M(t)M(-t),$$

using the independence of X and Y again. Next we note that, by the conditions of the problem, $M'(0) = \mathbf{E}(X) = 0$ and $M''(0) = \mathbf{E}(X^2) = 1$. Now differentiating (12) twice with respect to t, and then setting $t = 0$ gives

$$\begin{aligned}M(s)M''(s) - (M'(s))^2 &= (M(s))^2(M(0)M''(0) - (M'(0))^2)\\ &= (M(s))^2(\mathbf{E}(X^2) - (\mathbf{E}(X))^2) = (M(s))^2.\end{aligned}$$

Integrating this differential equation once gives

$$\frac{M'(s)}{M(s)} = s$$

and integrating again yields $M(s) = \exp(\frac{1}{2}s^2)$. Because this is the moment generating function of the $N(0, 1)$ density, it follows that X and Y have this density. ●

We are of course already aware from Chapter 6 that generating functions are of considerable value in handling sums of independent random variables.

(13) Example Let $(X_1, \ldots, X_n)$ be independent having the $N(0, 1)$ density. Show that $Y = \sum_{i=1}^n X_i^2$ has a $\chi^2(n)$ density.

Solution With a view to using moment generating functions, we first find

$$\mathbf{E}(e^{tX_1^2}) = \int_{-\infty}^{\infty} \frac{1}{(2\pi)^{\frac{1}{2}}} \exp\left(tx^2 - \frac{1}{2}x^2\right) dx = \frac{1}{(1-2t)^{\frac{1}{2}}}.$$

Hence,

(14)
$$\begin{aligned} \mathbf{E}(e^{tY}) &= (\mathbf{E}(e^{tX_1^2}))^n \quad \text{by independence} \\ &= \frac{1}{(1-2t)^{\frac{n}{2}}} \end{aligned}$$

and by (7.1.28) and (7.5.6), this is the m.g.f. of the $\chi^2(n)$ density. Hence, by Theorem 7.5.9, Y has a $\chi^2(n)$ density. ●

Many results about sums of random variables can now be established by methods which, if not trivial, are at least straightforward.

(15) Example Let $(X_k; k \geq 1)$ be independent and identically distributed exponential random variables, with parameter λ. Then for $S_n = \sum_{k=1}^n X_k$,

$$\begin{aligned} \mathbf{E}(e^{tS_n}) &= (\mathbf{E}(e^{tX_1}))^n \quad \text{by independence} \\ &= \left(\frac{\lambda}{\lambda - t}\right)^n \quad \text{by Example 7.5.4,} \end{aligned}$$

and so S_n has a gamma distribution by (7.5.6). ●

8.6 Conditional Density and Expectation

Suppose that X and Y have joint density $f(x, y)$, and we are given the value of Y. By analogy with the conditional mass function that arises when X and Y are discrete, we make the following definition.

(1) **Definition** *If X and Y have joint density $f(x, y)$, then the* conditional density *of X given $Y = y$ is given by*

(2)
$$f_{X|Y}(x|y) = \begin{cases} \dfrac{f(x, y)}{f_Y(y)} & \text{if } 0 < f_Y(y) < \infty \\ 0 & \text{elsewhere.} \end{cases}$$
▲

We observe immediately that $f_{X|Y}(x|y)$ is indeed a density, because it is nonnegative and

$$\int_{-\infty}^{\infty} f_{X|Y}(x|y)\,dx = \int_{-\infty}^{\infty} \frac{f(x, y)}{f_Y(y)}\,dx = \frac{1}{f_Y(y)} \cdot f_Y(y) = 1.$$

The corresponding *conditional distribution function* is

$$F_{X|Y}(x, y) = \int_{-\infty}^{x} f_{X|Y}(u|y)\,du = \mathbf{P}(X \le x|Y = y),$$

and we have the Key Rule

(3)
$$\mathbf{P}(X \in A|Y = y) = \int_{x \in A} f_{X|Y}(x|y)\,dx.$$

(4) **Example** Let (X, Y) be the coordinates of the point Q uniformly distributed on a circular disc of unit radius. What is $f_{Y|X}(y|x)$?

Solution Recall that for the marginal density $f_X(x) = (2/\pi)(1 - x^2)^{\frac{1}{2}}$. Hence, by definition,

$$\begin{aligned} f_{Y|X}(y|x) &= \frac{f(x, y)}{f_X(x)} = \frac{1}{\pi} \frac{\pi}{2(1 - x^2)^{\frac{1}{2}}} \\ &= \frac{1}{2}(1 - x^2)^{-\frac{1}{2}} \quad \text{for } |y| < (1 - x^2)^{\frac{1}{2}}. \end{aligned}$$

This conditional density is uniform on $(-(1 - x^2)^{\frac{1}{2}}, (1 - x^2)^{\frac{1}{2}})$, which is consistent with our earlier observations about conditioning of uniform densities. ●

(5) **Example** Let X and Y be independent and exponential with parameter λ. Show that the density of X conditional on $X + Y = v$ is uniform on $(0, v)$.

Solution To use (1), we need to take some preliminary steps. First note that the joint density of X and Y is $f(x, y) = \lambda^2 e^{-\lambda(x+y)}$ for $x > 0$, $y > 0$.

Next we need the joint density of X and $X + Y$ so we consider the transformation $u = x$ and $v = x + y$, with inverse $x = u$ and $y = v - u$, so that $J = 1$. Hence, by Theorem 8.2.1,

$$f_{U,V}(u, v) = \lambda^2 e^{-\lambda v} \quad \text{for } 0 < u < v < \infty.$$

It follows that

$$f_V(v) = \int_0^v \lambda^2 e^{-\lambda v}\, du = \lambda^2 v e^{-\lambda v},$$

and so by definition

$$f_{U|V}(u|v) = \frac{f(u, v)}{f_V(v)} = \frac{1}{v} \quad \text{for } 0 < u < v.$$

This is the required uniform density. ●

This striking result is related to the lack-of-memory property of the exponential density.

Now, because $f_{X|Y}(x|y)$ is a density it may have an expected value, which naturally enough is called *conditional expectation*.

(6) **Definition** *If* $\int_{\mathbb{R}} |x| f_{X|Y}(x|y)\, dx < \infty$, *then the* conditional expectation *of* X *given* $Y = y$ *is given by*

$$\mathbf{E}(X|Y = y) = \int_{\mathbb{R}} x f_{X|Y}(x|y) dx. \quad ▲$$

(7) **Example (5) Revisited** If X and Y are independent and exponential, then we showed that the density of X given $X + Y = v$ is uniform on $(0, v)$. Hence,

$$\mathbf{E}(X|X + Y = v) = \frac{1}{2}v. \quad ●$$

Actually, this is otherwise obvious because, for reasons of symmetry,

$$\mathbf{E}(X|X + Y = v) = \mathbf{E}(Y|X + Y = v),$$

and trivially $\mathbf{E}(X + Y|X + Y = v) = v$. Hence the result follows, provided it is true that for random variables X, Y and V, we have

(8) $$\mathbf{E}(X + Y|V = v) = \mathbf{E}(X|V = v) + \mathbf{E}(Y|V = v).$$

In fact, this is true as we now show.

(9) **Theorem** *Let* X, Y, *and* V *have joint density* $f(x, y, v)$. *Then (8) holds.*

Proof The joint density of $W = X + Y$ and V is $\int_{-\infty}^{\infty} f(w - u, u, v)\, du$.

Then, by definition,

$$\mathbf{E}(X + Y|V = v) = \frac{1}{f_V(v)} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} w f(w - u, u, v)\, du dw.$$

Now consider the transformation $x = w - u$ and $y = u$, with inverse $w = x + y$ and $u = y$, so that $J = 1$. Changing the variables in the double integral accordingly and using

standard results about such double integrals shows that

$$\begin{aligned}\mathbf{E}(X+Y|V=v) &= \frac{1}{f_V(v)}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x+y)f(x,y,v)\,dx\,dy \\ &= \frac{1}{f_V(v)}\int_{-\infty}^{\infty} x f_{X,V}(x,v)\,dx + \frac{1}{f_V(v)}\int_{-\infty}^{\infty} y f_{Y,V}(y,v)\,dy \\ &= \mathbf{E}(X|V=v)+\mathbf{E}(Y|V=v). \qquad \blacksquare\end{aligned}$$

Next, we make the important observation that by writing

$$\psi(y) = \mathbf{E}(X|Y=y) \tag{10}$$

we emphasize the fact that the conditional expectation of X given Y is a function of Y. If the value of Y is left unspecified, we write

$$\psi(Y) = \mathbf{E}(X|Y)$$

on the understanding that when $Y = y$, $\psi(Y)$ takes the value $\mathbf{E}(X|Y=y)$ defined above.

It is therefore natural to think of $\mathbf{E}(X|Y)$ as a random variable that is a function of Y. (A more rigorous analysis can indeed justify this assumption.) Just as in the discrete case, its expected value is $\mathbf{E}(X)$.

(11) Theorem *The expected value of $\psi(Y)$ is $\mathbf{E}(X)$; thus, $\mathbf{E}X = \mathbf{E}(\mathbf{E}(X|Y))$.*

Proof Because $\psi(Y)$ is a function of Y, we can calculate its expected value in the usual way as

$$\begin{aligned}\mathbf{E}(\psi(Y)) &= \int_{-\infty}^{\infty}\psi(y)f_Y(y)\,dy \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x f_{X|Y}(x|y)f_Y(y)\,dx\,dy \qquad \text{by Definition 6} \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x f(x,y)\,dx\,dy = \mathbf{E}(X). \qquad \blacksquare\end{aligned}$$

We recall an earlier example.

(12) Example Let X and Y have density $f(x,y) = 8xy$ for $0 < y < x < 1$. Find $\mathbf{E}(X|Y)$ and $\mathbf{E}(Y|X)$.

Solution Because

$$f_X(x) = \int_0^x 8xy\,dy = 4x^3,$$

and

$$f_Y(y) = \int_y^1 8xy\,dx = 4y(1-y^2),$$

we have $f_{X|Y}(x|y) = 2x/(1-y^2)$, $y < x < 1$. Hence,

$$\mathbf{E}(X|Y) = \int_Y^1 \frac{2x^2}{(1-Y)^2} dx = \frac{2}{3}\left(\frac{1-Y^3}{1-Y^2}\right).$$

Likewise, $f_{Y|X}(y|x) = 2y/x^2 \quad 0 < y < x$ and, therefore,

$$\mathbf{E}(Y|X) = \int_0^X \frac{2y^2}{X^2}\, dy = \frac{2}{3}X. \qquad \bullet$$

The identity in Theorem 11 can also be used to calculate probabilities by the simple device of letting X be the indicator of the event of interest.

(13) Example Let U and Y have density $f(u, y)$. What is $\mathbf{P}(U < Y)$?

Solution Let X be the indicator of $U < Y$. Then

$$\begin{aligned}\mathbf{P}(U < Y) = \mathbf{E}(X) &= \mathbf{E}(\mathbf{E}(X|Y)) \quad \text{by Theorem 11} \\ &= \mathbf{E}\left(\int_{-\infty}^{Y} f(u|Y)\, du\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{y} f(u, y)\, du dy \quad \text{by (2).}\end{aligned}$$

Of course, we could have written this down immediately by Theorem 8.1.9. $\bullet$

(14) Example: Bivariate Normal Let X and Y have the bivariate normal density

$$f(x, y) = \frac{1}{2\pi\sigma\tau(1-\rho^2)^{\frac{1}{2}}} \exp\left(-\frac{1}{2(1-\rho^2)}\left(\frac{x^2}{\sigma^2} - \frac{2\rho xy}{\sigma\tau} + \frac{y^2}{\tau^2}\right)\right). \tag{15}$$

(a) Find the conditional density of X given $Y = y$.
(b) Find $\mathbf{E}(e^{tXY})$, and hence find the density of $Z = X_1Y_1 + X_2Y_2$, where (X_1, Y_1) is independent of (X_2, Y_2) and each has the density (15).

Solution (a) From Example 8.1.19, we know that Y has the $N(0, \tau^2)$ density. Hence,

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{1}{\sigma(2\pi(1-\rho^2)^{1/2})} \exp\left(-\frac{1}{2(1-\rho^2)}\left(\frac{x}{\sigma} - \frac{\rho y}{\tau}\right)^2\right).$$

Hence, the conditional density of X given $Y = y$ is $N(\rho\sigma y/\tau, \sigma^2(1-\rho^2))$. Note that if $\rho = 0$, then this does not depend on y, which is to say that X is independent of Y.

(b) By part (a), the conditional moment generating function of X given Y is

$$M_{X|Y}(t) = \exp\left(\frac{\rho\sigma Y}{\tau}t + \frac{1}{2}\sigma^2(1-\rho^2)t^2\right).$$

Hence, by conditional expectation,

$$\begin{aligned}\mathbf{E}(e^{tXY}) &= \mathbf{E}(\mathbf{E}(e^{tXY}|Y)) = \mathbf{E}(M_{X|Y}(tY)) \\ &= \mathbf{E}\left(\exp\left[\left(\frac{\rho\sigma}{\tau}t + \frac{1}{2}\sigma^2(1-\rho^2)t^2\right)Y^2\right]\right) \\ &= \frac{1}{(1-2\rho\sigma\tau t - \sigma^2\tau^2(1-\rho^2)t^2)^{\frac{1}{2}}} \quad \text{using Example 8.5.13,}\end{aligned}$$

and so $X_1Y_1 + X_2Y_2$ has moment generating function,

$$\begin{aligned}M(t) &= (1-2\rho\sigma\tau t - \sigma^2\tau^2(1-\rho^2)t^2)^{-1} \\ &= \frac{1-\rho}{2}\frac{1}{1+\sigma\tau(1-\rho)t} + \frac{1+\rho}{2}\frac{1}{1-\sigma\tau(1+\rho)t}.\end{aligned}$$

Hence, $Z = X_1Y_1 + X_2Y_2$ has an asymmetric bilateral exponential density,

$$f(z) = \begin{cases} \frac{1+\rho}{2}\exp(-\sigma\tau(1+\rho)z) & \text{if } z > 0 \\ \frac{1-\rho}{2}\exp(\sigma\tau(1-\rho)z) & \text{if } z < 0. \end{cases} \tag{16}$$

●

We note without proof that $\psi(Y)$ has the useful properties that we recorded in the discrete case. Among the most important is that

$$\mathbf{E}(Xg(Y)|Y) = g(Y)\psi(Y) \tag{17}$$

for any function $g(Y)$ of Y.

Finally, we stress that conditional expectation is important in its own right, it should not be regarded merely as a stage on the way to calculating something else.

For example, suppose that X and Y are random variables, and we want to record the value of X. Unfortunately, X is inaccessible to measurement, so we can only record the value of Y. Can this help us to make a good guess at X?

First, we have to decide what a "good" guess $g(Y)$ at X is. We decide that $g_1(Y)$ is a better guess than $g_2(Y)$ if

$$\mathbf{E}[(g_1(Y) - X)^2] < \mathbf{E}[(g_2(Y) - X)^2]. \tag{18}$$

According to this (somewhat arbitrary) rating, it turns out that the best guess at X given Y is $\psi(Y) = \mathbf{E}(X|Y)$.

(19) Theorem *For any function $g(Y)$ of Y,*

$$\mathbf{E}[(X - g(Y))^2] \geq \mathbf{E}[(X - \psi(Y))^2].$$

Proof Using (17), we have

$$\mathbf{E}[(X-\psi)(\psi-g)] = \mathbf{E}[(\psi-g)\mathbf{E}(X-\psi|Y)] = 0. \tag{20}$$

Hence,

$$\begin{aligned}\mathbf{E}[(X-g)^2] &= \mathbf{E}[(X-\psi+\psi-g)^2] = \mathbf{E}[(X-\psi)^2] + \mathbf{E}[(\psi-g)^2] \text{ by (20)} \\ &\geq \mathbf{E}[(X-\psi)^2].\end{aligned}$$

■

We conclude this section by recording one more useful property of conditional densities, which may be called the *continuous partition rule*. The proof is an easy *exercise*. For continuous random variables X and Y,

$$f_X(x) = \int_{\mathbb{R}} f_{X|Y}(x|y) f_Y(y)\, dy. \tag{21}$$

8.7 Transformations: Order Statistics

We introduced the change of variable technique in Section 8.2. We return to this topic to consider a particularly important class of transformations, namely, those that are linear. Thus, let the random vector $(X_1, \ldots, X_n)$ have joint density $f(x_1, \ldots, x_n)$. Suppose that for $1 \leq i, j \leq n$, and some constants a_{ij},

$$Y_i = \sum_{j=1}^{n} a_{ij} X_i. \tag{1}$$

What can be said about the joint density of $(Y_1, \ldots, Y_n)$?

In Section 8.2, we required such transformations to be invertible, and we make the same restriction here. Therefore, we suppose that the matrix $A = (a_{ij})$ has an inverse $A^{-1} = (b_{ij}) = B$. A sufficient condition for this is that the determinant $\det A$ is not zero. Then we have the following useful result, which we state without proof.

(2) Theorem *Suppose that $(X_1, \ldots, X_n)$ has density $f_X(x_1, \ldots, x_n)$, and that $(Y_1, \ldots, Y_n)$ is related to $(X_1, \ldots, X_n)$ by*

$$Y_i = \sum_{j=1}^{n} a_{ij} X_j \quad \text{and} \quad X_i = \sum_{j=1}^{n} b_{ij} Y_j,$$

where $BA = I$, the identity matrix, and $\det A \neq 0$. Then the density of $(Y_1, \ldots, Y_n)$ is given by

$$\begin{aligned} f_Y(y_1, \ldots, y_n) &= \frac{1}{|\det A|} f_X(x_1(y_1, \ldots, y_n) \ldots x_n(y_1, \ldots, y_n)) \\ &= |\det B| f_X(x_1, \ldots, x_n). \end{aligned} \tag{3}$$

(4) Example: Normal Sample Let $(X_1, \ldots, X_n)$ be independent $N(0, 1)$ random variables, and define

$$Y_j = \sum_{j=1}^{n} x_j a_{ij} \quad \text{for } 1 \leq j \leq n, \tag{5}$$

where the matrix $A = (a_{ij})$ is an orthogonal rotation with $\det A = 1$ and, denoting the transpose of A by A^T,

$$AA^T = I = A^T A.$$

(a) Show that $(Y_1, \ldots, Y_n)$ are independent $N(0, 1)$ random variables.
(b) Deduce that the sample mean

$$\overline{X} = \frac{1}{n} \sum_{1}^{n} X_i$$

and the sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2$$

are independent, and that $(n-1)s^2$ has a χ^2 density. ●

Solution (a) It is convenient to use the standard notation for vectors and matrices in problems of this type. Thus, we write x^T for the transpose of x, where

$$x = (x_1, \ldots, x_n) = yA^T \quad \text{from (5).} \tag{6}$$

Furthermore,

$$\sum_{i=1}^{n} x_i^2 = xx^T = yA^T Ay^T = yy^T = \sum_{i=1}^{n} y_i^2.$$

Hence, by (3), $(Y_1, \ldots, Y_n)$ have density

$$f_Y = \frac{1}{(2\pi)^{\frac{n}{2}}} \exp\left(-\frac{1}{2} \sum_{i=1}^{n} y_i^2 \right). \tag{7}$$

Because this factorizes, $(Y_1, \ldots, Y_n)$ are independent with the $N(0, 1)$ density.

(b) Now let (a_{ij}) be any rotation such that $a_{1j} = n^{-\frac{1}{2}}$, giving

$$Y_1 = \sum_{j=1}^{n} \frac{1}{\sqrt{n}} X_j = \sqrt{n}\,\overline{X}.$$

Then

$$\begin{aligned}(n-1)s^2 &= \sum_{i=1}^{n} X_i^2 - 2\overline{X} \sum_{i=1}^{n} X_i + n\overline{X}^2 \\ &= \sum_{i=1}^{n} X_i^2 - n\overline{X}^2 \\ &= \sum_{i=1}^{n} Y_i^2 - Y_1^2 = \sum_{2}^{n} Y_i^2.\end{aligned}$$

Hence, s^2 is independent of $\overline{X}$, by the independence of Y_1 and $(Y_2, \ldots, Y_n)$. Finally, because each Y_i is $N(0, 1)$, $(n-1)s^2$ has a $\chi^2(n-1)$ density by Example 8.5.13. See Problem 8.43 for another way to do this. ●

A particularly important linear transformation is the one that places $(X_1, \ldots, X_n)$ in non-decreasing order. Thus,

$$\begin{aligned} Y_1 &= \text{ smallest of } X_1, \ldots, X_n \\ Y_2 &= \text{ second smallest of } X_1, \ldots, X_n \\ &\vdots \\ Y_n &= \text{ largest of } X_1, \ldots, X_n. \end{aligned}$$

We assume that each X_k has a density $f(x_k)$, so that the chance of ties is zero. It is customary to use the special notation

$$Y_k = X_{(k)},$$

and then $X_{(1)}, X_{(2)}, \ldots, X_{(n)}$ are known as the *order statistics* of $X_1, \ldots, X_n$. Now the above transformation is linear, but not one–one. It is in fact many–one; to see this, suppose that $y_1 < y_2 < \ldots < y_n$. Then the outcomes

$$X_1 = y_1, X_2 = y_2, \ldots, X_n = y_n$$

and

$$X_2 = y_1, X_1 = y_2, \ldots, X_n = y_n$$

both yield the same set of order statistics, namely,

$$X_{(1)} = y_1, X_{(2)} = y_2, \ldots, X_{(n)} = y_n.$$

However, if $(\pi(1), \ldots, \pi(n))$ is any one of the $n!$ distinct permutations of the first n integers and R_π is the region $x_{\pi(1)} < x_{\pi(2)} < \ldots < x_{\pi(n)}$, then the transformation

$$x_{(k)} = x_{\pi(k)}; \quad 1 \le k \le n$$

is one–one and linear. In the notation of (2), we have

$$a_{ij} = \begin{cases} 1 & \text{if } i = \pi(j) \\ 0 & \text{otherwise,} \end{cases}$$

and $|\det A| = 1$. Therefore, the density of $X_{(1)}, \ldots, X_{(n)}$ is $\prod_{i=1}^{n} f(y_i)$.

Now we observe that $X_1, X_2, \ldots, X_n$ lies in just one of the $n!$ regions R_π; hence, the order statistics have joint density

$$n! \prod_{i=1}^{n} f(y_i) \quad \text{for } y_1 < y_2 < \ldots < y_n. \tag{8}$$

Here are some applications of this useful result.

(9) Example Let $(X_1, \ldots, X_n)$ be independently and uniformly distributed on $(0, a)$. Then, by (8), their order statistics have the density

$$f = \frac{n!}{a^n} \quad \text{for } y_1 < y_2 < \ldots < y_n. \tag{10}$$

●

It follows from (8) that we may in principle obtain the marginal density of any subset of the order statistics by performing appropriate integrations. For small subsets, this is actually unnecessary.

(11) Example Show that $X_{(k)}$ has density

$$f_{(k)}(y) = k\binom{n}{k} f(y)(1 - F(y))^{n-k}[F(y)]^{k-1}.$$

Solution The event $X_{(k)} \leq y$ occurs if and only if at least k of the X_i lie in $(-\infty, y]$. Hence,

$$F_{(k)}(y) = \sum_{j=k}^{n} \binom{n}{j} [F(y)]^j (1 - F(y))^{n-j}. \tag{12}$$

Now, differentiating to obtain the density,

$$\begin{aligned} f_{(k)}(y) &= f(y) \sum_{j=k}^{n} \left[j\binom{n}{j} F^{j-1}(1 - F)^{n-j} - (j+1)\binom{n}{j+1} F^j (1 - F)^{n-(j+1)} \right] \\ &= f(y) k \binom{n}{k} F^{k-1}(1 - F)^{n-k} \end{aligned} \tag{13}$$

by successive cancellation in the sum. ●

8.8 The Poisson Process: Martingales

A recurring idea in previous chapters has been that of a series of events or happenings that may occur repeatedly at random times denoted by $T_1, T_2, \ldots$. For example, we have considered light bulbs that may fail and be replaced at $(T_n; n \geq 1)$, or machine bits that may wear out and be renewed at $(T_n; n \geq 1)$ and so on. Other practical problems may also have this structure; for example, the T_n may be the times at which my telephone rings, cars arrive at the toll booth, meteorites fall from the sky, or you get stung by a wasp.

You can think of many more such examples yourself, and it is clear that it would be desirable to have a general theory of such processes. This is beyond our scope, but we can now consider one exceptionally important special case of such processes.

The basic requirement is that the times between events should be independent and identically distributed random variables $(X_k; k \geq 1)$; we assume further that they have an exponential distribution.

(1) Definition *Let $(X_k; k \geq 1)$ be independent identically distributed exponential random variables with parameter λ. Let $T_0 = 0$, and set*

$$T_n = \sum_{1}^{n} X_k; \quad n \geq 1.$$

Define

(2) $$N(t) = \max\{n: T_n \le t\}; \quad t \ge 0.$$

Then $N(t)$ is a Poisson process with parameter λ. ▲

A couple of remarks are in order here. First, note that $N(t)$ is just the number of happenings or events by time t; $N(t)$ is constant until an event occurs, when it increases by 1. Second, the collection $(N(t); t \ge 0)$ is an uncountable collection of random variables. We have said nothing about such collections up to now, and so our analysis of $N(t)$ must of necessity be rather informal.

Our first result explains why $N(t)$ is called a Poisson process.

(3) Theorem *$N(t)$ has mass function*

$$f_N(k) = \frac{e^{-\lambda t}(\lambda t)^k}{k!}; \quad k \ge 0.$$

(4) *Proof* First, we note that from Definition 2 of $N(t)$, the event $N(t) \ge k$ occurs if and only if $T_k \le t$. It follows that

(5) $$\mathbf{P}(N(t) \ge k) = \mathbf{P}(T_k \le t),$$

and because T_k has a gamma density by (8.5.15) we have

$$\begin{aligned} f_N(k) &= \mathbf{P}(N(t) \ge k) - \mathbf{P}(N(t) \ge k+1) = \mathbf{P}(T_k \le t) - \mathbf{P}(T_{k+1} \le t) \\ &= \int_0^t \left(\frac{\lambda^k v^{k-1}}{(k-1)!} - \frac{\lambda^{k+1} v^k}{k!} \right) e^{-\lambda v}\, dv = \frac{e^{-\lambda t}(\lambda t)^k}{k!} \end{aligned}$$

after an integration by parts. As an alternative, we could argue straight from (5) and (6.1.7) that

$$\begin{aligned} \frac{1 - s\mathbf{E}(s^{N(t)})}{1-s} &= \sum_0^\infty s^k \mathbf{P}(N(t) \ge k) = 1 + \sum_{k=1}^\infty s^k \int_0^t \frac{\lambda^k v^{k-1}}{(k-1)!} e^{-\lambda v} dv \\ &= 1 + s\lambda \int_0^t e^{\lambda v s - \lambda v} dv = 1 - \frac{s}{1-s}[e^{\lambda v(s-1)}]_0^t \\ &= 1 + \frac{s}{1-s} - \frac{s e^{\lambda t(s-1)}}{1-s} = \frac{1 - s e^{\lambda t(s-1)}}{1-s}, \end{aligned}$$

and the Poisson mass function of $N(t)$ follows. ■

Our next result is one of the most striking and important properties of $N(t)$, from which many other results flow.

(6) Theorem: Conditional Property of the Poisson Process *Let $N(t)$ be a Poisson process as defined in Definition 1. Conditional on the event $N(t) = k$, the k random variables*

$T_1, \ldots, T_k$ have conditional density

$$f_{T|N=k}(t_1, \ldots, t_k) = \frac{k!}{t^k}; \quad 0 < t_1 < t_2 < \ldots < t_k \leq t. \tag{7}$$

Before proving (7) let us interpret it. From Example 8.7.9, we recognize that the density $k!/t^k$ is the density of the order statistics of k independent random variables, each uniform on $(0, t)$. Thus, Theorem 6 can be more dramatically expressed: given $N(t) = k$, the k events of the process are independently and uniformly distributed on $(0, t)$.

Proof of (6) Because $X_1, \ldots, X_k$ are independent and exponential, they have joint density

$$f(x_1, \ldots, x_k) = \lambda^k \exp(-\lambda(x_1 + \cdots + x_k)). \tag{8}$$

Next, observe that the transformation

$$t_n = \sum_{i=1}^{n} x_i; \quad 1 \leq n \leq k+1$$

is linear and invertible with $|J| = 1$. Hence, by Theorem 8.7.2, the random variables $T_n = \sum_1^n X_i; 1 \leq n \leq k+1$ have joint density

$$f(t_1, \ldots, t_{k+1}) = \lambda^{k+1} e^{-\lambda t_{k+1}}; \quad 0 < t_1 < \ldots < t_{k+1}. \tag{9}$$

Now

$$\begin{aligned} &\mathbf{P}(0 < T_1 < t_1 < T_2 < \ldots < T_k < t_k; N(t) = k) \\ &\qquad = \mathbf{P}(0 < T_1 < t_1 < \ldots < T_k < t_k < t < T_{k+1}) \\ &\qquad = \lambda^k t_1 (t_2 - t_1) \ldots (t_k - t_{k-1}) e^{-\lambda t} \end{aligned}$$

on integrating the density (9). Hence, the conditional distribution of $T_1, \ldots, T_k$ given $N(t) = k$ is given by

$$\begin{aligned} \mathbf{P}(T_1 \leq t_1 < \ldots < T_k \leq t_k | N(t) = k) &= \mathbf{P}(T_1 \leq t_1 < \ldots < t_k; N(t) = k)/\mathbf{P}(N(t) = k) \\ &= t_1(t_2 - t_1) \ldots (t_k - t_{k-1}) \frac{k!}{t^k}. \end{aligned} \tag{10}$$

Now differentiating (10) with respect to all of $t_1, \ldots, t_k$ gives (7), as required. ■

As we have remarked, this result finds many applications, see Example 8.17 for some of them. For the moment we content ourselves with showing that $N(t)$ has the so-called *independent increments* property.

(11) Theorem: The Poisson Process has Independent Increments *Let $N(t)$ be a Poisson process, as usual, and let $s < t \leq u < v$. Then $N(t) - N(s)$ is independent of $N(v) - N(u)$.*

Proof Let $W = N(t) - N(s)$ and $Z = N(v) - N(u)$. Then, by conditional expectation,

$$\mathbf{E}(w^W z^Z) = \mathbf{E}[\mathbf{E}(w^W z^Z | N(v))]. \tag{12}$$

However, conditional on $N(v) = k$, these events are independently and uniformly distributed in $(0, v)$, whence $(W, Z, k - W - Z)$ has a trinomial distribution with

$$\mathbf{E}(w^W z^Z | N(v) = k) = \left(\left(\frac{t-s}{v} \right) w + \left(\frac{v-u}{v} \right) z + \left(\frac{s+u-t}{v} \right) \right)^k. \tag{13}$$

Hence, combining (12) and (13) and Theorem (3) gives

$$\begin{aligned} \mathbf{E}(w^W z^Z) &= \exp(\lambda w(t-s) + \lambda z(v-u) + \lambda(s+u-t) - \lambda v) \\ &= \exp(\lambda(t-s)(w-1) + \lambda(v-u)(z-1)) = \mathbf{E}(w^W)\mathbf{E}(z^Z), \end{aligned} \tag{14}$$

as required. ■

We may also observe from (14) that because $\mathbf{E}(w^W) = \exp(\lambda(t-s)(w-1))$, it follows that W is Poisson with parameter $\lambda(t-s)$. That is to say, $N(t) - N(s)$ has the same mass function as $N(t-s)$. This property may be called *homogeneity* or the property of *stationary increments*. Note that it is possible, and often convenient, to *define* $N(t)$ as a process with stationary independent Poisson-distributed increments. It is then necessary to prove that interevent times are independent identically distributed exponential random variables.

Now looking back at Section 5.6, we see that the simple random walk also has the property of independent increments. In Section 5.7, we found that martingales were particularly useful in analysing the behaviour of such random walks. We may expect (correctly) that martingales will be equally useful here. First, we need to define what we mean by a martingale for an uncountably infinite family of random variables $(X(t); t \geq 0)$.

(15) Definition $(X(t); t \geq 0)$ *is a* martingale *if*

(a) $\mathbf{E}|X(t)| < \infty$, *for all* t.
(b) $\mathbf{E}(X(t)|X(t_1), X(t_2), \ldots, X(t_n), X(s)) = X(s)$, *for any* $0 \leq t_1 < t_2 < \ldots < t_n < s < t$.

It is customary and convenient to rewrite (b) more briefly as

$$(\mathrm{b})' \mathbf{E}(X(t)|X(u); u \leq s) = X(s),$$

but note that this is slightly sloppy shorthand; we cannot condition on an infinite number of values of $X(u)$.

A *stopping time* for $X(t)$ is a random variable T taking values in $[0, \infty)$, such that the event $(T \leq t)$ depends only on values of $X(s)$ for $s \leq t$. ▲

There are many technical details that should properly be dealt with here; we merely note that they can all be satisfactorily resolved and ignore them from now on.

Such martingales and their stopping times can behave much like those with discrete parameter. We state this useful theorem without proof, or technical details.

(16) Theorem: Optional Stopping *Let $X(t)$ be a continuous parameter martingale and T a stopping time for $X(t)$. Then $X(t \wedge T)$ is a martingale, so*

$$\mathbf{E}X(0) = \mathbf{E}X(t \wedge T) = \mathbf{E}X(t).$$

Furthermore, $\mathbf{E}X(T) = \mathbf{E}X(0)$ if any one of the following holds for some constant $K < \infty$:

(a) *T is bounded (i.e., $T \leq K < \infty$).*
(b) *$|X(t)| \leq K$ for all t, and $\mathbf{P}(T < \infty) = 1$.*
(c) *$\mathbf{E}(\sup X(t \wedge T) < \infty$, and $\mathbf{P}(T < \infty) = 1$.*
(d) *$\mathbf{E}|X(T)| \leq K$, and $\mathbf{P}(T < \infty) = 1$, and $\lim_{t\to\infty} \mathbf{E}(X(t)I(T > t)) = 0$.*

Note that similar appropriate definitions and theorems apply to sub- and supermartingales with continuous parameter, but we do not explore that further here. Note further that, just as in the discrete-time case, a popular technique when $\mathbf{P}(T < \infty) = 1$ is to apply part (a) of the theorem at $T \wedge t$, allow $t \to \infty$, and use either the Dominated or Monotone Convergence theorem, as appropriate.

(17) Example: Poisson Martingales If $(N(t); t \geq 0)$ is a Poisson process with parameter λ, then the following are all martingales:

(a) $U(t) = N(t) - \lambda t$.
(b) $V(t) = (U(t))^2 - \lambda t$.
(c) $W(t) = \exp\{-\theta N(t) + \lambda t(1 - e^{-\theta})\}, \theta \in \mathbb{R}$.

To see (a), we calculate

$$\begin{aligned}\mathbf{E}(U(t)|N(u); 0 \leq u \leq s) &= \mathbf{E}(N(t) - N(s) + N(s)|N(u), 0 \leq u \leq s)\\ &= U(s) + \mathbf{E}[N(t) - N(s)] - \lambda(t - s)\\ &= U(s),\end{aligned}$$

where we used the independent increments property. The proof of (b) proceeds likewise:

$$\begin{aligned}&\mathbf{E}(V(t)|N(u), 0 \leq u \leq s) =\\ &\quad = \mathbf{E}[N(t) - \lambda t - N(s) + \lambda s] + (N(s) - \lambda s)]^2|N(u); 0 \leq u \leq s] - \lambda t\\ &\quad = [U(s)]^2 + \lambda(t - s) - \lambda t = V(s),\end{aligned}$$

on using the independence of the increments and var $[N(t) - N(s)] = \lambda(t - s)$. Part (c) is left as an exercise, but see also Example (8.22). ●

8.9 Two Limit Theorems

Perhaps surprisingly (as we are toward the end of the book), this is an appropriate moment to reconsider our basic ideas about chance. Suppose we are given a number n of similar observations or measurements, denoted by $x_1, \ldots, x_n$. For example, the x_i may be the height of each of n men, or they may be the lifetimes of n light bulbs, or they may be the weight of potatoes yielded by each of n plants. By "similar" in this context, we mean that no measurement has any generic reason to be larger or smaller than the others; the

potatoes are of the same variety and grown in the same circumstances; the light bulbs are of the same make and type, and the men are of the same age and race.

It is convenient to have one number that gives an idea of the size of a typical x_i, and a popular candidate for this number is the average $\bar{x}$ given by

$$\bar{x}_n = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

One reason for the popularity of $\bar{x}$ is that it is empirically observed that, as n increases, the sequence $\bar{x}_n$ undergoes smaller and smaller fluctuations, and indeed exhibits behaviour of the kind we call convergent.

A special case of such measurements arises when each x_i takes the value 1 or 0 according to whether some event A occurs. Then $\bar{x}_n$ is the proportion of times that A occurs in n trials, and the fact that $\bar{x}_n$ fluctuates less and less as n increases is sometimes used as a basis to justify the axioms of probability.

Of course, in mathematical terms, we think of x_i as the outcome of some random variable X_i. It follows that, if our theory of probability is as relevant as we have claimed, then the sequence

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

ought also to exhibit a similar kind of regularity in the long run as $n \to \infty$. What kind might there be?

To gain some insight into the problem, consider the case when each x_i is 1 if A occurs, and 0 otherwise. In the mathematical formulation of this, X_i is the indicator of the event A, and we assume the X_i are independent. Then $\sum_1^n X_i$ is a binomial random variable with parameters n and p, and we have shown in Examples 4.18 and 7.5.11 that, as $n \to \infty$,

$$\mathbf{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n} X_i - p\right| > \epsilon\right) \to 0 \tag{1}$$

and

$$\mathbf{P}\left(\frac{1}{(npq)^{\frac{1}{2}}}\sum_{i=1}^{n}(X_i - p) \le x\right) \to \Phi(x), \tag{2}$$

where $\Phi(x)$ is the standard normal distribution. (Indeed, we proved something even stronger than (1) in Example 4.18 and Theorem 5.8.8.)

It seems that $n^{-1}\sum_1^n(X_i - \mathbf{E}(X_1))$ is settling down around $\mathbf{E}(X_i) = p$ and that the distribution of $(n \operatorname{var}(X_1))^{-\frac{1}{2}}\sum_1^n(X_i - \mathbf{E}(X_1))$ is getting closer to the standard normal distribution $\Phi(x)$. More generally, we showed in Theorem 5.8.6 that (1) holds for any collection of independent discrete random variables with the same mean and variance. We called this the weak law of large numbers.

This is deliberately vague and informal, but it should now seem at least plausible that the following results might be true.

(3) Theorem *Let $(X_k; k \geq 1)$ be independent and identically distributed random variables with mean μ, variance $\sigma^2 < \infty$, and moment generating function $M_X(t)$, $|t| < a$. Then we have:*

(i) **Weak Law of Large Numbers** For $\varepsilon > 0$, as $n \to \infty$,

$$\mathbf{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)\right| > \varepsilon\right) \to 0. \tag{4}$$

(ii) **Central Limit Theorem** As $n \to \infty$,

$$\mathbf{P}\left(\frac{1}{\sigma\sqrt{n}}\sum_{i=1}^{n}(X_i - \mu) \leq x\right) \to \Phi(x) = \int_{-\infty}^{x}(2\pi)^{-\frac{1}{2}}e^{-\frac{1}{2}y^2}\,dy. \tag{5}$$

It is a remarkable fact that both of these are indeed true, and we now prove them.

Proof of (4) The essential step here is to recall Chebyshov's inequality, for then we may write, using Theorem 7.4.16,

$$\begin{aligned}\mathbf{P}\left(\frac{1}{n}\left|\sum_{1}^{n}(X_i - \mu)\right| > \varepsilon\right) &\leq \varepsilon^{-2}\mathbf{E}\left(n^{-2}\left(\sum_{1}^{n}(X_i - \mu)\right)^2\right)\\ &= \varepsilon^{-2}n^{-2}\mathbf{E}\left(\sum_{1}^{n}(X_i - \mu)^2\right) \quad \text{by independence,}\\ &= \varepsilon^{-2}n^{-1}\sigma^2 \to 0 \quad \text{as } n \to \infty\end{aligned}$$

■

Note that the proof here is the same as that of Theorem 5.8.6. So it is not necessary for (4) that the X_i be identically distributed or have an m.g.f., it is sufficient that they have the same mean and variance.

Proof of (5) The essential step here is to recall the continuity theorem (7.5.10), for then we may write

$$\begin{aligned}\mathbf{E}\left(\exp\left[\frac{t}{\sigma\sqrt{n}}\sum_{1}^{n}(X_i - \mu)\right]\right) &= \left(\mathbf{E}\left(\exp\left[\frac{t}{\sigma\sqrt{n}}(X_1 - \mu)\right]\right)\right)^n, \quad \text{by independence}\\ &= (M_Y(t/(\sigma\sqrt{n})))^n \quad \text{where } Y = X - \mu,\\ &= \left(1 + \mathbf{E}(Y^2)\frac{t^2}{2\sigma^2 n} + o\left(\frac{t^2}{n}\right)\right)^2 \quad \text{by Theorem 7.5.9}\\ &\to e^{\frac{1}{2}t^2}, \text{ because } \mathbf{E}(Y^2) = \sigma^2.\end{aligned}$$

Now we recall that $e^{\frac{1}{2}t^2}$ is the moment generating function of the standard normal density $\phi(x)$, and (5) follows by the continuity theorem (7.5.10). ■

8.10 Review and Checklist for Chapter 8

In this chapter, we consider the joint behaviour of collections of continuous random variables having joint density functions. We also introduce the joint distribution function, and show how these yield the marginal densities and distributions. The change of variable technique is given and used to study important functions of sets of random variables (including sums, products, quotients, and order statistics). We look at expectation, independence, and conditioning, especially the key concept of conditional expectation. Finally, we discuss the Poisson process and its crucial properties, together with continuous parameter martingales and the optional stopping theorem, and prove simple forms of the weak law of large numbers and the central limit theorem.

We summarize most of these principal properties for the bivariate case (X, Y). The extension to larger collections of random variables $(X_1, X_2, X_3, \ldots;$ the multivariate case) is straightforward but typographically tedious.

SYNOPSIS OF FORMULAE:

The random vector (X, Y) is supposed to have joint density $f(x, y)$ and distribution $F(x, y)$.

Key rule:

$$\mathbf{P}[(X, Y) \in B] = \int_{(x,y)\in B} f(x, y)\, dxdy.$$

Basic rules:

$$F(x, y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f(u, v)\, dudv = \mathbf{P}(X \le x, Y \le y).$$

For small h and k,

$$\mathbf{P}(x < X \le x + h, y < Y \le y + k) \simeq f(x, y)\, hk,$$

and when F is differentiable,

$$\frac{\partial^2 F(x, y)}{\partial x \partial y} = f(x, y).$$

Marginals:

$$f_X(x) = \int_{\mathbb{R}} f(x, y)dy; \quad f_Y(y) = \int_{\mathbb{R}} f(x, y)dx,$$
$$F_X(x) = F(x, \infty); \quad F_Y(y) = F(\infty, y).$$

Functions:

$$\mathbf{P}(g(X, Y) \le z) = \int\int_{g\le z} f(x, y)\, dxdy.$$

In particular,

$$f_{X+Y}(z) = \int_{\mathbb{R}} f(x, z-x)\,dx.$$

Transformations:
More generally, suppose $(u, v) = (u(x, y), v(x, y))$ defines a one–one invertible function; let

$$J(u, v) = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial y}{\partial u}\frac{\partial x}{\partial v},$$

where the derivatives are continuous in the domain of (u, v). Then the random variables $(U, V) = (u(X, Y), v(X, Y))$ are jointly continuous with density

$$f_{U,V}(u, v) = f_{X,Y}(x(u, v), y(u, v))|J(u, v)|.$$

Independence: X and Y are independent, if and only if

$$F_{X,Y}(x, y) = F_X(x)F_Y(y), \qquad \text{for all } x, y,$$

or

$$f_{X,Y}(x, y) = f_X(x)f_Y(y), \qquad \text{for all } x, y.$$

Expectation: If $\int\int |g| f(x, y)dxdy < \infty$, then $g(X, Y)$ has an expected value given by

$$\mathbf{E}g(X, Y) = \int\int g(x, y)f(x, y)dxdy.$$

Moments: Joint moments, covariance, and correlation are defined as they were in the discrete case.
Joint generating functions: The joint probability generating function of integer-valued X and Y is

$$G_{X,Y}(x, y) = \mathbf{E}(x^X y^Y).$$

The joint moment generating function of X and Y is

$$M_{X,Y}(s, t) = \mathbf{E}(e^{sX+tY}) = G_{X,Y}(e^s, e^t).$$

Moments are obtained by appropriate differentiation, so

$$\operatorname{cov}(X, Y) = \begin{cases} \frac{\partial^2 G}{\partial x \partial y}(1, 1) - \frac{\partial G}{\partial x}(1, 1)\frac{\partial G}{\partial y}(1, 1) & \text{if } X, Y \text{ are discrete} \\ \frac{\partial^2 M}{\partial s \partial t}(0, 0) - \frac{\partial M}{\partial s}(0, 0)\frac{\partial M}{\partial t}(0, 0) & \text{in any case.} \end{cases}$$

Random variables X and Y are independent if and only if the moment generating function factorizes as a product of separate functions of s and t for all s, t; thus,

$$M(s, t) = M(s, 0)M(0, t).$$

Conditioning: The conditional density of X given $Y = y$ is

$$f_{X|Y}(x|y) = \begin{cases} \dfrac{f(x,y)}{f_Y(y)}, & 0 < f_Y(y) < \infty \\ 0, & \text{otherwise.} \end{cases}$$

The Key Rule is

$$\mathbf{P}(X \in A|Y = y) = \int_{x\in A} f_{X|Y}(x|y)\,dx,$$

the continuous partition rule is

$$f_X(x) = \int_{\mathbb{R}} f_{X|Y}(x|y) f_Y(y)\,dy,$$

and the conditional distribution function is

$$F_{X|Y}(x|y) = \int_{-\infty}^{x} f_{X|Y}(x|y)dx = \mathbf{P}(X < x|Y = y).$$

If $\int |x| f_{X|Y}(x|y)dx < \infty$, then the conditional expectation of X given $Y = y$ is

$$\mathbf{E}(X|Y = y) = \int_{\mathbb{R}} x f_{X|Y}(x|y)\,dx.$$

As Y varies, this defines a function $\mathbf{E}(X|Y)$, where $\mathbf{E}(X|Y) = \mathbf{E}(X|Y = y)$ when $Y = y$.
Key theorem for conditional expectation:

$$\mathbf{E}(\mathbf{E}(X|Y)) = \mathbf{E}X.$$

This has the same properties as the conditional expectation in the discrete case.

Limit Theorems: We established the Weak Law of Large Numbers and the Central Limit Theorem in (8.9.3).

Multivariate normal density: Jointly normal random variables are particularly important. Recall that the standard normal density is

$$\phi(x) = f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x^2\right), \quad -\infty < x < \infty.$$

If the random variables X and Y are jointly distributed, and $aX + bY$ has a normal distribution for all choices of the constants a and b, then X and Y are said to have a bivariate normal, or *binormal*, density. In particular, X and Y have the standard binormal density with correlation coefficient (or parameter) ρ, if

$$f(x,y) = (2\pi)^{-1}(1-\rho^2)^{-1/2} \exp\left\{\frac{-1}{2(1-\rho^2)}(x^2 - 2\rho xy + y^2)\right\}.$$

Thus X and Y are independent if and only if uncorrelated.

More generally, $\mathbf{X} = (X_1, \ldots, X_n)$ is said to be multivariate normal (or ***multinormal***) if $\sum_1^n a_i X_i$ has a normal distribution for all choices of $\mathbf{a} = (a_1, \ldots, a_n)$. It follows that the multinormal distribution is determined by the means and covariances of $(X_1, \ldots, X_n)$.

To see this, we simply calculate the joint moment generating function of $\mathbf{X}$:

$$M_{\mathbf{X}}(\mathbf{t}) = \mathbf{E}\exp\left(\sum_1^n t_r X_r\right).$$

This is easy because $\sum_1^n t_r X_r$ is normal with mean $\sum_i t_i \mathbf{E}X_i$ and variance $\sum_{i,j} \operatorname{cov}(X_i, X_j) t_i t_j$. Hence, by Example 7.5.7,

$$M_{\mathbf{X}}(\mathbf{t}) = \exp\left[\sum_i t_i \mathbf{E}X_i + \frac{1}{2}\sum_{i,j} \operatorname{cov}(X_i, X_j) t_i t_j\right],$$

and this determines the joint distribution by the basic property of m.g.f.s.

In particular, X_1, X_2, X_3 have the *standard trivariate normal* distribution (or trinormal) when $\operatorname{var} X_1 = \operatorname{var} X_2 = \operatorname{var} X_3 = 1$, $\mathbf{E}X_1X_2 = \rho_{12}$, $\mathbf{E}X_1X_3 = \rho_{23}$, $\mathbf{E}X_3X_1 = \rho_{31}$, and

$$f(x_1, x_2, x_3) = (2\pi)^{-3/2} A^{-\frac{1}{2}} \exp\left\{-\frac{1}{2A}\sum_j\sum_k a_{jk} x_j x_k\right\},$$

where $a_{11} = 1 - \rho_{23}^2$, $a_{22} = 1 - \rho_{31}^2$, $a_{33} = 1 - \rho_{12}^2$, $a_{12} = a_{21} = \rho_{31}\rho_{23} - \rho_{12}$, $a_{13} = a_{31} = \rho_{12}\rho_{23} - \rho_{31}$, $a_{23} = a_{32} = \rho_{12}\rho_{31} - \rho_{23}$, and $A = 1 - \rho_{12}^2 - \rho_{23}^2 - \rho_{31}^2 + 2\rho_{12}\rho_{23}\rho_{31}$. The joint moment generating function of X_1, X_2, X_3 is

$$M(t_1, t_2, t_3) = \exp\left\{\frac{1}{2}\left(t_1^2 + t_2^2 + t_3^2 + \rho_{12}t_1t_2 + \rho_{23}t_2t_3 + \rho_{31}t_3t_1\right)\right\}.$$

Checklist of Terms for Chapter 8

8.1 joint distribution function
joint density function
marginals
bivariate normal density
8.2 change of variable formula
8.3 independence and factorization
8.4 sums, products, and quotients
8.5 expectation
Cauchy–Schwarz inequality
independence and expectation
covariance and correlation
joint moment generating function
normal sample
8.6 conditional density
conditional distribution
conditional expectation
8.7 normal sample
order statistics
8.8 Poisson process
conditional property

independent increments
martingales
optional stopping theorem
8.9 weak law of large numbers
central limit theorem

WORKED EXAMPLES AND EXERCISES

8.11 Example: Bivariate Normal Density

Let X and Y have the standard bivariate normal joint density

$$f(x, y) = \frac{1}{2\pi(1-\rho^2)^{\frac{1}{2}}} \exp\left(-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right).$$

Show that the joint moment generating function of X and Y is

$$\exp\left(\frac{1}{2}(s^2 + 2\rho st + t^2)\right).$$

Solution We are asked to find

$$M = \mathbf{E}(e^{sX+tY}) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)e^{sx+ty}\,dxdy. \tag{1}$$

After a little thought, we observe that the terms in the exponents in the integrand can be rearranged to give

$$M = \frac{1}{2\pi(1-\rho^2)^{\frac{1}{2}}} \int\int \exp\left(-\frac{x^2}{2} + x(s+t\rho) - \frac{1}{2}\left(\frac{y-\rho x}{(1-\rho^2)^{\frac{1}{2}}}\right)^2 + \frac{y-\rho x}{(1-\rho^2)^{\frac{1}{2}}} t(1-\rho^2)^{\frac{1}{2}}\right) dxdy.$$

This suggests that we make the change of variables $u = x$, $v = (y - \rho x)/(1-\rho^2)^{\frac{1}{2}}$ in the integral. This map is one–one, and $J = (1-\rho^2)^{\frac{1}{2}}$. Hence,

$$M = \frac{1}{2\pi}\int_{-\infty}^{-\infty}\int_{-\infty}^{-\infty} \exp\left(-\frac{1}{2}u^2 + (s+t\rho)u - \frac{1}{2}v^2 + t(1-\rho^2)^{\frac{1}{2}}v\right) dudv.$$

Because the integrand factorizes, we now recognize the right-hand side as being equal to

$$\mathbf{E}(e^{(s+t\rho)U})\mathbf{E}(e^{t(1-\rho^2)^{\frac{1}{2}}V}),$$

where U and V are standard normal random variables. But we know the m.g.f. $\mathbf{E}(e^{tV})$ of a standard normal random variable V to be $e^{\frac{1}{2}t^2}$. Hence,

$$M = e^{\frac{1}{2}(s+t\rho)^2} e^{\frac{1}{2}t^2(1-\rho^2)}, \tag{2}$$

as required.

but

$$|\cos tX + i \sin tX| = \cos^2 tX + \sin^2 tX = 1,$$

and so $|\phi(t)| \leq 1$. If X is exponential,

$$\mathbf{E}e^{itX} = \int_0^\infty \lambda e^{-\lambda x} \cos tx \, dx + i \int_0^\infty \lambda e^{-\lambda x} \sin tx \, dx.$$

Integrating by parts gives

$$\int_0^\infty \lambda e^{-\lambda x} \cos tx \, dx = 1 - \int_0^\infty te^{-\lambda x} \sin tx \, dx = 1 - \int_0^\infty \frac{t^2}{\lambda^2} \lambda e^{-\lambda x} \cos tx \, dx.$$

Hence,

$$\int_0^\infty \lambda e^{-\lambda x} \cos tx \, dx \left(1 - \frac{t^2}{\lambda^2}\right) = 1$$

Likewise,

$$\int_0^\infty \lambda e^{-\lambda x} \sin tx \, dx \left(1 + \frac{t^2}{\lambda^2}\right) = \frac{t}{\lambda}.$$

Hence,

$$\mathbf{E}e^{itX} = \frac{1 + it/\lambda}{1 + t^2/\lambda^2} = \frac{\lambda}{\lambda - it}.$$

Remark $\phi(t)$ is called the *characteristic function* of X. We have occasionally been frustrated by the nonexistence of a moment generating function. This example shows that we can use the characteristic function in such cases because it always exists. For example, it can be shown that the Cauchy density $f(x) = \{\pi(1 + x^2)\}^{-1}$, which has no moments at all, has characteristic function $\phi(t) = e^{-|t|}$.

Furthermore, for any random variable X that does have a moment generating function $M(\theta) = \mathbf{E}e^{\theta X}$, for $|\theta| < \varepsilon > 0$, the characteristic function of X is given by $\phi(t) = M(it)$.

(1) **Exercise** Show that the random variable uniform on $[-1, 1]$ has characteristic function $\phi(t) = \frac{\sin t}{t}$.

(2) **Exercise** If $X_1, \ldots, X_n$ are independent Cauchy random variables, show that $\overline{X} = \frac{1}{n} \sum_1^n X_i$ has the same Cauchy density as the X_i.

(3) **Exercise** Find the characteristic function of the random variable X with density

$$f(x) = \frac{1}{2} e^{-|x|}, \quad x \in \mathbb{R}.$$

PROBLEMS

1 When is $f(x, y) = xy + ax + by + 1$ a joint density function on $0 \leq x, y \leq 1$? Can it be the joint density of random variables X and Y that are independent?

2 Find cov (X, Y) for the joint density of Problem 1.

3 Let X and Y have joint density $f(x, y) = c\exp(-x - y)$ for $x > 0$, $y > 0$. Find (a) c, (b) $\mathbf{P}(X + Y > 1)$, and (c) $\mathbf{P}(X < Y)$.

4 Let X and Y have joint density $f(x, y) = g(x + y)$ for $x \geq 0$, $y \geq 0$. Find the density of $Z = X + Y$.

5 Let X and Y have joint density $f = c(1 + x^2 + y^2)^{-\frac{3}{2}}$ for all x and y. (a) What is c? (b) Find the marginal density of X.

6 Let X and Y have the joint density of Problem 5, and define $W = X^2 + Y^2$, $Z = Y/X$. Show that W and Z are independent.

7 Let U and V be independently and uniformly distributed on (0, 1). Find the joint density of $X = U^{\frac{1}{2}}/(U^{\frac{1}{2}} + V^{\frac{1}{2}})$ and $Y = U^{\frac{1}{2}} + V^{\frac{1}{2}}$. By considering $\mathbf{P}(X \leq x | Y \leq 1)$, devise a rejection sampling procedure for simulating a random variable with density $6x(1 - x); 0 < x < 1$.

8 Let U_1, U_2, and U_3 be independently and uniformly distributed on (0, 1), with order statistics $U_{(1)} < U_{(2)} < U_{(3)}$. Show that the density of $U_{(2)}$ is $6x(1 - x); 0 < x < 1$.

9 Let U_1, U_2, U_3, and U_4 be independently and uniformly distributed on (0, 1). What is the density of $X = \log(U_1U_2)/\log(U_1U_2U_3U_4)$? (*Hint*: One way uses Example 8.15).

10 Let X and Y be independent normal random variables, and set $U = X + Y$, $V = X - Y$. Show that U and V are independent if and only if $\text{var}(X) = \text{var}(Y)$.

11 **Simulation Using Bivariate Rejection** Let U and V be independent and uniform on (0, 1). Define the random variables

$$\begin{aligned} Z &= (2U - 1)^2 + (2V - 1)^2, \\ X &= (2U - 1)(2Z^{-1}\log Z^{-1})^{\frac{1}{2}} \\ Y &= (2V - 1)(2Z^{-1}\log Z^{-1})^{\frac{1}{2}}. \end{aligned}$$

Show that the conditional joint density of X and Y given $Z < 1$, is $(2\pi)^{-1}\exp(-\frac{1}{2}(x^2 + y^2))$. Explain how this provides a method for simulating normal random variables.

12 Let X and Y be independent exponential random variables with respective parameters λ and μ. Find $\mathbf{P}(\max\{X, Y\} \leq aX)$ for $a > 0$.

13 Let X and Y have joint density $f = cye^{-y(x+1)}$, $0 \leq x < y < \infty$ for some constant c. What is the conditional density of X given $Y = y$?

14 A spherical melon has radius 1. Three gravid insects alight independently (for oviposition) at A, B, and C, where A, B, and C are uniformly distributed on the surface of the melon. For any two insects, if the distance between them (along a great circle of the melon) is less than $\pi/2$, then they detect each other's presence and will both fly off to seek an unoccupied melon. Show that the probability that exactly one insect is left in possession of the melon is $3(\pi - 1)/4\pi$, and that the probability that all three remain on the melon is $1/4\pi$.

15 Let X and Y have joint density $f(x, y) = c\sin(x + y)$, $0 < x, y < \pi/2$.
Show that $c = \frac{1}{2}$; $\text{cov}(X, Y) = \frac{1}{2}(\pi - 2) - \frac{1}{16}\pi^2$, and $\rho(X, Y) = (8(\pi - 2) - \pi^2)/(\pi^2 + 8\pi - 32)$.

16 Let X and Y be independent exponential with parameters λ and μ, respectively. Now define $U = X \wedge Y$ and $V = X \vee Y$. Find $\mathbf{P}(U = X)$, and show that U and $V - U$ are independent.

17 Let $(X_i; i \geq 1)$ be independent with the uniform density on $(-1, 1)$. Let the density of $\sum_{i=1}^{n} X_i$ be $f_n(x)$. Show that

$$f_n(x) = \frac{1}{2}\int_{x-1}^{x+1} f_{n-1}(u)\,du \quad \text{for } n \geq 2,$$

and deduce that for any integer k, the density $f_n(x)$ is a polynomial in x for $x \in [k, k + 1)$.

18 Let X and Y have the bivariate normal density of Examples 8.4.3 and 8.20.
(a) If $\sigma = \tau$, what is $\mathbf{E}(X|X + Y)$?
(b) If $\sigma \neq \tau$, what are $\mathbf{E}(X|X + Y)$ and $\mathbf{E}(Y|X + Y)$?

19 Let $(X_n; n \geq 1)$ be independent and uniformly distributed on (0, 1). Define

$$T = \min\left\{n: \sum_{i=1}^{n} X_i > 1\right\}.$$

Show that $\mathbf{P}(T \geq j+1) = 1/j!$ for $j \geq 1$. Deduce that $\mathbf{E}(T) = e$.

20 Let (X_1, X_2, X_3) be independent and uniformly distributed on (0, 1). What is the probability that the lengths X_1, X_2, X_3 can form a triangle?

21 Let $(U_i; i \geq 1)$ be independently and uniformly distributed on (0, 1), and define $M_n = \max\{U_1, \ldots, U_n\}$. Show that, as $n \to \infty$, the distribution of $Z_n = n(1 - M_n)$ converges to an exponential distribution.

22 Let $(X_i; i \geq 1)$ be independent exponential random variables each with parameter μ. Let N be independent of the X_i having mass function $f_N(n) = (1-p)p^{n-1}; n \geq 1$. What is the density of $Y = \sum_{i=1}^{N} X_i$?

23 Let $N(t)$ be a Poisson process, $C(t)$ its current life at t, and X_1 the time of the first event. Show that $\operatorname{cov}(X_1, C(t)) = \frac{1}{2}t^2 e^{-\lambda t}$.

24 **Simulating Gamma** Let U and X be independent, where U is uniform on (0, 1) and X is exponential with parameter $\alpha^{-1} \leq 1$. Show that the density of X conditional on

$$\left(\frac{eX}{\alpha}\right)^{\alpha-1} \exp\left(-\left(\frac{\alpha-1}{\alpha}\right)X\right) \geq U$$

is $x^{\alpha-1}e^{-x}/\Gamma(\alpha)$. Why is this of value?

25 (a) Let X be exponential with parameter 1. Show that X/λ is exponential with parameter λ.

(b) Let $(X_i; 1 \leq i \leq n)$ be independent and exponential with parameter 1. Use the lack-of-memory property of the exponential density to show that $\max\{X_1, \ldots, X_n\}$ has the same distribution as $X_1 + X_2/2 + \ldots + X_n/n$.

26 Let X_1, X_2, X_3, and X_4 be independent standard normal random variables. Show that $W = X_1/X_2$ has the Cauchy density, and $Z = |X_1X_2 + X_3X_4|$ has an exponential density.

27 Let X and Y be independent Poisson random variables each with parameter n. Show that, as $n \to \infty$, $\mathbf{P}(X - Y \leq \sqrt{2n}x) \to \Phi(x)$.

28 Let $(U_i; i \geq 1)$ be a collection of independent random variables each uniform on (0, 1). Let X have mass function $f_X(x) = (e-1)e^{-x}; x \geq 1$ and let Y have mass function $f_Y(y) = 1/\{(e-1)y!\}$ $y \geq 1$. Show that $Z = X - \max\{U_1, \ldots, U_Y\}$ is exponential. (Assume X and Y are independent of each other and of the U_i).

29 Let X and Y be independent gamma with parameters $(\alpha, 1)$ and $(\beta, 1)$, respectively. Find the conditional density of X given $X + Y = z$.

30 Let X and Y be independent standard normal random variables. Show that the pair X and Z, where $Z = \rho X + (1-\rho^2)^{\frac{1}{2}}Y$, $|\rho| \leq 1$, has a standard bivariate normal density.

31 Let X and Y have joint moment generating function $M(s,t)$ and define $K(s,t) = \log M(s,t)$. Show that $K_s(0,0) = \mathbf{E}(X)$, $K_{ss}(0,0) = \operatorname{var}(X)$, and $K_{st}(0,0) = \operatorname{cov}(X,Y)$.

32 A sorcerer has hidden a ring in one of an infinite number of boxes numbered $\ldots, -2, -1, 0, 1, 2, 3, \ldots$. You only have time to look in 11 boxes. The sorcerer gives you a hint. He tosses 100 fair coins and counts the number of heads. He does not tell you this number, nor does he tell you the number of the box with the ring in it, but he tells you the sum of these two numbers.

(a) If the sum is 75, which 11 boxes should you look in?

(b) Give an approximation to the probability of finding the ring.

$$\left(\frac{1}{\sqrt{2\pi}}\int_0^{\frac{11}{10}} e^{-\frac{u^2}{2}}\,du = 0.36.\right)$$

33 **Multivariate Normal Density** Let $(Y_1, \ldots, Y_n)$ be independent, each having the $N(0, 1)$ density. If

$$X_i = \sum_j a_{ij} Y_j + b_i \quad \text{for } 1 \le i, j \le n,$$

then $(X_1, \ldots, X_n)$ are said to have a multivariate normal density. Find the joint m.g.f. $M_n(t_1, \ldots, t_n) = \mathbf{E}(\exp(\sum_{i=1}^n t_i X_i))$. Deduce that the following three statements are equivalent:
(a) The random variables $(X_1, \ldots, X_n)$ are independent.
(b) $(X_1, \ldots, X_n)$ are pairwise independent.
(c) $\operatorname{cov}(X_i, X_j) = 0 \quad \text{for } 1 \le i \ne j \le n$.

34 A sequence of random variables $X_1, X_2, \ldots$ is said to obey the Central Limit Theorem (CLT) if and only if the distribution of $(S_n - \mathbf{E}(S_n))/\sqrt{\operatorname{var}(S_n)}$ tends to the standard normal distribution, where $S_n = \sum_{i=1}^n X_i$.

State sufficient conditions on (X_n) for the sequence to obey the CLT and say which of your conditions are necessary.

Let $(U_n(\lambda_n))$ be a sequence of independent random variables having the Poisson distribution with nonzero means (λ_n). In each of the following cases, determine whether the sequence (X_n) obeys the CLT:

(i) $X_n = U_n(1)$.
(ii) $X_n = U_n(1) + n$.
(iii) $X_n = U_n(\frac{1}{2})!$.
(iv) $X_n = U_{2n}(1)/(1 + U_{2n-1}(1))$.
(v) $X_n = U_n(n)$.

35 Let X_1 and X_2 be independent with the same density $f(x)$. Let U be independent of both and uniformly distributed on $(0, 1)$. Let $Y = U(X_1 + X_2)$. Find $f(x)$ such that Y can also have density $f(x)$.

36 **Molecules** A molecule M has velocity $v = (v_1, v_2, v_3)$ in Cartesian corrdinates. Suppose that v_1, v_2, and v_3 have joint density:

$$f(x, y, z) = (2\pi\sigma^2)^{-\frac{3}{2}} \exp\left(-\frac{1}{2\sigma^2}(x^2 + y^2 + z^2)\right).$$

Show that the density of the magnitude $|v|$ of v is

$$f(w) = \left(\frac{2}{\pi}\right)^{\frac{1}{2}} \sigma^{-3} w^2 \exp\left(-\frac{1}{2\sigma^2} w^2\right), \quad w > 0.$$

37 Let C be a circle radius r with centre O. Choose two points P and Q independently at random in C. Show that the probability that the triangle OPQ contains an obtuse angle is $\frac{3}{2}$.
(*Note*: No integration is required.)

38 Given a fixed line AB, a point C is picked at random such that $\max\{AC, BC\} \le AB$. Show that the probability that the triangle ABC contains an obtuse angle is

$$\frac{1}{8}\pi\left(\frac{1}{3}\pi - \frac{1}{4}\sqrt{3}\right)^{-1}.$$

[*Note*: No integration is required. This is a version of a problem given by Lewis Carroll. To combat insomnia, he solved mathematical problems in his head; this one was solved on the night of 20 January 1884. He collected a number of these mental exercises in a book entitled *Pillow problems* (Macmillan, 1895).]

39 Let X be a nonnegative random variable such that $\mathbf{P}(X > x) > 0$ for all $x > 0$. Show that $\mathbf{P}(X > x) \le \mathbf{E}X^n/x^n$ for all $n \ge 0$. Deduce that $s = \sum_{n=0}^{\infty} \frac{1}{\mathbf{E}X^n} < \infty$.

Let N be a random variable with the mass function $\mathbf{P}(N = n) = \frac{1}{s\mathbf{E}X^n}$.

Show that: (a) for all $x > 0$, $\mathbf{E}x^N < \infty$; (b) $\mathbf{E}X^N = \infty$.

40 Let X and Y be independent and uniform on $[0, 1]$, and let Z be the fractional part of $X + Y$. Show that Z is uniform on $[0, 1]$ and that X, Y, Z are pairwise independent but not independent.

41 Let $(X_{(k)}; 1 \leq k \leq n)$ be the order statistics derived from n independent random variables, each uniformly distributed on $[0, 1]$. Show that

(a) $\mathbf{E}X_{(k)} = \dfrac{k}{n+1}$.

(b) $\operatorname{var} X_{(k)} = \dfrac{k(n-k+1)}{(n+1)^2(n+2)}$.

42 Let $(X_{(k)}; 1 \leq k \leq n)$ be the order statistics derived from n independent random variables each uniformly distributed on $[0, 1]$. Show that they have the same distribution as $(Y_k; 1 \leq k \leq n)$, where $Y_0 = 0$ and, given Y_{j-1}, Y_j, has the density $(1 - y)^{n-j}; Y_{j-1} \leq y \leq 1$, for $1 \leq j \leq n$.

43 **Normal Sample:** Example 8.7.4 Revisited Let $(X_r; 1 \leq r \leq n)$ be independent $N(\mu, \sigma^2)$ random variables.

(a) By considering the joint moment generating function of $\overline{X}$ and $(X_r - \overline{X}; 1 \leq r \leq n)$, show that $\overline{X} = \frac{1}{n}\sum_1^n X_r$ and

$$S^2 = \frac{1}{n-1}\sum_1^n (X_r - \overline{X})^2$$

are independent.

(b) Show that $\overline{X}$ and $X_r - \overline{X}$ are uncorrelated, and deduce that $\overline{X}$ and S^2 are independent.

44 (i) Suppose that the random variable Q has density $\sin q$, on $[0, \pi]$. Find the distribution function of Q, and deduce that $\sin^2\{Q/2\}$ has the uniform density on $[0, 1]$.

(ii) Suppose that the random vector (X, Y) is uniformly distributed on the unit circle, and set $R^2 = X^2 + Y^2$. Show that R^2 has the uniform density on $[0, 1]$. Deduce that the random vector (U, V, W) is uniformly distributed on the unit sphere, where $U = 2X(1 - R^2)^{\frac{1}{2}}$, $V = 2Y(1 - R^2)^{\frac{1}{2}}$, and $W = 1 - 2R^2$.